Beyond-Birthday-Bound Security with HCTR2: Cascaded Construction and Tweak-based Key Derivation*

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Abstract. The block cipher (BC) mode for realizing a variable-input-length strong tweakable pseudorandom permutation (VIL-STPRP), also known as the accordion mode, is a rapidly growing research field driven by NIST's standardization project, which considers AES as a primitive. Widely used VIL-STPRP modes, such as HCTR2, have birthday-bound security and provide only 64-bit security with AES. To provide higher security, NIST is considering two directions: to develop new modes with beyond-birthday-bound (BBB) security and to use Rijndael-256-256 with HCTR2. This paper pursues the first direction while maintaining compatibility with HCTR2. In particular, we provide two solutions to achieve BBB security for two different approaches: (i) general cases without any conditions on the tweak and (ii) under the condition that the same tweak is not repeated too often as adopted in *bbb-ddd-AES* recently presented at Eurocrypt 2025. For the first approach, we propose a new mode, CHCTR, that iterates HCTR2 with two independent keys, which achieves $2n/3$-bit security in the multi-user (mu) setting and satisfies NIST's requirements. For the second approach, we prove mu security of HCTR2, which allows us to apply the tweak-based key derivation (TwKD) to HCTR2 in a provable manner. When the number of BC calls processed by a single tweak is upper-bounded by $2^{n/3}$, HCTR2-TwKD achieves $2n/3$-bit mu security. By benchmarking optimized software implementations, we show that CHCTR with AES-256 outperforms HCTR2 with Rijndael-256-256, in all the twelve processor models examined. Similarly, HCTR2-TwKD outperforms *bbb-ddd-AES* in general cases, and it is even comparable to *bbb-ddd-AES* rigorously optimized for tweak-repeating use cases using precomputation.

Keywords: Mode · Accordion · Beyond-Birthday-Bound security · Multi-user security · HCTR2 · Cascaded HCTR · Backward Compatibility

1 Introduction

A block cipher (BC) converts a fixed-length plaintext into a fixed-length ciphertext. BCs are used inside modes of operation to provide various functionalities.

In 2023, the National Institute of Standards and Technology (NIST) announced the development of a new BC mode: an "accordion mode" [32]. It corresponds to a variable-input-length strong tweakable pseudorandom permutation (VIL-STPRP) or tweakable wide-block encryption [31]. A VIL-STPRP provides an STPRP for a plaintext of any length, e.g., changing a single bit of the plaintext (resp. ciphertext) or the tweak randomizes the ciphertext (resp. plaintext).

In a VIL-STPRP, the encryption of a plaintext results in a ciphertext of the same length. This property is crucial in many applications. For example, in storage encryption, ciphertext expansion is not allowed because the data is divided into equal-sized data units, and the ciphertext is stored instead of the plaintext. To avoid ECB-style leakage, the sector

*This paper is the full version of [7] published in Asiacrypt 2025 by the same authors.

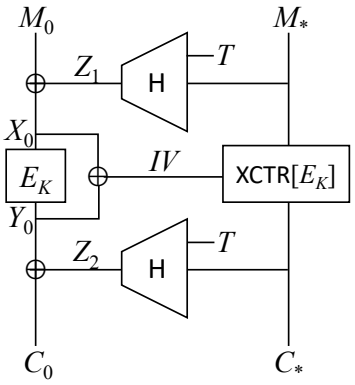


Fig. 1. HCTR2. The hash key is defined as $E_K(0)$ and the mask value of XCTR $L = E_K(1)$ is included in XCTR.

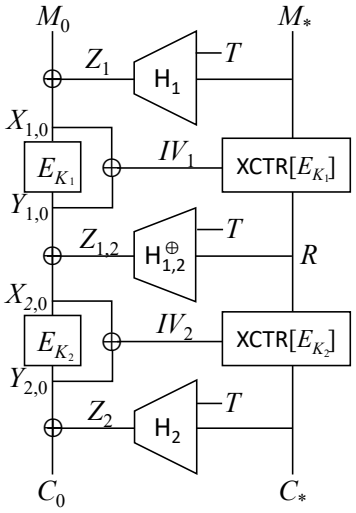


Fig. 2. CHCTR2. The second-round hash function is $Z_{1,2} = \mathsf{H}^{\oplus}_{1,2}(T, R) = \mathsf{H}_1(T, R) \oplus \mathsf{H}_2(T, R)$.

number is used as a tweak. Hence, the mode needs to be tweakable. Moreover, if a tweak of any length is acceptable, the encode-then-encipher paradigm [3] can be used: bits of zeros are padded to the plaintext and are checked during decryption. This provides robust authentication encryption (AE) [22], which mitigates nonce misuse in nonce-based AE.

Several BC-based VIL-STPRP modes have been designed, such as CMC [20], EME [21], HCH [4], TET [19], and HCTR2 [8]. Among them, HCTR2 has a wide adoption, e.g., the Linux kernel [35] and file encryption in Android [2]. HCTR2 is based on HCTR [37] and is proved to be secure up to the birthday bound. It adopts the hash-encrypt-hash structure depicted in Fig. 1: a plaintext is split into n-bit M_L and remaining M_R, where n is the block size of a BC E_K with a k-bit key K. First, a polynomial hash function H, i.e., POLYVAL, a variant of GHASH, processes M_R and a tweak T, and the output is XORed to M_L. Next, the left n-bit state is updated using E_K, an IV is extracted for encryption, and the XCTR mode (a variant of the CTR mode) encrypts M_R. Finally, H processes C_R and T, and its output is XORed into the left n bits to produce C_L. The cost per block is roughly 1 BC call and 2 n-bit field multiplications (FMULs).

In NIST's standardization, NIST requires using AES [30] as a primitive [32]. In the NIST Workshop [32], a concern was raised that existing VIL-STPRP modes, such as HCTR2, only provide birthday security and only provide 64-bit security when combined with AES. NIST then proposed two directions [6]; one is to develop VIL-STPRP modes with beyond-birthday-bound (BBB) security, and the other is to standardize a BC with a 256-bit block size such as Rijndael-256-256 [27,36] and use it in a birthday-secure mode. A concern for the second direction is that Rijndael-256-256 has not been standardized yet, and the transition to a new BC takes a long time. Consequently, this paper pursues the first direction of designing a new BBB mode, which can take full advantage of established security evaluation and implementation techniques of AES. **

In general, there are two approaches to achieve BBB. The first approach typically enlarges the state size so that the number of queries required to cause undesirable events,

**The initial preliminary draft of NIST SP 800-197A [33], released after the initial submission of this paper, proposes to develop BBBAcc, a beyond-birthday-bound accordion based on AES and HCTR2 technique.

such as collisions, can increase. The second approach, originally proposed for AES-GCM-SIV, derives keys based on the tweak or nonce, assuming that each tweak or nonce is not repeated many times [17].

The first approach has the advantage of achieving BBB unconditionally for query values. NIST prefers this property, noted as *"NIST expects the BBB security to hold for all queries, including multiple queries with the same tweak"* [6]. Consequently, several designs presented at the NIST Workshop in 2024 [32], including ACCOR-L [13] and DbHCTR [25], followed this approach. However, ACCOR-L has the limitation that only AES-256 can be used as a primitive, and the construction relies on the assumption of the related-key security of AES-256. The presentation of DbHCTR was limited to the introductory idea, and a complete specification, a concrete bound, and an accompanying proof were not presented. Currently, there are no BBB-secure VIL-STPRP modes with proofs that meet NIST's requirements.

A large computational overhead is the major drawback of the BBB-secure modes, and the second approach can provide better performance, even comparable to HCTR2. However, the assumption on the tweak/nonce repetition is valid for particular use cases only. The assumption could be valid for nonce-based AEAD where the tweak includes a nonce, while it may be invalid for other applications, particularly storage encryption where the same disk unit is updated without a nonce. At the NIST workshop [32], beyond-birthday-bound docked-double-decker AES (*bbb-ddd-AES*) was presented to achieve $2n/3$-bit security when *"... assuming it is not used with the same tweak too often ..."*, which has recently been presented at Eurocrypt 2025 [10, 11]. *bbb-ddd-AES* requires 1 AES call to derive tweak-dependent mask values for encryption in addition to almost the same cost as HCTR2. Hence, without any precomputation, the cost per block is 2 BC calls and 2 FMULs. The designers [11] suggest the implementation technique to precompute all the mask values, which works in the limited use cases where the tweak value does not change. This reduces the cost to 1 BC call and 2 FMULs, the same as that of HCTR2. However, precomputation is unavailable for some applications, such as disk encryption that uses a sector number as a tweak.

Multi-user (mu) security is considered important by NIST [6]. In the classic single-user (su) security, an attacker could only query one user with a fixed key, but in the mu security, an attacker can query multiple users with independent keys, and the goal is to break any of them. This reflects real-world usage where multiple users with independent keys exist for a single service. Meanwhile, to the best of our knowledge, there are no existing VIL-STPRP modes with proved mu security except those obtained by the classical conversion from su bounds. Dedicated mu bounds are important because *"NIST prefers to standardize an accordion with better multi-user security than is provided by generic bounds"* [6].

Note that a VIL-STPRP with more than 64-bit security could be achieved by using primitives other than AES, e.g., modes with tweakable BCs [9, 12] or permutation-based modes [18] with a large permutation. Such an approach is outside the scope of this paper, because it is not the direction of NIST.

1.1 Our Contributions

First, with a novel idea of cascading HCTR2, we propose a new VIL-STPRP mode CHCTR2 achieving BBB security in the mu setting without imposing additional conditions on the tweak, which meets NIST's requirements at a high level. Second, we prove mu security of HCTR2, which further allows us to apply the tweak-based key derivation (TwKD) to HCTR2 in a provable manner. When the same tweak is not iterated many times, as recently discussed with *bbb-ddd-AES*, HCTR2-TwKD achieves BBB security. More details are explained in the following.

Table 1. Comparison of VIL-STPRP modes. #Primitive refers to the number of BC and 128-bit field multiplication (FMUL) calls to process each 128-bit block.

Mode	Model	Security Level		#Keys		#Primitives		Ref.
		General	Conditional	Hash	BC	BC	FMUL	
HCTR2	su	$n/2$	$n/2$	0	1	1	2	[8]
HCTR2†	mu	$n/2$	$n/2$	0	1	1	2	Ours
CHCTR2	mu	$2n/3$	$2n/3$	0	2	2	3	Ours
bbb-ddd-AES w/o PC	su	$n/2$	$2n/3$	1	1	2	2	[11]
bbb-ddd-AES w/ PC‡	su	$n/2$	$2n/3$	1	1	1	2	[11]
HCTR2-TwKD*	mu	$n/2$	$2n/3$	0	1	1	2	Ours

†The security level holds when $\sigma_{\mathsf{u}} \approx \sigma$. If $\sigma_{\mathsf{u}} \ll \sigma$, HCTR2 has BBB security.
‡Precomputation (PC) available under certain prerequisites reduces the number of BCs by 1 in *bbb-ddd-AES*. Section 8 provides more details.
*The security level holds when $\ell\sigma_{\mathsf{u}}^2 \approx \sigma^2$. If $\ell\sigma_{\mathsf{u}}^2 \ll \sigma^2$, CHCTR2 has beyond $2n/3$-bit security.

CHCTR2: A BBB-Secure VIL-STPRP Mode. We consider a cascaded construction that sequentially computes HCTR2 twice with two independent keys. CHCTR2 can call HCTR2 as a block box, and thus maintains backward compatibility with existing implementations of HCTR2. With such black-box calls, the structure is hash-encrypt-hash-hash-encrypt-hash, which costs two HCTR2 calls. Further optimization is possible in dedicated CHCTR2 implementations: considering that POLYVAL has a homomorphic property, the middle two hash layers can be implemented as a single hash function whose key is the XOR of two keys, as shown in Fig. 2. Consequently, the cost per block is 2 BC calls and 3 FMULs. Regarding security, we prove the following mu-VIL-STPRP security bound of CHCTR2 in the standard mu-SPRP assumption: $O\left(\frac{\sigma}{2^n} + \frac{\ell\sigma_{\mathsf{u}}^2\sigma}{2^{2n}}\right)$, where σ is the number of n-bit data blocks (tweak and plaintext blocks) in all queries, σ_{u} is the maximum number of n-bit data blocks per user, and ℓ is the maximum number of n-bit data blocks per query. Assuming $\ell\sigma_{\mathsf{u}}^2 \le \sigma^2$, CHCTR2 achieves $2n/3$-bit mu-VIL-STPRP security. Furthermore, if $\ell\sigma_{\mathsf{u}}^2 \ll \sigma^2$, CHCTR2 has the beyond $2n/3$-bit mu-VIL-STPRP security.

mu-Security Proof of HCTR2. We next derive the first mu-VIL-STPRP-security bound of HCTR2 in the mu-SPRP assumption. Our bound is $O\left(\frac{\sigma_{\mathsf{u}}\sigma}{2^n}\right)$, showing that HCTR2 achieves $(n - \log_2 \sigma_{\mathsf{u}})$-bit security. If $\sigma \approx \sigma_{\mathsf{u}}$, i.e., when the adversary sends the most queries to a particular user, which is rather closer to the su setting, then HCTR2 achieves birthday-bound security, and if $\sigma_{\mathsf{u}} \ll 2^{n/2}$, it achieves BBB security. Note that XCTR of HCTR2 adds not only a counter but also a secret mask $L = E_K(1)$ to IV, while our proof targets a generalized construction such that the counter addition can be instantiated both with L and without L. Our achieved bound suggests that L is unnecessary and can be removed from the specification without degrading mu-VIL-STPRP security.

HCTR2-TwKD: HCTR2 with Tweak-Based Key Derivation. The mu-security proof of HCTR2 allows us to apply the tweak-based key derivation (TwKD) to HCTR2 in a provable manner, in which HCTR2's key is derived from a PRF that takes the tweak as an input. Similar to CHCTR2, this method calls HCTR2 as a block box, and thus maintains backward compatibility with existing implementations of HCTR2. Let σ_{tw} be the maximum number of data blocks under the same tweak and user. The resulting mu bound is $\frac{\sigma_{\mathsf{tw}}\sigma}{2^n}$. Under the condition $\sigma_{\mathsf{tw}} \le 2^{\frac{n}{3}}$, HCTR2-TwKD achieves $2n/3$-bit mu security.

Compared to *bbb-ddd-AES* that achieves $2n/3$-bit su security under the condition that the number of tweak repetitions is $2^{n/3}$ or less, HCTR2-TwKD achieves $2n/3$-bit mu security under the condition that the number of BC invocations for a single tweak is upper-bounded by $2^{n/3}$, therefore the security of HCTR2-TwKD is comparable in quantity and better in quality. Regarding performance, the overhead of HCTR2-TwKD over HCTR2 is negligible, i.e., the cost per block stays 1 BC and 2 FMULs, which outperforms *bbb-ddd-AES* with 2 BCs and 2 FMULs in the general case. Although *bbb-ddd-AES* can reduce the cost to 1 BC and 2 FMULs by precomputing tweak-dependent mask values in advance, this rigorous optimization is possible for limited use cases that repeat the same tweak; HCTR2-TwKD achieves the same performance without such restrictions and the cost of precomputation.

Software Performance Evaluation. We compare the software performance of the proposed methods with the state-of-the-art modes by instantiating them using BCs with a 256-bit key, as required by NIST [6]. Each implementation is optimized with dedicated CPU instructions [14], and their cycle-per-byte (cpb) performances are evaluated on 12 processors from four different vendors, including x86 and ARM architectures, and in local and cloud environments [1]. The first category evaluates CHCTR2 instantiated with AES-256 that achieves BBB security without any restriction on the tweak. The counterpart is HCTR2 instantiated with Rijndael-256-256 and a 256-bit universal hash function that concatenates two POLYVAL instances. While Rijndael-256-256 benefits from the AES instructions, the overhead of additional instructions needed for Rijndael-256-256 [15] is significant and CHCTR2 outperforms the other in all cases.

The second comparison compares HCTR2-TwKD and *bbb-ddd-AES*, both instantiated with AES-256, that achieve the same BBB security when the same tweak is not repeated too often. We implement two versions of *bbb-ddd-AES* with and without precomputation. As a result, HCTR2-TwKD completely outperforms the other without precomputation because *bbb-ddd-AES* needs one additional AES-256 for each 128-bit block. Moreover, HCTR2-TwKD achieves a similar performance compared to *bbb-ddd-AES* with precomputation. In summary, the relative performances of HCTR2-TwKD and *bbb-ddd-AES* follow those suggested by the numbers of primitive calls in Table 1, and HCTR2-TwKD achieves better performance in general cases.

1.2 Paper Organization

Section 2 gives basic notation. Section 3 defines the specifications of HCTR2, CHCTR2, and HCTR2-TwKD. Section 4 summarizes security models. Section 5 shows the security bound of CHCTR2, which is proved in Section 6. Section 7 proves the mu security of HCTR2. Section 8 shows the implementation results.

2 Basic Notations

Let ε be the empty string and $\emptyset$ the empty set. Let $\{0,1\}^i$ be the set of all i-bit strings and $\{0,1\}^0 \coloneqq \{\varepsilon\}$. Let $\{0,1\}^{\leq n} \coloneqq \cup_{i\in[n]}\{0,1\}^i$. Let $\{0,1\}^{n*} \coloneqq \cup_{i\in\mathbf{N}}\{0,1\}^{in}$. Let $\mathsf{msb}_i(X)$ and $\mathsf{lsb}_i(X)$ be the most and least significant i bits of a bit string X, respectively. Let $\mathtt{bin}_n(i)$ be an n-bit representation of an integer $i \geq 0$. For integers $0 \leq i \leq j$, let $[i,j] \coloneqq \{i, i{+}1, \ldots, j\}$, and $[j] \coloneqq [1,j]$. If $j < i$ then $[i,j] \coloneqq \emptyset$. Let $|D|_n \coloneqq \lceil |D|/n \rceil$ be an n-bit block length of D. For a value or a set X, $Y \leftarrow X$ means that X is assigned to Y. $(D_1, D_2) \xleftarrow{n,|D|-n} D$ means that a bit string D is partitioned into the n-bit string D_1 and the $(|D|-n)$-bit string D_2 such that $D = D_1 \| D_2$, and $(D_1, \ldots, D_\ell) \xleftarrow{n} D$ means that D is partitioned into the n-bit

Algorithm 1 HCTR2

Encryption $\mathsf{HCTR2}[E_K, \mathsf{H}](T, M)$

1: $H \leftarrow E_K(\mathtt{bin}_n(0))$; $(M_0, M_*) \xleftarrow{n,|M|-n} M$
2: $Z_1 \leftarrow \mathsf{H}_H(T, M_*)$; $X_0 \leftarrow Z_1 \oplus M_0$
3: $Y_0 \leftarrow E_K(X_0)$; $IV \leftarrow X_0 \oplus Y_0$; $C_* \leftarrow \mathsf{XCTR}[E_K](IV, M_*)$;
4: $Z_2 \leftarrow \mathsf{H}_H(T, C_*)$; $C_0 \leftarrow Z_2 \oplus Y_0$
5: $C \leftarrow C_0 \| C_*$; **return** C

Decryption $\mathsf{HCTR2}^{-1}[E_K^{\pm}, \mathsf{H}](T, C)$

1: $H \leftarrow E_K(\mathtt{bin}_n(0))$; $(C_0, C_*) \xleftarrow{n,|M|-n} C$
2: $Z_2 \leftarrow \mathsf{H}_H(T, C_*)$; $Y_0 \leftarrow Z_2 \oplus C_0$
3: $X_0 \leftarrow E_K^{-1}(Y_0)$; $IV \leftarrow X_0 \oplus Y_0$; $M_* \leftarrow \mathsf{XCTR}[E_K](IV, C_*)$;
4: $Z_1 \leftarrow \mathsf{H}_H(T, M_*)$; $M_0 \leftarrow Z_1 \oplus X_0$
5: $M \leftarrow M_0 \| M_*$; **return** M

Subroutine $\mathsf{XCTR}[E_K](IV, D)$

1: **for** $i = 1, \ldots, |D|_n$ **do** $Y_i \leftarrow E_K(\mathsf{add}(IV, i))$ **end for**
2: $Y \leftarrow \mathsf{msb}_{|M|}(Y_1 \| \cdots \| Y_{|D|_n})$; $D' \leftarrow D \oplus Y$; **return** D'

strings $D_1, \ldots, D_\ell$ such that $|D_i| = n$ for $i = 1, \ldots, \ell - 1$, $|D_\ell| \leq n$, and $D = D_1 \| \cdots \| D_\ell$. $T \xleftarrow{\$} \mathcal{T}$ means that an element is chosen uniformly at random from a non-empty set $\mathcal{T}$ and assigned to T. For two sets $\mathcal{T}_1$ and $\mathcal{T}_2$, $\mathcal{T}_1 \xleftarrow{\cup} \mathcal{T}_2$ means that $\mathcal{T}_1 \leftarrow \mathcal{T}_1 \cup \mathcal{T}_2$.

3 Specifications of HCTR2, CHCTR2, and HCTR2-TwKD

3.1 Primitives

Block Cipher (BC). We fix the key and block sizes of the underlying BCs to k and n. Let $\mathcal{BC}$ be the set of all block ciphers with k-bit keys and n-bit blocks. For a key $K \in \{0,1\}^k$ and $E \in \mathcal{BC}$, the BC is denoted by $E_K(\cdot) = E(K, \cdot)$. The decryption is denoted by E_K^{-1}. Let $E_K^{\pm} \coloneqq (E_K, E_K^{-1})$.

Keyed Hash Function. HCTR2 and its extensions CHCTR2 and HCTR2-TwKD use a family of keyed hash functions where the key space is $\{0,1\}^n$. Let $\mathcal{T} \subset \{0,1\}^*$ and $\mathcal{M} \subset \{0,1\}^*$ be the sets of tweaks and plaintexts/ciphertexts, respectively. Let $\{\mathsf{H}_H\}_{H \in \{0,1\}^n} : \mathcal{T} \times \mathcal{M} \to \{0,1\}^n$ be a family of keyed hash functions with n-bit keys H in $\{0,1\}^n$.

In this paper, a keyed hash function is decomposed into poly_H and pad defined as follows.

- $\{\mathsf{poly}_H\}_{H \in \mathcal{H}}$ is a family of keyed hash functions with the key space $\mathcal{H}$ such that $\mathsf{poly}_H : \{0,1\}^{n*} \to \{0,1\}^n$ accepts an input of length multiple of n and returns an n-bit hash value.
- $\mathsf{pad} : \mathcal{T} \times \mathcal{M} \to \{0,1\}^{n*}$ is an injective padding function such that $\forall (T, D) \in \mathcal{T} \times \mathcal{M} : (1 \leq |\mathsf{pad}(T, D)|_n \leq |T|_n + |D|_n + 1) \wedge (\mathsf{pad}(T, D) \notin \cup_{i \in \mathbf{N}} \{0^{in}\})$.

Then, H_H is defined as follows: for a key $H \in \mathcal{H}$ and an input pair $(T, D) \in \mathcal{T} \times \mathcal{M}$, $\mathsf{H}_H(T, D) \coloneqq \mathsf{poly}_H(\mathsf{pad}(T, D))$.

3.2 HCTR2

HCTR2 is a tweakable wide-block encryption mode based on a BC E_K and a keyed hash function H_H. Let $K \in \{0,1\}^k$ be a BC key. The hash key is defined as $H = E_K(\mathtt{bin}_n(0))$.

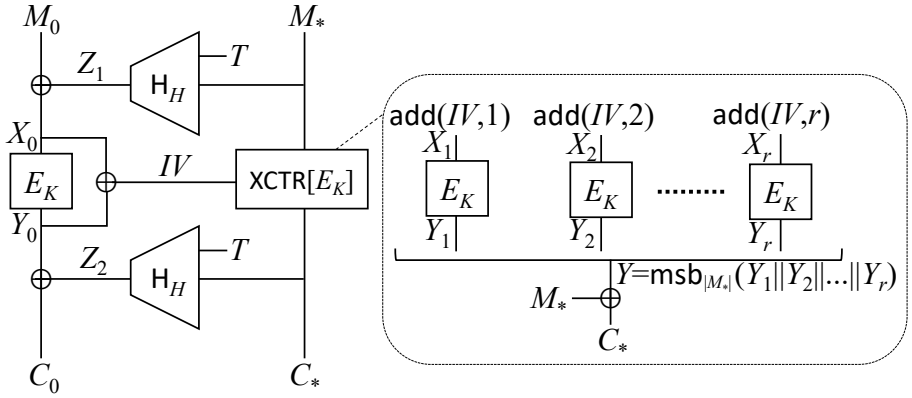


Fig. 3. HCTR2 and XCTR. The hash key is $H = E_K(\mathtt{bin}_n(0))$. $r = |M_*|_n$.

HCTR2 has three rounds. In the first round, H_H is performed. In the second round, E_K is performed once to generate IV of the encryption scheme $\mathsf{XCTR}[E_K]$, and then $\mathsf{XCTR}[E_K]$ is performed. In the third round, H_H is performed.

Let $r_{\mathsf{max}} + 1$ be the maximum n-bit block length of plaintexts. The set of plaintexts of HCTR2 is defined as $\mathcal{M} := \{M \in \{0,1\}^* \mid n \le |M| \le (r_{\mathsf{max}} + 1)n\}$. Hence, HCTR2 accepts plaintexts of length at least n bits. Let $\mathsf{add} : \{0,1\}^n \times [r_{\mathsf{max}}] \to \{0,1\}^n$ be a function used in XCTR that takes an n-bit initial value IV and a counter i, and returns the plaintext at the i-th BC call in XCTR. We assume that add is a linear function defined as $\mathsf{add}(IV, i) = IV \oplus A_i$ where A_i is an n-bit value associated with i such that $\forall IV, i, j$ s.t. $i \neq j$: $\mathsf{add}(IV, i) \neq \mathsf{add}(IV, j)$. Note that XCTR uses the BC's key-dependent value defined as $A_i = E_K(\mathtt{bin}_n(1)) \oplus \mathtt{bin}_n(i)$, which is covered by the definition of add.

HCTR2 consists of encryption and decryption algorithms. These algorithms and the underlying mode XCTR are defined in Algorithm 1 and depicted in Fig. 3. HCTR2 is the encryption algorithm that takes a plaintext $M \in \mathcal{M}$ and a tweak $T \in \mathcal{T}$, and returns the ciphertext C. $\mathsf{HCTR2}^{-1}$ is the decryption one.

3.3 CHCTR2

We present a new tweakable wide-block encryption mode CHCTR2 for BBB security. CHCTR2 has a cascaded construction of HCTR2. It has two BC keys K_1 and K_2, which are respectively used for the first and second HCTR2 calls. The encryption CHCTR2 and the decryption $\mathsf{CHCTR2}^{-1}$ are defined as

$$\mathsf{CHCTR2}[E_{K_1}, E_{K_2}, \mathsf{H}](T, M) := \mathsf{HCTR2}[E_{K_2}, \mathsf{H}]\left(T, \mathsf{HCTR2}[E_{K_1}, \mathsf{H}](T, M)\right),$$
$$\mathsf{CHCTR2}^{-1}[E^{\pm}_{K_1}, E^{\pm}_{K_2}, \mathsf{H}](T, C) := \mathsf{HCTR2}^{-1}[E^{\pm}_{K_1}, \mathsf{H}]\left(T, \mathsf{HCTR2}^{-1}[E^{\pm}_{K_2}, \mathsf{H}](T, C)\right).$$

Let $(X_{1,0}, Y_{1,0})$ and $(X_{2,0}, Y_{2,0})$ be respectively the pairs of plaintext and ciphertext blocks of E_{K_1} and E_{K_2} in the left part. For $j \in [r]$, let $(X_{1,j}, Y_{1,j})$ and $(X_{2,j}, Y_{2,j})$ be, respectively, the j-th pairs of E_{K_1} and E_{K_2} in the XCTR invocation of the right part. Let $Z_1, Z_{1,2}$, and Z_2 be respectively the hash values of the first, second, and third hash functions. Note that $Z_{1,2}$ is the result of the sum of two hash values. The detailed construction of CHCTR2 is given in Fig. 4.

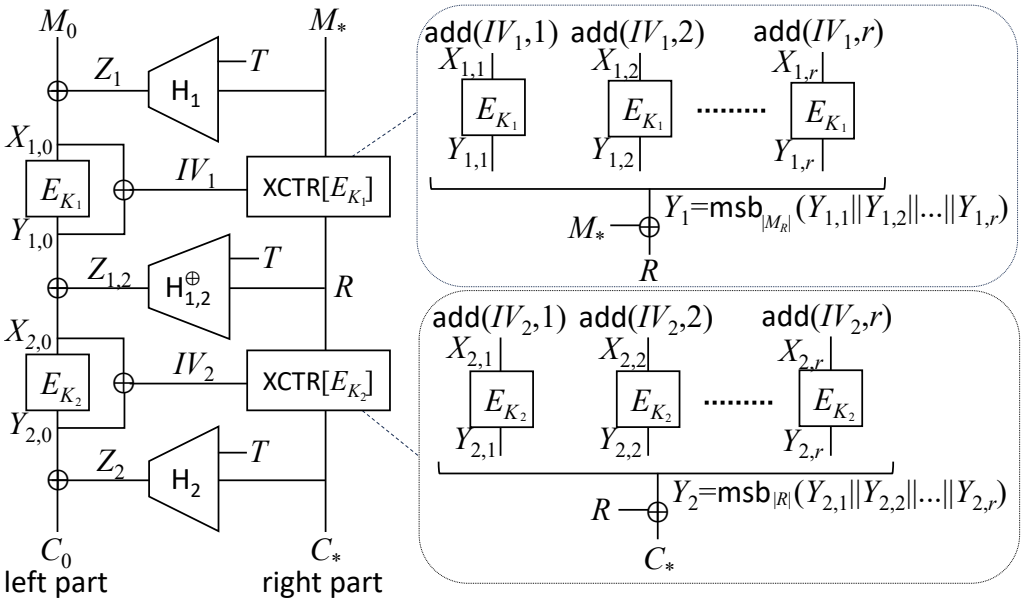


Fig. 4. CHCTR2. $r = |M_*|_n$. $\mathsf{H}_1 \coloneqq \mathsf{H}_{H_1}$ and $\mathsf{H}_2 \coloneqq \mathsf{H}_{H_2}$. The hash keys are defined as $H_1 = E_{K_1}(\mathtt{bin}_n(0))$ and $H_2 = E_{K_2}(\mathtt{bin}_n(0))$. The second hash value is defined as $Z_{1,2} = \mathsf{H}^{\oplus}_{1,2}(T,R) = \mathsf{H}_1(T,R) \oplus \mathsf{H}_2(T,R)$.

3.4 HCTR2-TwKD

For specific applications wherein the number of repetitions of the same tweak is limited, we consider HCTR2-TwKD which is HCTR2 with a tweak-based key derivation function (KDF). Let $F : \mathcal{L} \times \{0,1\}^{t_0} \to \{0,1\}^k$ be a KDF wherein $\mathcal{L}$ is the key space of the function. In HCTR2-TwKD, a tweak $T \in \mathcal{T}$ such that $|T| \geq t_0$ is partitioned into T_0 and T_* where the number of repetitions of the same pair of a key L and a tweak T_0 is limited. Then, for a key $L \in \mathcal{L}$, F with L is denoted by F_L. The encryption HCTR2-TwKD and the decryption HCTR2-TwKD^{-1} are defined as

$$\mathsf{HCTR2\text{-}TwKD}[F_L, E, \mathsf{H}](T, M) \coloneqq \mathsf{HCTR2}[E_{F_L(T_0)}, \mathsf{H}]\,(T_*, M)\,,$$
$$\mathsf{HCTR2\text{-}TwKD}^{-1}[F_L, E^{\pm}, \mathsf{H}](T, C) \coloneqq \mathsf{HCTR2}^{-1}[E^{\pm}_{F_L(T_0)}, \mathsf{H}]\,(T_*, C)\,.$$

4 Security Model and Tool

4.1 Distinguishing Advantage

We consider distinguishing-type security notions for BCs and VIL-STPRPs. For the security notions, we define the following distinguishing advantage of an adversary $\mathbf{A}$ that has access to either $\mathcal{O}_1$ or $\mathcal{O}_2$ and returns a decision bit. For $i \in [2]$, let $\mathbf{A}^{\mathcal{O}_i} = 1$ be an event that $\mathbf{A}$ with $\mathcal{O}_i$ returns 1. The distinguishing advantage function of $\mathbf{A}$ is defined as

$$\mathbf{Adv}^{\mathsf{dist}}_{\mathcal{O}_1,\mathcal{O}_2}(\mathbf{A}) \coloneqq \Pr\left[\mathbf{A}^{\mathcal{O}_1} = 1\right] - \Pr\left[\mathbf{A}^{\mathcal{O}_2} = 1\right].$$

4.2 Security Model for CHCTR2, HCTR2, and HCTR2-TwKD

We prove the security of CHCTR2, HCTR2, and HCTR2-TwKD in the sense of multi-user variable-input-length tweakable SPRP (mu-VIL-STPRP) security.

Let u be the number of users. Let $\Sigma \in \{\mathsf{CHCTR2}, \mathsf{HCTR2}, \mathsf{HCTR2\text{-}TwKD}\}$ be the target scheme. Let $\mathcal{K}$ be the key space of Σ. Let Σ_K be the encryption of the target scheme with a key $K \in \mathcal{K}$ and Σ_K^{-1} be the decryption. Let $\widetilde{\mathsf{WPerm}}$ be the set of all tweakable variable-length permutations with the same interface as Σ_K. In the mu-VIL-STPRP game, an adversary interacts with either the real-world or ideal-world oracles: $\mathcal{O}_{\mathsf{real}} \coloneqq (\Sigma_{K^{[\nu]}}, \Sigma_{K^{[\nu]}}^{-1})_{\nu\in[u]}$ and $\mathcal{O}_{\mathsf{ideal}} \coloneqq (\widetilde{\Psi}_\nu, \widetilde{\Psi}_\nu^{-1})_{\nu\in[u]}$, where $\forall \nu \in [u]$: $K^{[\nu]} \xleftarrow{\$} \mathcal{K}$ (real world) and $\widetilde{\Psi}_\nu \xleftarrow{\$} \widetilde{\mathsf{WPerm}}$ (ideal world). $K^{[\nu]}$ is the ν-th user's key and $\widetilde{\Psi}_\nu$ is the ν-th user's VIL-tweakable RP. At the end of this game, $\mathbf{A}$ returns a decision bit in $\{0,1\}$. Let $\mathbf{A}^{\mathcal{O}} \in \{0,1\}$ be an output of $\mathbf{A}$ with access to a set of oracles $\mathcal{O}$. Then, the mu-VIL-STPRP advantage function of $\mathbf{A}$ is defined as

$$\mathbf{Adv}_{\Sigma}^{\mathsf{mu\text{-}vil\text{-}stprp}}(\mathbf{A}) = \Pr\left[\mathbf{A}^{\mathcal{O}_{\mathsf{real}}} = 1\right] - \Pr\left[\mathbf{A}^{\mathcal{O}_{\mathsf{ideal}}} = 1\right].$$

4.3 Proof Tool: Coefficient-H Technique

We refer to the set of query-response tuples (and additional values) that an adversary obtains in a security game as a "transcript." Let T_R be a transcript obtained by sampling in the real world, i.e., sampling of hash keys and RPs. Let T_I be a transcript obtained by sampling in the ideal world, i.e., sampling of $\widetilde{\Psi}_1, \ldots, \widetilde{\Psi}_u$ (and additional values). We call a transcript τ *valid* if $\Pr[\mathsf{T}_I = \tau] > 0$. Let $\mathcal{T}$ be the set of all valid transcripts such that $\forall \tau \in \mathcal{T} : \Pr[\mathsf{T}_R = \tau] \leq \Pr[\mathsf{T}_I = \tau]$. Then, we have $\mathsf{SD}(\mathsf{T}_R, \mathsf{T}_I) \coloneqq \sum_{\tau\in\mathcal{T}}(\Pr[\mathsf{T}_I = \tau] - \Pr[\mathsf{T}_R = \tau])$. By using the following lemma [34], we can derive the upper-bound of the statistical distance.

Lemma 1. *Let $\mathcal{T}_{\mathsf{good}}$ and $\mathcal{T}_{\mathsf{bad}}$ be the good and bad transcripts into which $\mathcal{T}$ can be partitioned. If $\forall \tau \in \mathcal{T}_{\mathsf{good}} : \frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]} \geq 1-\varepsilon$ s.t. $0 \leq \varepsilon \leq 1$, then $\mathsf{SD}(\mathsf{T}_R, \mathsf{T}_I) \leq \Pr[\mathsf{T}_I \in \mathcal{T}_{\mathsf{bad}}] + \varepsilon$.*

We (1) define good and bad transcripts; (2) upper-bound $\Pr[\mathsf{T}_I \in \mathcal{T}_{\mathsf{bad}}]$; (3) lower-bound $\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]}$. Then, putting these bounds into the above lemma, we obtain the upper-bound.

4.4 Multi-user SPRP (mu-SPRP) Assumption

In our proofs, the underlying BCs are assumed to be mu-SPRP secure. Under the assumption, BC instantiations with independent keys are securely replaced with independent random permutations (RPs). Let $\mathsf{Perm}(n)$ be the set of n-bit permutations. Let u be the number of users. In the mu-SPRP game, an adversary $\mathbf{A}$ interacts with either the real-world oracles $(E_{K^{[1]}}^{\pm}, \ldots, E_{K^{[u]}}^{\pm})$ or the ideal-world oracles $(\pi_1^{\pm}, \ldots, \pi_u^{\pm})$, where $\forall \nu \in [u] : K^{[\nu]} \xleftarrow{\$} \{0,1\}^k$, $\pi_\nu \xleftarrow{\$} \mathsf{Perm}(n)$, and $\pi^{\pm} \coloneqq (\pi_\nu, \pi_\nu^{-1})$. At the end of this game, $\mathbf{A}$ returns a decision bit in $\{0,1\}$. The mu-SPRP advantage function of $\mathbf{A}$ is defined as

$$\mathbf{Adv}_{E}^{\mathsf{mu\text{-}sprp}}(\mathbf{A}) \coloneqq \mathbf{Adv}^{\mathsf{dist}}_{(E_{K_1}^{\pm},\ldots,E_{K_u}^{\pm}),(\pi_1^{\pm},\ldots,\pi_u^{\pm})}(\mathbf{A}).$$

For all possible adversaries $\mathbf{A}$ that have access to u users, make at most q queries, and run in time t, the maximum advantage is defined as

$$\mathbf{Adv}_{E}^{\mathsf{mu\text{-}sprp}}(u, q, t) \coloneqq \max_{\mathbf{A}} \mathbf{Adv}_{E}^{\mathsf{mu\text{-}sprp}}(\mathbf{A}).$$

4.5 Multi-user PRF (mu-PRF) Assumption

In our proof of HCTR2-TwKD, we assume that the KDF F is mu-PRF secure. Let u be the number of users. Let $\mathsf{Func}(t_0, k)$ be the set of all functions from $\{0,1\}^{t_0}$ to $\{0,1\}^k$. In the mu-PRF-security game, an adversary $\mathbf{A}$ has access to either real-world oracles $(F_{L^{[1]}}, \ldots, F_{L^{[u]}})$ or ideal-world ones $(\mathcal{R}_1, \ldots, \mathcal{R}_u)$, where $\forall \nu \in [u] : L^{[\nu]} \xleftarrow{\$} \mathcal{L}$ and $\mathcal{R}_\nu \xleftarrow{\$} \mathsf{Func}(n, k)$. At the end of this game, $\mathbf{A}$ return a decision bit. Then, the advantage function of $\mathbf{A}$ is defined as

$$\mathbf{Adv}^{\mathsf{mu\text{-}prf}}_F(\mathbf{A}) \coloneqq \mathbf{Adv}^{\mathsf{dist}}_{(F_{L^{[\nu]}})_{\nu\in[u]},(\mathcal{R}_\nu)_{\nu\in[u]}}(\mathbf{A}).$$

For all possible adversaries $\mathbf{A}$ that have access to u users, make at most q queries, and run in time t, the maximum advantage is defined as

$$\mathbf{Adv}^{\mathsf{mu\text{-}prf}}_F(u, q, t) \coloneqq \max_{\mathbf{A}} \mathbf{Adv}^{\mathsf{mu\text{-}prf}}_F(\mathbf{A}).$$

4.6 Universal Hash Function

In our proofs, we use the following assumptions for the hash functions.

Definition 1 (Regular Hash, Almost XOR Universal Hash). *Let $\omega \geq 1$. $\{\mathsf{H}_H\}_{H\in\mathcal{H}}$ is said to be ω-regular if for any $Y \in \{0,1\}^n$ and $(T, D) \in \mathcal{T} \times \mathcal{M}$, and $b \in \{0,1\}$,*

$$\Pr[H \xleftarrow{\$} \{0,1\}^n, \mathsf{H}_H(T, D) = Y \oplus bH] \leq \frac{\omega|\mathsf{pad}(T, D)|_n}{2^n}.$$

$\{\mathsf{H}_H\}_{H\in\mathcal{H}}$ is said to be ω-AXU if for any $Y \in \{0,1\}^n$ and distinct input pairs $(T, D), (T', D') \in \mathcal{T} \times \mathcal{M}$,

$$\Pr[H \xleftarrow{\$} \{0,1\}^n; \mathsf{H}_H(T, D) \oplus \mathsf{H}_H(T', D') = Y] \leq \frac{\omega \max\{|\mathsf{pad}(T, D)|_n, |\mathsf{pad}(T', D')|_n\}}{2^n}.$$

The HCTR2's hash with POLYVAL and GHASH are 1-retular and 1-AXU.

5 Security of CHCTR2

5.1 Security Bound

In the proof of CHCTR2, we use the following assumptions for the hash functions H_{H_1} and H_{H_2}.

Definition 2 (XOR-Invertible Hash). *$\{\mathsf{H}_H\}_{H\in\mathcal{H}}$ is XOR-invertible if for any integers $i, j \geq 0$, $Y_1 \in \{0,1\}^{in}$, $Y_2', Y_2^* \in \{0,1\}^n, Y_3 \in \{0,1\}^j$, $T \in \mathcal{T}$, and $H_1, H_2 \in \{0,1\}^n$ such that $Y_2' \neq Y_2^*$, the following condition holds.*

$$\mathsf{H}_H(T, Y_1\|Y_2'\|Y_3) \oplus \mathsf{H}_{H_2}(T, Y_1\|Y_2'\|Y_3) \neq \mathsf{H}_{H_1}(T, Y_1\|Y_2^*\|Y_3) \oplus \mathsf{H}_{H_2}(T, Y_1\|Y_2^*\|Y_3).$$

Polynomial hash functions such as POLYVAL and GHASH have this property. The following theorem shows the mu-VIL-STPRP-security bound of CHCTR2.

Theorem 1. *Assume that $\{\mathsf{H}_H\}_{H\in\mathcal{H}}$ is XOR-invertible, ω-regular, and ω-AXU. For any adversary $\mathbf{A}$ running in time t and making q queries to u users such that the total number of n-bit data (tweak and plaintext/ciphertext) blocks in all queries is σ, the maximum number of n-bit data blocks per user is σ_{u}, and the maximum number of n-bit data blocks per query is ℓ, we have*

$$\mathbf{Adv}^{\mathsf{mu\text{-}vil\text{-}stprp}}_{\mathsf{CHCTR}}(\mathbf{A}) \leq \frac{4\omega^2\ell\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{16\omega^2\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{6\omega\sigma}{2^n} + \mathbf{Adv}^{\mathsf{mu\text{-}sprp}}_E(2u, \sigma, t + O(\sigma)).$$

An overview of the proof is given below. The proof is given in Section 6.

5.2 Proof Overview

We prove the mu-VIL-STPRP security of CHCTR2 by using the coefficient-H technique with the resampling method [5, 29]. The resampling method is a novel way of defining dummy internal values in the ideal world. In the proofs with this method, the (dummy) internal values and (dummy) keys are revealed to an adversary after finishing all queries. We explain the resampling method with the sum of two RPs SoP [26], since the right part of CHCTR2 can be seen as an iteration of SoP. Let $\dot{\pi}$ and $\ddot{\pi}$ be independent RPs. For an input $X \in \{0,1\}^n$, the output T of SoP is defined as $T = Y_1 \oplus Y_2$ where $Y_1 = \dot{\pi}(X)$ and $Y_2 = \ddot{\pi}(X)$. We consider the single-user-PRF-security of SoP (i.e., $u = 1$), i.e., indistinguishability between a single instance of SoP and an n-bit RF $\mathcal{R}$.

In the ideal world where an adversary interacts with an n-bit RF, when all queries are finished, or put differently, when all input-output pairs $((X^{(1)}, T^{(1)}), \ldots, (X^{(q)}, T^{(q)}))$ are fixed, we sample each dummy ciphertext of $\dot{\pi}$. For each α-th query $X^{(\alpha)}$, we first sample the (dummy) ciphertext $Y_1^{(\alpha)}$ of $\dot{\pi}$ as $Y_1^{(\alpha)} \xleftarrow{\$} \{0,1\}^n$ and then the other one $Y_2^{(\alpha)}$ is defined as $Y_2^{(\alpha)} = T^{(\alpha)} \oplus Y_1^{(\alpha)}$ according to the structure of SoP. Since all outputs of $\mathcal{R}$ are independently chosen, a collision may occur for the ciphertexts, i.e., $\exists \beta \in [\alpha - 1], i \in [2]$ s.t. $X^{(\alpha)} \neq X^{(\beta)} \wedge Y_i^{(\alpha)} = Y_i^{(\beta)}$. The collision is incompatible with the property of RPs, and the collision probability is about $O(\frac{\alpha}{2^n})$. Hence, only with the first sampling, the ciphertexts are distinguishable from those in the real world with the birthday bound $\sum_{\alpha \in [q]} \frac{\alpha}{2^n} = O(\frac{q^2}{2^n})$.

The resampling method can improve the birthday bound by introducing the second sampling. After the first sampling fails for the α-th query, we discard the ciphertexts of the first sampling and freshly sample $Y_1^{(\alpha)}$ again. Then, the other one $Y_2^{(\alpha)}$ is defined as $Y_2^{(\alpha)} = T^{(\alpha)} \oplus Y_1^{(\alpha)}$. For each $\alpha \in [q]$, the collision probability of the second sampling is $\frac{2(\alpha-1)}{2^n}$, giving a $2n/3$-bit security bound $\sum_{\alpha \in [q]} (\frac{2(\alpha-1)}{2^n})^2 = O(\frac{q^3}{2^{2n}})$.[††]

For good transcripts τ, we lower-bound $\frac{\Pr[\mathsf{T}_R = \tau]}{\Pr[\mathsf{T}_I = \tau]}$, the ratio of the real-world and ideal-world probabilities. Although the resampling method reduces the collision probability explained above, each dummy ciphertext is sampled at most twice, and the number of chances to satisfy the target good transcript is increased, i.e., $\Pr[\mathsf{T}_I = \tau]$ is increased. In the real world, all pairs of plaintext and ciphertext are defined by RPs $(\dot{\pi}, \ddot{\pi})$. In the ideal world, responses to construction queries are defined by an RF $\mathcal{R}$, and each dummy ciphertext is chosen uniformly at random from $\{0,1\}^n$. Hence, the sampling space of the ideal world is larger than that of the real world, which we call a "budget" of the ideal world. As in the proofs of triple encryption [28] and of key-alternating ciphers [5], the impact of the increase can be canceled out by using the budget of the ideal world, and providing the nice lower-bound $\frac{\Pr[\mathsf{T}_R = \tau]}{\Pr[\mathsf{T}_I = \tau]} \geq 1$. The detail is given in Section 5.3.

In our proof of CHCTR2, we update the sampling method according to the structure of CHCTR2. In CHCTR2, the right part consists of two XCTR invocations, but the IVs IV_1 and IV_2 are defined by using two encrypted keyed hash functions. Hence, we need to consider collisions of the hash functions. Moreover, one IV IV_i is defined by using the other one IV_{3-i} via XCTR and the hash function in the middle layer of CHCTR2. Hence, we carefully define the sampling algorithm by considering collisions of the hash functions and

[††]Note that if the number of samplings is three (or more) times, we can improve the collision probability to $O(\frac{q^4}{2^{3n}})$. However, the $2n/3$-bit term $\frac{q^3}{2^{2n}}$ appears in the bound of the ratio for good transcripts. In Section 5.3, we show that if the number of samplings is at most 2, the ratio is ≥ 1. However, when the third sampling is introduced, the probability that the first and second samplings fail is introduced, which is about $\sum_{\alpha \in [q]} \left(\frac{2(\alpha-1)}{2^n}\right)^2 = O(\frac{q^3}{2^{2n}})$, which cannot be canceled out by the budget of the ideal world.

the dependency between IV_1 and IV_2. The proof is given in Section 6, where we successfully define the sampling algorithm that offers the $2n/3$-bit security bound of CHCTR2.

5.3 Good Transcript Evaluation for PRF-Security of SoP

Let τ be a target good transcript. We evaluate $\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]}$, the ratio of the real-world and ideal-world probabilities.

Real-world probability. For each $\alpha \in [q]$, $Y_1^{(\alpha)}$ and $Y_2^{(\alpha)}$ are respectively chosen uniformly at random from $2^n - (\alpha - 1)$ elements in $\{0,1\}^n$ by RPs $\dot{\pi}$ and $\ddot{\pi}$. Hence, the probability that $Y_1^{(\alpha)}$ and $Y_2^{(\alpha)}$ satisfy τ is at most $\left(\frac{1}{2^n-(\alpha-1)}\right)^2$.

Ideal-world probability. For each $\alpha \in [q]$, the probability that the α-th output $T^{(\alpha)}$ satisfies τ is $\frac{1}{2^n}$. Regarding $Y_1^{(\alpha)}$, since there are two chances of sampling $Y_1^{(\alpha)}$, we evaluate the probability for each case.

- The probability that $Y_1^{(\alpha)}$ satisfies τ is $\frac{1}{2^n}$. The probability that $T^{(\alpha)}$ and $Y_1^{(\alpha)}$ defined by the first sampling satisfy τ is $\left(\frac{1}{2^n}\right)^2$.
- The probability that the first sampling fails (a collision occurs at the first sampling on either $Y_1^{(\alpha)}$ or $Y_2^{(\alpha)}$) is $\frac{2(\alpha-1)}{2^n}$. Hence, the probability that $T^{(\alpha)}$ and $Y_1^{(\alpha)}$ by the second sampling satisfy τ is $\left(\frac{1}{2^n}\right)^2 \cdot \frac{2(\alpha-1)}{2^n}$.

So, the probability that $T^{(\alpha)}$ and $Y_1^{(\alpha)}$ (and $Y_2^{(\alpha)}$) satisfy τ is $\left(\frac{1}{2^n}\right)^2 \cdot \left(1 + \frac{2(\alpha-1)}{2^n}\right)$.

Lower-Bound of $\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]}$. By using the above probabilities, the ratio of the real-world and ideal-world probabilities for good transcripts is

$$\begin{aligned}
\prod_{\alpha\in[q]} \frac{\left(\frac{1}{2^n-(\alpha-1)}\right)^2}{\left(\frac{1}{2^n}\right)^2 \cdot \left(1 + \frac{2(\alpha-1)}{2^n}\right)} &= \prod_{\alpha\in[q]} \left(\left(1 - \frac{\alpha-1}{2^n}\right)^2 \cdot \left(1 + \frac{2(\alpha-1)}{2^n}\right)\right)^{-1} \\
&= \prod_{\alpha\in[q]} \left(\left(1 - \frac{2(\alpha-1)}{2^n} + \left(\frac{\alpha-1}{2^n}\right)^2\right) \cdot \left(1 + \frac{2(\alpha-1)}{2^n}\right)\right)^{-1} \\
&= \prod_{\alpha\in[q]} \left(1 - 3\left(\frac{\alpha-1}{2^n}\right)^2 + 2\left(\frac{\alpha-1}{2^n}\right)^3\right)^{-1}.
\end{aligned}$$

The term $2\left(\frac{\alpha-1}{2^n}\right)^3$ is introduced due to the second sampling that increases the number of chances to satisfy a good transcript. On the other hand, the term "$3\left(\frac{\alpha-1}{2^n}\right)^2$" can be used as the "budget" of the ideal world. Since $3\left(\frac{\alpha-1}{2^n}\right)^2 \geq 2\left(\frac{\alpha-1}{2^n}\right)^3$, the term $2\left(\frac{\alpha-1}{2^n}\right)^3$ is canceled out by the budget, and we obtain the lower bound $\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]} \geq 1$.

6 Proof of Theorem 1

Without loss of generality, we assume that the adversary is deterministic and makes no repeated or trivial query.

We first replace the BCs $(E^{\pm}_{K_1^{[\nu]}}, E^{\pm}_{K_2^{[\nu]}})_{\nu\in[u]}$ with RPs $(\dot{\pi}^{\pm}_{\nu}, \ddot{\pi}^{\pm}_{\nu})_{\nu\in[u]}$ where $\forall \nu \in [u]$: $\dot{\pi}_\nu \xleftarrow{\$} \mathsf{Perm}(n)$, $\ddot{\pi}_\nu \xleftarrow{\$} \mathsf{Perm}(n)$, $\dot{\pi}^{\pm}_{\nu} \coloneqq (\dot{\pi}_\nu, \dot{\pi}_\nu^{-1})$, $\ddot{\pi}^{\pm}_{\nu} \coloneqq (\ddot{\pi}_\nu, \ddot{\pi}_\nu^{-1})$, introducing the mu-SPRP

advantage function $\mathbf{Adv}^{\mathsf{mu\text{-}sprp}}_{E}(2u, \sigma, t+O(\sigma))$ in the bound. Hence, in the following evaluation, the oracles $\mathcal{O}_{\mathsf{real}} = \left(\mathsf{CHCTR2}[\dot{\pi}_\nu, \ddot{\pi}_\nu, \mathsf{H}], \mathsf{CHCTR2}^{-1}[\dot{\pi}^{\pm}_\nu, \ddot{\pi}^{\pm}_\nu, \mathsf{H}]\right)_{\nu\in[u]}$ are the real-world ones.

6.1 Definitions

Let q_ν be the number of queries to the ν-th user. Without loss of generality, we assume that $\forall \nu \in [u] : q_\nu \geq 1$. Let σ_ν be the total number of n-bit data blocks in queries to the ν-th user. Note that $\sigma = \sum_{\nu\in[u]} \sigma_\nu$. We call the stage that an adversary makes queries the "query stage" and the stage after the query stage the "decision stage". The values $X_{1,0}$ and $Y_{2,0}$ in the left part of $\mathsf{CHCTR2}$ are called "edge values."

For $\alpha \in [q]$, values corresponding to the α-th query are denoted by using the superscript symbol (α), e.g., $M^{(\alpha)}$, $T^{(\alpha)}$, $X^{(\alpha)}_{i,j}$, $Y^{(\alpha)}_{i,j}$, etc. Let $t_\alpha := |T^{(\alpha)}|_n$ be the length of the tweak $T^{(\alpha)}$. Let $m_\alpha := |M^{(\alpha)}|$ be the length of the plaintext $M^{(\alpha)}$. Let $r_\alpha = \lceil |M^{(\alpha)}_*|/n \rceil$ be the block length of $M^{(\alpha)}_*$, i.e., the number of n-bit blocks in $M^{(\alpha)}_*$. Let $m^+_\alpha := n(r_\alpha+1) - m_\alpha$ be the number of truncated bits in XCTR in the α-th query and $\ell_\alpha := t_\alpha + (r_\alpha+1)$ the number of n-bit data blocks in the α-th query. For a plaintext $M^{(\alpha)}_*$ (resp. ciphertext $C^{(\alpha)}_*$) in the right part, let $M^{(\alpha)}_i$ (resp. $C^{(\alpha)}_i$) be the i-th n-bit plaintext (resp. ciphertext) block, i.e., $M^{(\alpha)}_* = M^{(\alpha)}_1 \| \cdots \| M^{(\alpha)}_{r_\alpha}$ and $C^{(\alpha)}_* = C^{(\alpha)}_1 \| \cdots \| C^{(\alpha)}_{r_\alpha}$. Note that $|M^{(\alpha)}_{r_\alpha}| \leq n$ and $|C^{(\alpha)}_{r_\alpha}| \leq n$. Let $\mathsf{u}_\alpha \in [u]$ be the user index corresponding to the α-th query, i.e., if the query is to the ν-th user, then $\mathsf{u}_\alpha = \nu$. For $\nu \in [u]$, let $\mathcal{Q}^{[\nu]} := \{\alpha \in [q] \mid \mathsf{u}_\alpha = \nu\}$ be query indexes for the ν-th user. For $\alpha \in [q]$, let $\mathsf{query}^{(\alpha)}$ be a predicate for the α-th query: if the α-th query is an encryption one, then $\mathsf{query}^{(\alpha)} := \mathsf{enc}$, and else $\mathsf{query}^{(\alpha)} := \mathsf{dec}$. Let

$$\mathcal{Q}^{[\nu]}_{<\alpha} := \{\beta \in \mathcal{Q}^{[\nu]} \mid \text{the internal pairs of the } \beta\text{-th query is defined before those of the } \alpha\text{-th query}\}.$$

For $\nu \in [u]$, let $\mathcal{P}^{[\nu]}_1 := \{(X^{(\alpha)}_{1,j}, Y^{(\alpha)}_{1,j}) \mid \alpha \in \mathcal{Q}^{[\nu]}, j \in [0, r_\alpha]\}$ and $\mathcal{P}^{[\nu]}_2 := \{(X^{(\alpha)}_{2,j}, Y^{(\alpha)}_{2,j}) \mid \alpha \in \mathcal{Q}^{[\nu]}, j \in [0, r_\alpha]\}$ be the sets of all pairs of plaintext and ciphertext blocks of $\dot{\pi}^{\pm}_\nu$ and $\ddot{\pi}^{\pm}_\nu$ in queries to the ν-th user, respectively. For $i \in [2]$, if $(X, Y) \in \mathcal{P}^{[\nu]}_i$ then $\mathcal{P}^{[\nu]}_i(X) := Y$ and $\mathcal{P}^{-1[\nu]}_i(Y) := X$. If $(X, *) \notin \mathcal{P}^{[\nu]}_i$ (resp. $(*, Y) \notin \mathcal{P}^{[\nu]}_i$), then $\mathcal{P}^{[\nu]}_i(X) := \varepsilon$ (resp. $\mathcal{P}^{-1[\nu]}_i(Y) := \varepsilon$). For $i \in [2]$, let $\mathcal{X}^{[\nu]}_i := \{X \mid (X, *) \in \mathcal{P}^{[\nu]}_i\}$ and $\mathcal{Y}^{[\nu]}_i := \{Y \mid (*, Y) \in \mathcal{P}^{[\nu]}_i\}$. For $\nu \in [u]$, let $\mathsf{H}^{[\nu]}_1 := \mathsf{H}_{H^{[\nu]}_1}$ and $\mathsf{H}^{[\nu]}_2 := \mathsf{H}_{H^{[\nu]}_2}$.

6.2 Collision Set

For each $\nu \in [u]$, we define collision sets for edge values $(X^{(\alpha)}_{1,0}, Y^{(\alpha)}_{2,0})_{\alpha\in\mathcal{Q}^{[\nu]}}$. For each $\nu \in [u]$, a (γ-th) collision set $\mathcal{G}^{[\nu]}_{\mathsf{coll}}[\gamma]$ includes query indexes of the ν-th user with the same collisions for either $X_{1,0}$ or $Y_{2,0}$.

A (γ-th) collision set $\mathcal{G}^{[\nu]}_{\mathsf{coll}}[\gamma]$ is a set of pairs of query and round indexes with a collision of an edge value. For example, for distinct query indexes $\alpha_1, \alpha_2 \in \mathcal{Q}^{[\nu]}$ with the collision $X^{(\alpha_1)}_{1,0} = X^{(\alpha_2)}_{1,0}$, the collision set is defined as $\mathcal{G}^{[\nu]}_{\mathsf{coll}}[\gamma] := \{(\alpha_1, 1), (\alpha_2, 1)\}$. If a multi-collision occurs, then the collision set keeps all query indexes with the multi-collision. For example, for distinct query indexes $\alpha_1, \alpha_2, \ldots, \alpha_j \in \mathcal{Q}^{[\nu]}$ with the multi-collision $Y^{(\alpha_1)}_{2,0} = Y^{(\alpha_2)}_{2,0} = \cdots = Y^{(\alpha_j)}_{2,0}$, the collision set is defined as $\mathcal{G}^{[\nu]}_{\mathsf{coll}}[\gamma] := ((\alpha_1, 2), (\alpha_2, 2), \ldots, (\alpha_j, 2))$. See Fig. 5.

For a collision set $\mathcal{G}^{[\nu]}_{\mathsf{coll}}[\gamma] := \{(\alpha_1, i), (\alpha_2, i), \ldots, (\alpha_j, i)\}$ where $i \in [2]$, the (j-multi-)collision ensures that if $i = 1$, then the (j-multi-)collision $Y^{(\alpha_1)}_{1,0} = Y^{(\alpha_2)}_{1,0} = \cdots = Y^{(\alpha_j)}_{1,0}$

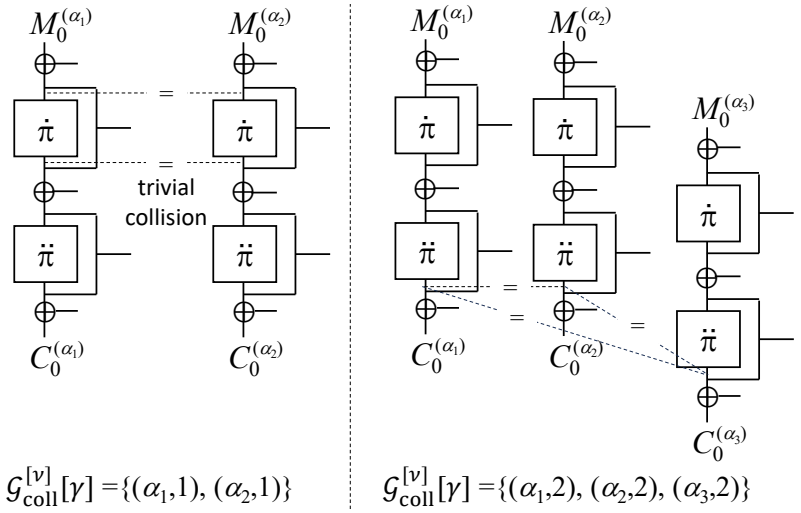


Fig. 5. Collision sets with a single collision (left) and with a 3-multi-collision (right) where the right parts of CHCTR2 are omitted.

occurs; if $i = 2$, then the (j-multi-)collision $X_{2,0}^{(\alpha_1)} = X_{2,0}^{(\alpha_2)} = \cdots = X_{2,0}^{(\alpha_j)}$ occurs. Such collisions are called "trivial collisions".

Let $\mathcal{G}^{[\nu]}_{\mathsf{coll}}$ be the family of collision sets of the ν-th user. For a collision set $\mathcal{G}^{[\nu]}_{\mathsf{coll}}$ and $(\alpha_1, *) \in \mathcal{G}^{[\nu]}_{\mathsf{coll}}$, we say α_1 is a longest query index if the block length of the plaintext $M^{(\alpha_1)}$ is maximum in the collision set, i.e.,$\forall(\alpha, *) \in \mathcal{G}^{[\nu]}_{\mathsf{coll}} \backslash \{(\alpha_1, *)\} : r_\alpha \leq r_{\alpha_1}$. We say the other queries are non-longest queries. Note that query indexes with both $X_{1,0}$ and $Y_{2,0}$ collisions are considered in the bad event 2coll defined later.

Let $\mathcal{Q}^{[\nu]}_{\mathsf{coll}}[\gamma] \coloneqq \{\alpha \mid (\alpha, *) \in \mathcal{G}^{[\nu]}_{\mathsf{coll}}[\gamma]\}$ be query indexes in $\mathcal{G}^{[\nu]}_{\mathsf{coll}}[\gamma]$. Let $\mathcal{Q}^{[\nu]}_{\mathsf{coll}} \coloneqq \cup_{\gamma \in [|\mathcal{G}^{[\nu]}_{\mathsf{coll}}|]} \mathcal{Q}^{[\nu]}_{\mathsf{coll}}[\gamma]$. Let $\mathcal{Q}^{[\nu]}_{\mathsf{non\text{-}coll}} \subseteq \mathcal{Q}^{[\nu]}$ be query indexes of the ν-th user without collisions of the edge values. Let $\mathcal{Q}^{[\nu]}_{\mathsf{coll},1} \subseteq \mathcal{Q}^{[\nu]}_{\mathsf{coll}}$ be the set of all longest queries in $\mathcal{Q}^{[\nu]}_{\mathsf{coll}}$, and $\mathcal{Q}^{[\nu]}_{\mathsf{coll},2} \coloneqq \mathcal{Q}^{[\nu]}_{\mathsf{coll}} \backslash \mathcal{Q}^{[\nu]}_{\mathsf{coll},1}$ the set of non-longest queries.

6.3 Sampling Algorithm for CHCTR2

By using Algorithms 2, 3, and 4, dummy hash keys, and dummy internal pairs of RPs are defined in the decision stage of the ideal world. Algorithm 2 is the main algorithm, Algorithms 3 is the sampling algorithm, and Algorithm 4 is the assignment algorithm for query indexes with collisions.

Algorithm 2. This algorithm is a main algorithm of our sampling procedure. We explain the algorithm below.

Lines 2-10. The preliminary procedure is given in these lines. First, the ν-th user's hash keys $H_1^{[\nu]}$ and $H_2^{[\nu]}$ are defined, fixing all edge values $(X_{1,0}^{(\alpha)}, Y_{2,0}^{(\alpha)})_{\alpha \in \mathcal{Q}^{[\nu]}}$ of the ν-th user and a family of collision sets $\mathcal{G}^{[\nu]}_{\mathsf{coll}}$. For each $\alpha \in \mathcal{Q}^{[\nu]}$, a counter sample_α is introduced to count the number of collisions for the α-th query. In this algorithm, a plaintext $M^{(\alpha)}$ and ciphertext $C^{(\alpha)}$ are respectively extended by appending the m_α^+-bit zeros $0^{m_\alpha^+}$ and an m_α^+-bit random value $C_+^{(\alpha)}$ if $\mathsf{query}^{(\alpha)} = \mathsf{enc}$; an m_α^+-bit random value $M_+^{(\alpha)}$ and the m_α^+-bit zeros $0^{m_\alpha^+}$ if $\mathsf{query}^{(\alpha)} = \mathsf{dec}$. $\hat{M}^{(\alpha)}$ (resp. $\hat{C}^{(\alpha)}$) is the extended plaintext (resp. ciphertext).

Algorithm 2 Main Algorithm for CHCTR2

1: **for** $\nu \in [u]$ **do**
2: $H_1^{[\nu]} \xleftarrow{\$} \{0,1\}^n$; $H_2^{[\nu]} \xleftarrow{\$} \{0,1\}^n$; $\mathcal{P}_1^{[\nu]} \leftarrow \emptyset$; $\mathcal{P}_2^{[\nu]} \leftarrow \emptyset$
3: **for** $\alpha \in \mathcal{Q}^{[\nu]}$ **do**
4: **for** $i \in [2]$; $j \in [0, r_{\mathsf{max}}]$ **do** $(X_{i,j}^{(\alpha)}, Y_{i,j}^{(\alpha)}) \leftarrow (\varepsilon, \varepsilon)$ **end for**
5: $Z_1^{(\alpha)} \leftarrow \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, M_*^{(\alpha)})$; $X_{1,0}^{(\alpha)} \leftarrow Z_1^{(\alpha)} \oplus M_0^{(\alpha)}$
6: $Z_2^{(\alpha)} \leftarrow \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, C_*^{(\alpha)})$; $Y_{2,0}^{(\alpha)} \leftarrow Z_2^{(\alpha)} \oplus C_0^{(\alpha)}$
7: **if** $\mathsf{query}^{(\alpha)} = \mathsf{enc}$ **then** $C_+^{(\alpha)} \xleftarrow{\$} \{0,1\}^{m_\alpha^+}$; $\hat{C}^{(\alpha)} \leftarrow C^{(\alpha)} \| C_+^{(\alpha)}$; $\hat{M}^{(\alpha)} \leftarrow M^{(\alpha)} \| 0^{m_\alpha^+}$
8: **else** $M_+^{(\alpha)} \xleftarrow{\$} \{0,1\}^{m_\alpha^+}$; $\hat{C}^{(\alpha)} \leftarrow C^{(\alpha)} \| 0^{m_\alpha^+}$; $\hat{M}^{(\alpha)} \leftarrow M^{(\alpha)} \| M_+^{(\alpha)}$ **end if**
9: $\hat{M}_1^{(\alpha)}, \ldots, \hat{M}_{r_\alpha}^{(\alpha)} \xleftarrow{n} \hat{M}_*^{(\alpha)}$; $\hat{C}_1^{(\alpha)} \| \ldots \| \hat{C}_{r_\alpha}^{(\alpha)} \xleftarrow{n} \hat{C}_*^{(\alpha)}$
10: **end for**
11: **for** $\gamma \in [|\mathcal{G}_{\mathsf{coll}}^{[\nu]}|]$ **do** ▷ Main Sampling for collision sets $\mathcal{G}_{\mathsf{coll}}^{[\nu]}$
12: Let $(\alpha_1, s) \in \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$ s.t. $r_{\alpha_1} = \max\{r_\beta \mid \beta \in \mathcal{Q}_{\mathsf{coll}}^{[\nu]}[\gamma]\}$
13: $\mathsf{sample}_{\alpha_1} \leftarrow 0$; **run** CHCTR2-Sample$_1(\alpha_1)$
14: **for** $(\alpha, s) \in \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma] \backslash \{(\alpha_1, s)\}$ **do** $\mathsf{sample}_\alpha \leftarrow 1$; **run** CHCTR2-Sample$_2(\nu, s, \alpha_1, \alpha)$ **end for**
15: **end for**
16: **for** $\alpha \in \mathcal{Q}_{\mathsf{non\text{-}coll}}^{[\nu]}$ **do** ▷ Main sampling for non-collision queries
17: $\mathsf{sample}_\alpha \leftarrow 0$; **run** CHCTR2-Sample$_1(\alpha, \emptyset)$
18: **end for**
19: **end for**
20: **if** $\exists \alpha \in [q]$ s.t. $\mathsf{sample}_\alpha \geq 2$ **then**
for $(\alpha, i, j) \in [q] \times [2] \times [0, r_{\mathsf{max}}]$ **do** $(X_{i,j}^{(\alpha)}, Y_{i,j}^{(\alpha)}) \leftarrow (\varepsilon, \varepsilon)$ **end for end if**

Lines 11-15. In these lines, for each collision set $\mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$, the pairs $(X_{i,j}^{(\alpha)}, Y_{i,j}^{(\alpha)})_{\alpha \in \mathcal{Q}_{\mathsf{coll}}^{[\nu]}[\gamma], i \in [2], j \in [0, r_\alpha]}$ are defined by using the sampling algorithm CHCTR2-Sample$_1$ and the assignment algorithm CHCTR2-Sample$_2$ that are respectively given in Algorithms 3 and 4. First, CHCTR2-Sample$_1$ is performed for the longest query index α_1. For the other query indexes in the collision set, CHCTR2-Sample$_2$ is performed, where the internal pairs of the α_1-th query at s-th round are assigned to those of the other queries due to the collision of the edge value. For the longest query index α_1, $\mathsf{sample}_{\alpha_1}$ is initialized to 0, since all internal values in the collision set are not defined before running CHCTR2-Sample$_1$. For each non-longest query index α, sample_α is initialized to 1.

Lines 16-18. For each $\alpha \in \mathcal{Q}_{\mathsf{non\text{-}coll}}^{[\nu]}$, the counter sample_α is initialized to 0 and the pairs $(X_{i,j}^{(\alpha)}, Y_{i,j}^{(\alpha)})_{i \in [2], j \in [0, r_\alpha]}$ are defined by using CHCTR2-Sample$_1$.

Line 20. For some $\alpha \in [q]$, if two (or more) collisions occur in the algorithm, the counter sample_α becomes 2 (or more), and our algorithm does not ensure that the pairs are consistent with respect to permutations. In this case, all internal values are overwritten with ε. In other words, if for all $\alpha \in [q]$, $\mathsf{sample}_\alpha \leq 1$, all internal pairs are consistently defined.

Algorithm 3: CHCTR2-Sample$_1$. For $\alpha \in \mathcal{Q}_{\mathsf{coll},1}^{[\nu]} \cup \mathcal{Q}_{\mathsf{non\text{-}coll}}^{[\nu]}$, the algorithm defines internal pairs $(X_{i,j}^{(\alpha)}, Y_{i,j}^{(\alpha)})_{i \in [2], j \in [0, r_\alpha]}$ and checks collisions of the pairs. In this algorithm, for each collision, the counter sample_α is incremented by 1. For a collision with one of the sampled values $Y_{1,0}^{(\alpha)}, X_{2,0}^{(\alpha)}$, and $\{Y_{i,j} \mid (i,j) \in [2] \times [r_\alpha]\}$, the resampling procedure is performed. On a collision with the other values $X_{1,0}^{(\alpha)}, Y_{2,0}^{(\alpha)}$, and $\{X_{i,j} \mid (i,j) \in [2] \times [r_\alpha]\}$, the resampling procedure is not performed to obtain the BBB bound for the good transcript.

Lines 1-7. In Lines 1-3, the values of the left part $(Y_{1,0}^{(\alpha)}, X_{2,0}^{(\alpha)})$ are defined, fixing the internal pairs of the left part. In Lines 4 and 5, by these values, the IVs and the plaintext

Algorithm 3 Sampling Algorithm

Subroutine $\mathsf{CHCTR2\text{-}Sample}_1(\nu, \alpha)$

1: $Y_{1,0}^{(\alpha)} \xleftarrow{\$} \{0,1\}^n$; $X_{2,0}^{(\alpha)} \xleftarrow{\$} \{0,1\}^n$; $\hat{\mathsf{i}} \leftarrow 1$
2: **if** $X_{1,0}^{(\alpha)} \in \mathcal{X}_1^{[\nu]}$ **then** $Y_{1,0}^{(\alpha)} \leftarrow \mathcal{P}_1^{[\nu]}(X_{1,0}^{(\alpha)})$; $\mathsf{sample}_\alpha \leftarrow \mathsf{sample}_\alpha + 1$ **end if**
3: **if** $Y_{2,0}^{(\alpha)} \in \mathcal{Y}_2^{[\nu]}$ **then** $X_{2,0}^{(\alpha)} \leftarrow \mathcal{P}_2^{-1[\nu]}(Y_{2,0}^{(\alpha)})$; $\hat{\mathsf{i}} \leftarrow 2$; $\mathsf{sample}_\alpha \leftarrow \mathsf{sample}_\alpha + 1$ **end if**
4: $IV_1^{(\alpha)} \leftarrow X_{1,0}^{(\alpha)} \oplus Y_{1,0}^{(\alpha)}$; $IV_2^{(\alpha)} \leftarrow X_{2,0}^{(\alpha)} \oplus Y_{2,0}^{(\alpha)}$
5: **for** $j \in [r_\alpha]$ **do** $X_{1,j}^{(\alpha)} \leftarrow \mathsf{add}(IV_1^{(\alpha)}, j)$; $X_{2,j}^{(\alpha)} \leftarrow \mathsf{add}(IV_2^{(\alpha)}, j)$ **end for**
6: **if** $\exists j \in [r_\alpha]$ s.t. $X_{2,j}^{(\alpha)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$ **then** $\hat{\mathsf{i}} \leftarrow 2$ **end if**
7: $j_{\mathrm{free}} \leftarrow \max\left(\{0\} \cup \left\{j \in [r_\alpha - 1] \mid X_{1,j}^{(\alpha)} \notin \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\} \vee X_{2,j}^{(\alpha)} \notin \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}\right\}\right)$
8: **for** $j \in [r_\alpha] \backslash \{j_{\mathrm{free}}\}$ **do**
9: $\quad Y_{\hat{\mathsf{i}},j}^{(\alpha)} \xleftarrow{\$} \{0,1\}^n$; **if** $X_{\hat{\mathsf{i}},j}^{(\alpha)} \in \mathcal{X}_{\hat{\mathsf{i}}}^{[\nu]}$ **then** $Y_{\hat{\mathsf{i}},j}^{(\alpha)} \leftarrow \mathcal{P}_{\hat{\mathsf{i}}}^{[\nu]}(X_{\hat{\mathsf{i}},j}^{(\alpha)})$ **end if**
10: $\quad$ **if** $X_{\hat{\mathsf{i}},j}^{(\alpha)} = X_{\hat{\mathsf{i}},0}^{(\alpha)}$ **then** $Y_{\hat{\mathsf{i}},j}^{(\alpha)} \leftarrow Y_{\hat{\mathsf{i}},0}^{(\alpha)}$ **end if**
11: $\quad Y_{3-\hat{\mathsf{i}},j}^{(\alpha)} \leftarrow Y_{\hat{\mathsf{i}},j}^{(\alpha)} \oplus \hat{M}_j^{(\alpha)} \oplus \hat{C}_j^{(\alpha)}$
12: **end for**
13: **if** $j_{\mathrm{free}} \neq 0$ **then**
14: $\quad$ Derive $Y_{1,j_{\mathrm{free}}}^{(\alpha)}$ such that $Y_{1,0}^{(\alpha)} \oplus X_{2,0}^{(\alpha)} = \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$ and $R^{(\alpha)} = \mathsf{msb}_{|M_*^{(\alpha)}|}(Y_{1,1}^{(\alpha)} \| \cdots \| Y_{1,r_\alpha}^{(\alpha)}) \oplus M_*^{(\alpha)}$
15: $\quad Y_{2,j_{\mathrm{free}}}^{(\alpha)} \leftarrow Y_{1,j_{\mathrm{free}}}^{(\alpha)} \oplus \hat{M}_{j_{\mathrm{free}}}^{(\alpha)} \oplus \hat{C}_{j_{\mathrm{free}}}^{(\alpha)}$
16: **end if**
17: $R^{(\alpha)} \leftarrow \mathsf{msb}_{|M_*^{(\alpha)}|}(Y_{1,1}^{(\alpha)} \| \cdots \| Y_{1,r_\alpha}^{(\alpha)}) \oplus M_*^{(\alpha)}$
18: **if** $j_{\mathrm{free}} = 0$ **then**
19: $\quad$ **if** $\hat{\mathsf{i}} = 1$ **then** $X_{2,0}^{(\alpha)} \leftarrow Y_{1,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$ **end if**
20: $\quad$ **if** $\hat{\mathsf{i}} = 2$ **then** $Y_{1,0}^{(\alpha)} \leftarrow X_{2,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$ **end if**
21: $\quad IV_{3-\hat{\mathsf{i}}}^{(\alpha)} \leftarrow X_{3-\hat{\mathsf{i}},0}^{(\alpha)} \oplus Y_{3-\hat{\mathsf{i}},0}^{(\alpha)}$; **for** $j \in [r_\alpha]$ **do** $X_{3-\hat{\mathsf{i}},j}^{(\alpha)} \leftarrow \mathsf{add}(IV_{3-\hat{\mathsf{i}}}^{(\alpha)}, j)$ **end for**
22: **end if**
23: **if** $\left(X_{1,0}^{(\alpha)} \notin \mathcal{X}_1^{[\nu]} \wedge Y_{1,0}^{(\alpha)} \in \mathcal{Y}_1^{[\nu]}\right) \vee \left(Y_{2,0}^{(\alpha)} \notin \mathcal{Y}_2^{[\nu]} \wedge X_{2,0}^{(\alpha)} \in \mathcal{X}_2^{[\nu]}\right)$ **then**
24: $\quad \mathsf{sample}_\alpha \leftarrow \mathsf{sample}_\alpha + 1$; **if** $\mathsf{sample}_\alpha \leq 1$ **then goto Step** 1 **end if**
25: **end if**
26: **if** $\exists j \in [r_\alpha]$ s.t. $X_{1,j}^{(\alpha)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$ **then** $\mathsf{sample}_\alpha \leftarrow \mathsf{sample}_\alpha + 1$ **end if**
27: **if** $\exists j \in [r_\alpha]$ s.t. $X_{2,j}^{(\alpha)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$ **then** $\mathsf{sample}_\alpha \leftarrow \mathsf{sample}_\alpha + 1$ **end if**
28: **if** $\exists i \in [2], j \in [r_\alpha]$ s.t. $(X_{i,j}^{(\alpha)} \notin \mathcal{X}_i^{[\nu]} \cup \{X_{i,0}^{(\alpha)}\}) \wedge (Y_{i,j}^{(\alpha)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\})$ **then**
29: $\quad \mathsf{sample}_\alpha \leftarrow \mathsf{sample}_\alpha + 1$; **if** $\mathsf{sample}_\alpha \leq 1$ **then goto Step** 1 **end if**
30: **end if**
31: **for** $i \in [2]$ **do** $\mathcal{P}_i^{[\nu]} \xleftarrow{\cup} \left\{(X_{i,j}^{(\alpha)}, Y_{i,j}^{(\alpha)}) \mid j \in [0, r_\alpha]\right\}$ **end for**

blocks of the right part are fixed. followed by the collision check for the plaintext blocks in Line 6. In the collision check, an index $\hat{\mathsf{i}}$ is used for recording a round index with a collision. Note that if collisions occur at both first and second rounds, then $\mathsf{sample}_\alpha \geq 2$. This event is analyzed in the bad events $\mathsf{fail}_{\mathsf{sample}}$ and $\mathsf{2coll}$, and the index $\hat{\mathsf{i}}$ does not care the two collisions. In Line 7, an index $j_{\mathrm{free}} \in [0, r_\alpha - 1]$ is introduced, which is the minimum block index of XCTR such that both $X_{1,j_{\mathrm{free}}}^{(\alpha)}$ and $X_{2,j_{\mathrm{free}}}^{(\alpha)}$ are new. If there is no such index, then $j_{\mathrm{free}} \coloneqq 0$. Note that the freeness of the j_{free}-th ciphertext blocks is used to define compatible internal pairs in the sense of the CHCTR2 structure.

Lines 8-17. The ciphertext blocks $(Y_{1,j}^{(\alpha)}, Y_{2,j}^{(\alpha)})_{j \in [r_\alpha] \backslash \{j_{\mathrm{free}}\}}$ are defined according to the structure of CHCTR2 in Lines 8-12, i.e., $\hat{C}_j^{(\alpha)} = Y_{1,j}^{(\alpha)} \oplus Y_{2,j}^{(\alpha)} \oplus \hat{M}_j^{(\alpha)}$. In Lines 13-16, the j_{free}-th pair of ciphertext blocks $(Y_{1,j_{\mathrm{free}}}^{(\alpha)}, Y_{2,j_{\mathrm{free}}}^{(\alpha)})$ are defined so that the middle of the left part is consistent, i.e., $Y_{1,0}^{(\alpha)} \oplus X_{2,0}^{(\alpha)} = Z_{1,2}^{(\alpha)}$. Such consistent pairs can be defined by using the freeness of the j_{free}-th pair and the XOR-invertible property. Note that j_{free} must not

Algorithm 4 Assignment Algorithm

Subroutine CHCTR2-Sample$_2(\nu, s, \alpha_1, \alpha)$

1: **for** $j \in [0, r_\alpha]$ **do**
2: $\quad (X^{(\alpha)}_{s,j}, Y^{(\alpha)}_{s,j}) \leftarrow (X^{(\alpha_1)}_{s,j}, Y^{(\alpha_1)}_{s,j})$
3: $\quad$ **if** $j \neq 0$ **then** $Y^{(\alpha)}_{3-s,j} \leftarrow Y^{(\alpha)}_{s,j} \oplus \hat{M}^{(\alpha)}_j \oplus \hat{C}^{(\alpha)}_j$ **end if**
4: **end for**
5: $R^{(\alpha)} \leftarrow \mathsf{msb}_{|\hat{M}^{(\alpha)}_*|}(Y^{(\alpha)}_{1,1} \| \cdots \| Y^{(\alpha)}_{1,r_\alpha}) \oplus M^{(\alpha)}_*$
6: $Z^{(\alpha)}_{1,2} \leftarrow \mathsf{H}^{[\nu]}_1(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}^{[\nu]}_2(T^{(\alpha)}, R^{(\alpha)})$
7: **if** $s = 1$ **then** $X^{(\alpha)}_{2,0} \leftarrow Y^{(\alpha)}_{1,0} \oplus Z^{(\alpha)}_{1,2}$ **else** $Y^{(\alpha)}_{1,0} \leftarrow X^{(\alpha)}_{2,0} \oplus Z^{(\alpha)}_{1,2}$ **end if**
8: $IV^{(\alpha)}_1 \leftarrow X^{(\alpha)}_{1,0} \oplus Y^{(\alpha)}_{1,0}$; $IV^{(\alpha)}_2 \leftarrow X^{(\alpha)}_{2,0} \oplus Y^{(\alpha)}_{2,0}$;
9: **for** $j \in [r_\alpha]$ **do** $X^{(\alpha)}_{3-s,j} \leftarrow \mathsf{add}(IV^{(\alpha)}_{3-s}, j)$ **end for**
10: **if** $X^{(\alpha)}_{3-s,0} \in \mathcal{X}^{[\nu]}_{3-s} \vee Y^{(\alpha)}_{3-s,0} \in \mathcal{Y}^{[\nu]}_{3-s}$ **then** $\mathsf{sample}_\alpha \leftarrow \mathsf{sample}_\alpha + 1$ **end if**
11: **if** $\exists j \in [r_\alpha]$ s.t. $(X^{(\alpha)}_{3-s,j} \in \mathcal{X}^{[\nu]}_{3-s} \cup \{X^{(\alpha)}_{3-s,j'} \mid j' \in [0, j-1]\})$
$\qquad \vee (Y^{(\alpha)}_{3-s,j} \in \mathcal{Y}^{[\nu]}_{3-s} \cup \{Y^{(\alpha)}_{3-s,j'} \mid j' \in [0, j-1]\})$ **then**
12: $\quad \mathsf{sample}_\alpha \leftarrow \mathsf{sample}_\alpha + 1$
13: **end if**
14: $\mathcal{P}^{[\nu]}_{3-s} \xleftarrow{\cup} \left\{(X^{(\alpha)}_{3-s,j}, Y^{(\alpha)}_{3-s,j}) \mid j \in [0, r_\alpha]\right\}$ **end for**

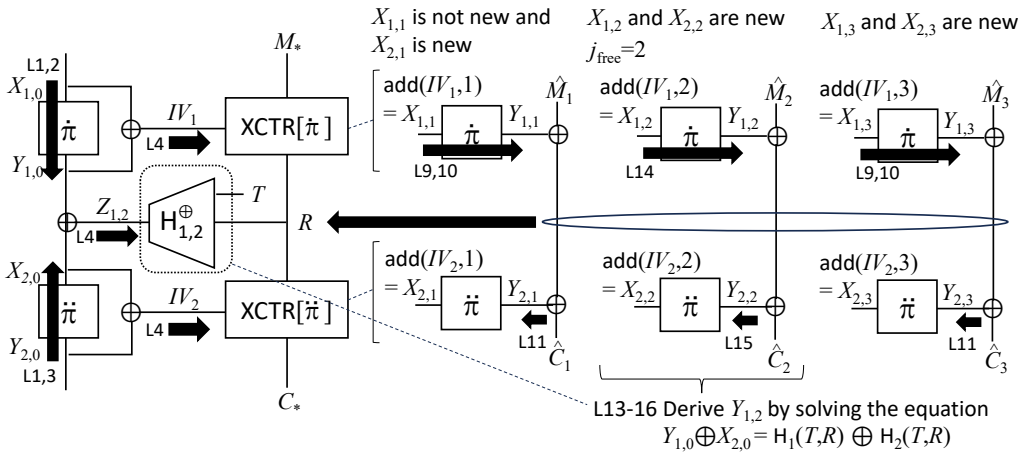


Fig. 6. CHCTR2-Sample$_1$ with $r_\alpha = 3$, $j_{\text{free}} = 2$ and $\hat{\mathsf{i}} = 1$. The arrows (or procedure) with "Li" show the lines with the corresponding procedures in Algorithm 3.

be r_α, since if the last block $M^{(\alpha)}_{r_\alpha}$ is not full n bits, the freeness of the ciphertext blocks $Y^{(\alpha)}_{1,r_\alpha}$ and $Y^{(\alpha)}_{2,r_\alpha}$ cannot be used. Fig. 6 shows a diagram of the procedure.

Lines 18-22. If $j_{\text{free}} = 0$, then there is no freeness of the ciphertext blocks in the right part. In this case, we use the freeness of the ciphertext block $Y^{(\alpha)}_{1,0}$ or the plaintext block $X^{(\alpha)}_{2,0}$ in the left part. In Lines 8-12 with $j_{\text{free}} = 0$, all ciphertext blocks $(Y^{(\alpha)}_{1,j}, Y^{(\alpha)}_{2,j})_{j \in [r_\alpha]}$ are defined. Then, the hash values $Z^{(\alpha)}_{1,2}$ are fixed. In Lines 18-22, by using the freeness of $Y^{(\alpha)}_{1,0}$ or $X^{(\alpha)}_{2,0}$ (assuming that $X^{(\alpha)}_{1,0}$ or $Y^{(\alpha)}_{2,0}$ is new), the free value is defined such that the middle of the left part is consistent, i.e., $Y^{(\alpha)}_{1,0} \oplus X^{(\alpha)}_{2,0} = Z^{(\alpha)}_{1,2}$. Fig. 7 shows a diagram of the procedure, where the plaintext blocks $(X^{(\alpha)}_{1,j})_{j \in [r_\alpha - 1]}$ except for the last block are not new, but the other plaintext blocks $(X^{(\alpha)}_{2,j})_{j \in [r_\alpha - 1]}$ are new.

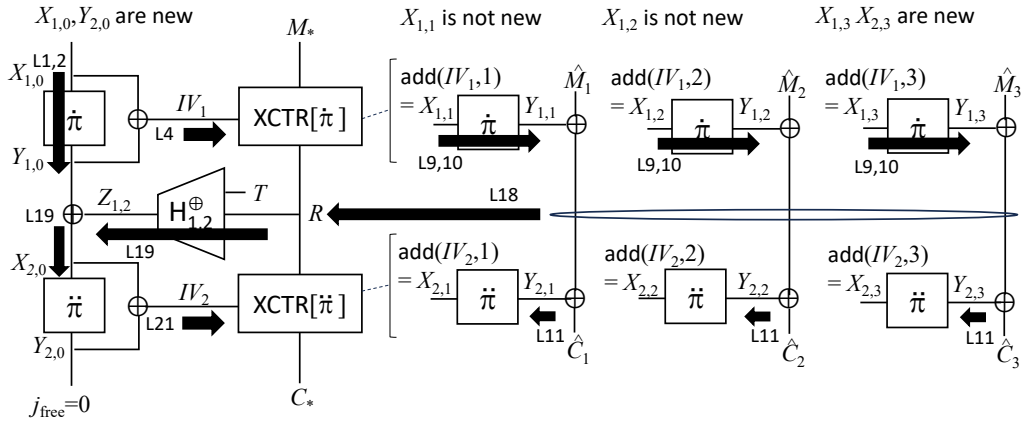


Fig. 7. CHCTR2-Sample$_1$ with $r_\alpha = 3$ and $j_{\text{free}} = 0$.

Lines 23-30. After all pairs are defined, collisions with the sampled values $Y_{1,0}^{(\alpha)}, X_{2,0}^{(\alpha)}$, $\{X_{i,j} \mid (i,j) \in [2] \times [r_\alpha]\}$, and $\{Y_{i,j} \mid (i,j) \in [2] \times [r_\alpha]\}$ are checked. If a collision with one of the values occurs, sample_α is incremented by 1, and if $\mathsf{sample}_\alpha \leq 1$ and the collision is for $Y_{1,0}^{(\alpha)}, X_{2,0}^{(\alpha)}$ or $Y_{i,j}$, then the resampling procedure is performed.

Line 31. For each $i \in [2]$, all internal pairs of the α-th query are added to the table $\mathcal{P}_i^{[\nu]}$.

Algorithm 4: CHCTR2-Sample$_2$. CHCTR2-Sample$_2$ defines internal pairs of the α-th query with a collision of the edge value such that α is not the index of the longest query. The input s is the round index with a collision in the left part: if $X_{1,0}^{(\alpha)} = X_{1,0}^{(\alpha_1)}$ (resp. $Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\alpha_1)}$), $s = 1$ (resp. $s = 2$). In CHCTR2-Sample$_2$, the internal pairs of the α-th query are defined in Lines 1-9, followed by collision checks in Lines 10-13 and a finalization in Line 14.

Lines 1-9. In Lines 1-4, the internal pairs of the s-th round of the α_1-th (longest) query are assigned to those of the s-th round of the α_1-th query due to the collision of the s-th round in the left part.[‡‡] In Lines 5-9, the internal pairs of the $(3-s)$-th round of the α-th query are fixed according to the structure of CHCTR2.

Lines 10-13. The lines check collisions for the pairs $(X_{3-s,j}^{(\alpha)}, Y_{3-s,j}^{(\alpha)})_{i \in [2], j \in [0, r_\alpha]}$ of the α-th query. In the lines, for a collision with one of the internal values, sample_α is incremented by 1, resulting in $\mathsf{sample}_\alpha \geq 2$.

Line 14. All internal pairs of the $(3-s)$-th round of the α-th query are added to the table $\mathcal{P}_{3-s}^{[\nu]}$.

Summary. Algorithms 2, 3, and 4 define dummy internal pairs of plaintext and ciphertext blocks according to the structure of CHCTR2. The number of collisions for the α-th query is counted by sample_α. The algorithms consistently define the internal pairs with respect to the property of RPs as long as $\mathsf{sample}_\alpha \leq 1$. On the other hand, if $\mathsf{sample}_\alpha \geq 2$, the algorithms do not ensure the consistency of the internal pairs. This case is evaluated as a bad event in our proof.

[‡‡]Note that since $r_\alpha \leq r_{\alpha_1}$, there is no internal value that is sampled freshly in these lines.

6.4 Adversary's View

In the real world, the extended plaintexts $(\hat{M}^{(\alpha)})_{\alpha\in[q]}$ and the extended ciphertexts $(\hat{C}^{(\alpha)})_{\alpha\in[q]}$ are defined as follows. For each $\alpha \in [q]$, if $\mathsf{query}^{(\alpha)} = \mathsf{enc}$, then $\hat{M}^{(\alpha)} := M^{(\alpha)}\|0^{m_\alpha^+}$ and $\hat{C}^{(\alpha)} := C^{(\alpha)}\|\mathsf{lsb}_{m_\alpha^+}\big(Y_{1,r_\alpha}^{(\alpha)} \oplus Y_{2,r_\alpha}^{(\alpha)}\big)$, where $Y_{1,r_\alpha}^{(\alpha)} \oplus Y_{2,r_\alpha}^{(\alpha)}$ is the sum of the truncated ciphertext blocks that is not used for the encryption. If $\mathsf{query}^{(\alpha)} = \mathsf{dec}$, then $\hat{M}^{(\alpha)} := M^{(\alpha)}\|\mathsf{lsb}_{m_\alpha^+}\big(Y_{1,r_\alpha}^{(\alpha)} \oplus Y_{2,r_\alpha}^{(\alpha)}\big)$ and $\hat{C}^{(\alpha)} := C^{(\alpha)}\|0^{m_\alpha^+}$. In the ideal world, the extended plaintexts and ciphertexts are defined by Algorithm 2.

In the decision stage, all hash keys and all internal pairs are revealed to the adversary $\mathbf{A}$. Then, $\mathbf{A}$'s transcript τ consists of (dummy) hash keys $\mathcal{H} = \{(H_1^{[\nu]}, H_2^{[\nu]}) \mid \nu \in [u]\}$, query-response tuples $\{(T^{(\alpha)}, \hat{M}^{(\alpha)}, \hat{C}^{(\alpha)}) \mid \alpha \in [q]\}$, and (dummy) internal pairs $(\mathcal{P}_1^{[\nu]})_{\nu\in[u]}$ and $(\mathcal{P}_2^{[\nu]})_{\nu\in[u]}$.

6.5 Definitions of Good and Bad Transcripts

Conditions on bad transcripts are defined below. The set of bad transcripts $\mathcal{T}_{\mathsf{bad}}$ is a subset of $\mathcal{T}$ such that one of the conditions holds. The set of good transcripts is defined as $\mathcal{T}_{\mathsf{good}} := \mathcal{T}\backslash\mathcal{T}_{\mathsf{bad}}$, which is the subset of $\mathcal{T}$ such that the conditions do not hold.

hit_0. The condition is defined as

$$\mathsf{hit}_0 \Leftrightarrow \exists\alpha \in [q], i \in [2], j \in [0, r_\alpha] \text{ s.t. } X_{i,j}^{(\alpha)} = \mathtt{bin}_n(0) \vee Y_{i,j}^{(\alpha)} = H_i^{[\mathsf{u}_\alpha]}.$$

The condition implies that for some α and i, the hash key $H_i^{[\mathsf{u}_\alpha]}$ is used in the process of $\mathsf{CHCTR2}$, except for the hash key generation. In other words, assuming that hit_0 does not occur, all hash keys can be independent of the other RP invocations in $\mathsf{CHCTR2}$.

2coll. The condition is defined as

$$\mathsf{2coll} \Leftrightarrow \exists\nu \in [u], \alpha, \beta, \gamma \in \mathcal{Q}^{[\nu]} \text{ s.t. } \beta \neq \alpha \neq \gamma \wedge X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta)} \wedge Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\gamma)}.$$

The condition implies that for some α, collisions with both of the edge values $X_{1,0}^{(\alpha)}$ and $Y_{2,0}^{(\alpha)}$ in the left part occur. Assuming that $\mathsf{2coll}$ does not occur, for any query index α in $\mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$, no collision occurs for at least one of the edge values.

$\mathsf{fail}_{\mathsf{sample}}$. This condition is defined as

$$\mathsf{fail}_{\mathsf{sample}} \Leftrightarrow X_{1,0}^{(1)} = \varepsilon.$$

The condition implies that for some $\alpha \in [q]$, two collisions occur and $\mathsf{sample}_\alpha \geq 2$. If $\mathsf{sample}_\alpha \leq 1$, our algorithms ensure the consistency of the internal values.

6.6 Evaluation for Bad Transcript: Upper-Bounding $\mathbf{Pr}[\mathsf{T}_I \in \mathcal{T}_{\mathsf{bad}}]$

For each $\mathsf{E} \in \{\mathsf{hit}_0, \mathsf{2coll}, \mathsf{fail}_{\mathsf{sample}}\}$, let E^* be an event that E occurs before the other events in $\{\mathsf{hit}_0, \mathsf{2coll}, \mathsf{fail}_{\mathsf{sample}}\}\backslash\{\mathsf{E}\}$ occur. We then have

$$\begin{aligned}
\Pr[\mathsf{T}_I \in \mathcal{T}_{\mathsf{bad}}] &\leq \Pr[\mathsf{hit}_0^*] + \Pr[\mathsf{2coll}^*] + \Pr[\mathsf{fail}_{\mathsf{sample}}^*] \\
&\leq \frac{6\omega\sigma}{2^n} + \frac{0.5\omega^2\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{4\omega^2\ell\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{11\omega^2\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} \\
&\leq \frac{4\omega^2\ell\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{12\omega^2\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{6\omega\sigma}{2^n}. \qquad (1)
\end{aligned}$$

These bounds are given below.

Evaluation of $\Pr[\mathsf{hit}_0^*]$. Fix $\alpha \in [q]$. We assume that hit_0 does not occur before the α-th query. For the collisions with $X_{1,0}^{(\alpha)}$, $Y_{1,0}^{(\alpha)}$, $X_{2,0}^{(\alpha)}$, or $Y_{2,0}^{(\alpha)}$, the regular hash functions ensure that

$$\Pr[X_{1,0}^{(\alpha)} = \mathtt{bin}_n(0)] \le \frac{\omega\ell_\alpha}{2^n}, \qquad \Pr[Y_{1,0}^{(\alpha)} = H_1^{[\mathsf{u}_\alpha]}] \le \frac{\omega\ell_\alpha}{2^n},$$
$$\Pr[X_{2,0}^{(\alpha)} = \mathtt{bin}_n(0)] \le \frac{\omega\ell_\alpha}{2^n}, \qquad \Pr[Y_{2,0}^{(\alpha)} = H_2^{[\mathsf{u}_\alpha]}] \le \frac{\omega\ell_\alpha}{2^n}.$$

We next consider the other collisions for hit_0. $X_{i,j}^{(\alpha)}$ values has the following structures:

$$X_{2,0}^{(\alpha)} = Y_{1,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}),$$
$$X_{i,j}^{(\alpha)} = X_{i,0}^{(\alpha)} \oplus Y_{i,0}^{(\alpha)} \oplus A_j.$$

Since $Y_{1,0}^{(\alpha)}$ or $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, for each $(i,j) \in [2] \times [r_\alpha]$, we have the following bounds:

$$\Pr[X_{i,j}^{(\alpha)} = \mathtt{bin}_n(0)] \le \frac{1}{2^n}, \qquad \Pr[Y_{i,j}^{(\alpha)} = H_i^{[\nu]} \mid X_{i,j}^{(\alpha)} \neq \mathtt{bin}_n(0)] \le \frac{1}{2^n}.$$

Summing these bounds for all possible cases, we have the following bound.

$$\begin{aligned}\Pr[\mathsf{hit}_0] &\le \sum_{\nu\in[u]} \sum_{\alpha\in\mathcal{Q}^{[\nu]}} \left(\frac{4\omega\ell_\alpha}{2^n} + \frac{2\ell_\alpha}{2^n}\right) \\ &\le \sum_{\nu\in[u]} \frac{6\omega\sigma_\nu}{2^n} \\ &\le \frac{6\omega\sigma}{2^n}.\end{aligned}$$

Evaluation of $\Pr[\mathsf{2coll}^*]$. For each $\nu \in [u]$, the number of triples $(\alpha,\beta,\gamma) \in (\mathcal{Q}^{[\nu]})^3$ such that $\beta \neq \alpha \neq \gamma$, the AXU hash functions $\mathsf{H}_1^{[\nu]}$ and $\mathsf{H}_2^{[\nu]}$ ensure that

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta)} \wedge Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\gamma)}] \le \frac{\omega\max\{\ell_\alpha,\ell_\beta\}}{2^n} \cdot \frac{\omega\max\{\ell_\alpha,\ell_\gamma\}}{2^n}.$$

Summing the bound for each $\nu \in [u], \alpha,\beta,\gamma \in \mathcal{Q}^{[\nu]}$, we have

$$\begin{aligned}\Pr[\mathsf{2coll}^*] &\le \sum_{\nu\in[u]} \sum_{\substack{\alpha,\beta,\gamma\in\mathcal{Q}^{[\nu]} \\ \text{s.t. } \beta\neq\alpha\neq\gamma}} \left(\frac{\omega\max\{\ell_\alpha,\ell_\beta\}}{2^n} \cdot \frac{\omega\max\{\ell_\alpha,\ell_\gamma\}}{2^n}\right) \\ &\le \sum_{\nu\in[u]} \frac{\omega^2\sigma_\nu^2 q_\nu}{2\cdot 2^{2n}} \\ &\le \frac{0.5\omega^2 q_{\mathsf{u}}\sigma_{\mathsf{u}}\sigma}{2^{2n}}.\end{aligned}$$

Evaluation of $\Pr[\mathsf{fail}^*_{\mathsf{sample}}]$. Our algorithm ensures that if $\mathsf{fail}_{\mathsf{sample}}$ holds, then $\exists\alpha \in [q]$ s.t. $\mathsf{sample}_\alpha \ge 2$. Hence, we have

$$\Pr[\mathsf{fail}^*_{\mathsf{sample}}] \le \Pr[(\exists\alpha \in [q] \text{ s.t. } \mathsf{sample}_\alpha \ge 2) \wedge \neg(\mathsf{hit}_0 \vee \mathsf{2coll})].$$

By $\neg$2coll, $\mathcal{Q}^{[\nu]} = \mathcal{Q}^{[\nu]}_{\mathsf{coll}} \cup \mathcal{Q}^{[\nu]}_{\mathsf{non\text{-}coll}}$. For each $\alpha \in \mathcal{Q}^{[\nu]}_{\mathsf{coll},2}$, initially, $\mathsf{sample}_\alpha = 1$, and for each $\alpha \in \mathcal{Q}^{[\nu]}_{\mathsf{coll},1} \cup \mathcal{Q}^{[\nu]}_{\mathsf{non\text{-}coll}}$, initially, $\mathsf{sample}_\alpha = 0$. We thus use the following conditions:

$$\mathsf{fail}^{*1}_{\mathsf{sample}} \Leftrightarrow (\exists \nu \in [\mathsf{u}], \alpha \in \mathcal{Q}^{[\nu]}_{\mathsf{coll},2} \text{ s.t. } \mathsf{sample}_\alpha \geq 2) \wedge \neg(\mathsf{hit}_0 \vee \mathsf{2coll}),$$
$$\mathsf{fail}^{*2}_{\mathsf{sample}} \Leftrightarrow (\exists \nu \in [\mathsf{u}], \alpha \in \mathcal{Q}^{[\nu]}_{\mathsf{coll},1} \cup \mathcal{Q}^{[\nu]}_{\mathsf{non\text{-}coll}} \text{ s.t. } \mathsf{sample}_\alpha \geq 2) \wedge \neg(\mathsf{hit}_0 \vee \mathsf{2coll}).$$

By using these conditions, we have

$$\begin{aligned}
\Pr[\mathsf{fail}^{*}_{\mathsf{sample}}] &\leq \Pr[(\exists \alpha \in [q] \text{ s.t. } \mathsf{sample}_\alpha \geq 2) \wedge \neg(\mathsf{hit}_0 \vee \mathsf{2coll})] \\
&\leq \Pr[\mathsf{fail}^{*1}_{\mathsf{sample}}] + \Pr[\mathsf{fail}^{*2}_{\mathsf{sample}}] \\
&\leq \frac{8\omega^2\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{3\omega^2\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{4\omega^2\ell\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} \\
&\leq \frac{4\omega^2\ell\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{11\omega^2\sigma_{\mathsf{u}}^2\sigma}{2^{2n}}.
\end{aligned}$$

Overviews of the evaluations for deriving the bounds are given below. The detailed evaluations are given in Section 9.

Evaluation of $\Pr[\mathsf{fail}^{*1}_{\mathsf{sample}}]$. For each $\nu \in [u], \alpha \in \mathcal{Q}^{[\nu]}_{\mathsf{coll}}$, initially, $\mathsf{sample}_\alpha = 1$ i.e,

$$\mathsf{C}_0 \Leftrightarrow \exists \nu \in [u], \alpha, \beta_1 \in \mathcal{Q}^{[\nu]} \text{ s.t. } \alpha \neq \beta_1 \wedge r_\alpha \leq r_{\beta_1} \wedge \left(X^{(\alpha)}_{1,0} = X^{(\beta_1)}_{1,0} \vee Y^{(\alpha)}_{2,0} = Y^{(\beta_1)}_{2,0}\right).$$

Note that the query index β_1 is a longest query. Fix ν and α. For each $\beta_1 \in \mathcal{Q}^{[\nu]}$, since $X^{(\alpha)}_{1,0}$ (resp. $Y^{(\alpha)}_{2,0}$) is defined by using the AXU hash function $\mathsf{H}^{[\nu]}_1$ (resp. $\mathsf{H}^{[\nu]}_2$) or the plaintext block $M^{(\alpha)}_0$ (resp. $C^{(\alpha)}_0$) that is an (almost) n-bit random value if $\mathsf{query}^{(\alpha)} = \mathsf{enc}$ (resp. $\mathsf{query}^{(\alpha)} = \mathsf{dec}$), the probabilities of the collisions $X^{(\alpha)}_{1,0} = X^{(\beta_1)}_{1,0}$ and $Y^{(\alpha)}_{2,0} = Y^{(\beta_1)}_{2,0}$ can be bounded by using the AXU bounds or the n-bit randomnesses.

We next consider a collision with one of the internal values of the α-th query which increments sample_α by 1 in Algorithm 4. Let s be the round index with a collision in the left part: if $X^{(\alpha)}_{1,0} = X^{(\beta_1)}_{1,0}$ (resp. $Y^{(\alpha)}_{2,0} = Y^{(\beta_1)}_{2,0}$), $s = 1$ (resp. $s = 2$). Then, our algorithm increments sample_α by 1 if one of the following collision occurs.

$$\begin{aligned}
&\mathsf{C}_{1,1} \Leftrightarrow X^{(\alpha)}_{3-s,0} \in \mathcal{X}^{[\nu]}_{3-s}. \\
&\mathsf{C}_{1,2} \Leftrightarrow Y^{(\alpha)}_{3-s,0} \in \mathcal{Y}^{[\nu]}_{3-s}. \\
&\mathsf{C}_{1,3} \Leftrightarrow \exists j \in [r_\alpha] \text{ s.t. } .X^{(\alpha)}_{3-s,j} \in \mathcal{X}^{[\nu]}_{3-s} \cup \{X^{(\alpha)}_{3-s,0}\} \\
&\mathsf{C}_{1,4} \Leftrightarrow \exists j \in [r_\alpha] \text{ s.t. } Y^{(\alpha)}_{3-s,j} \in \mathcal{Y}^{[\nu]}_{3-s} \cup \{Y^{(\alpha)}_{3-s,j'} \mid j' \in [0, j-1]\}.
\end{aligned}$$

Since each of $X_{i,j}, Y_{i,j}$ is defined by using the AXU/regular hash function, an n-bit random value, a plaintext block M_0, or a ciphertext block C_0, the collision probabilities can be bounded by using the AXU/regular bound or the n-bit randomness.

Hence, we have the following bounds by using the AXU/regular hash functions, n-bit random values, or the randomnesses of M_0/C_0.

- $\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,1}$: For each $\nu \in [u], \alpha, \beta_1 \in \mathcal{Q}^{[\nu]}, X^{(\beta_2)}_{3-s,j} \in \mathcal{X}^{[\nu]}_{3-s}$, we have the following bounds:

$$\Pr[X^{(\alpha)}_{1,0} = X^{(\beta_1)}_{1,0} \wedge X^{(\alpha)}_{2,0} = X^{(\beta_2)}_{2,j}] \leq \frac{2\omega \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}} \qquad \text{if } s = 1,$$
$$\Pr[Y^{(\alpha)}_{2,0} = Y^{(\beta_1)}_{2,0} \wedge X^{(\alpha)}_{1,0} = X^{(\beta_2)}_{1,j}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^{2n}} \qquad \text{if } s = 2.$$

– $\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,2}$: For each $\nu \in [u], \alpha, \beta_1 \in \mathcal{Q}^{[\nu]}, Y_{3-s,j}^{(\beta_2)} \in \mathcal{Y}_{3-s}^{[\nu]}$, we have the following bounds:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)} \wedge X_{2,0}^{(\alpha)} = X_{2,j}^{(\beta_2)}] \leq \frac{2\omega \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}} \qquad \text{if } s = 1,$$

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)} \wedge X_{1,0}^{(\alpha)} = X_{1,j}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega^2 \max\{\ell_\alpha, \ell_{\beta_1}\} \max\{\ell_\alpha, \ell_{\beta_2}\}}{2^{2n}} & \text{if } s = 2 \wedge \beta_1 = \beta_2 \wedge j = 0, \\ \dfrac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^{2n}} & \text{if } s = 2 \wedge (\beta_1 \neq \beta_2 \vee j \neq 0). \end{cases}$$

– $\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,3}$: For each $\nu \in [u], \alpha, \beta_1 \in \mathcal{Q}^{[\nu]}$, $j \in [r_\alpha], X_{3-s,j^\dagger}^{(\beta_2)} \in \mathcal{X}_{3-s}^{[\nu]} \cup \{X_{3-s,0}^{(\alpha)}\}$, we have the following bounds:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)} \wedge X_{2,j}^{(\alpha)} = X_{2,j^\dagger}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega^2 \ell_\alpha \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^{2n}} & \text{if } s = 1 \wedge \alpha = \beta_2 \wedge j^\dagger = 0, \\ \dfrac{2\omega \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}} & \text{if } s = 1 \wedge (\alpha \neq \beta_2 \vee j^\dagger \neq 0), \end{cases}$$

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)} \wedge X_{1,j}^{(\alpha)} = X_{1,j^\dagger}^{(\beta_2)}] \leq \frac{2\omega \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}} \qquad \text{if } s = 2.$$

– $\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,4}$: For each $\nu \in [u], \alpha, \beta_1 \in \mathcal{Q}^{[\nu]}$, $j \in [r_\alpha], Y_{3-s,j^\dagger}^{(\beta_2)} \in \mathcal{Y}_{3-s}^{[\nu]} \cup \{Y_{3-s,j'}^{(\alpha)} \mid j' \in [0, j-1]\}$, we have the following bounds:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)} \wedge Y_{2,j}^{(\alpha)} = Y_{2,j^\dagger}^{(\beta_2)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^{2n}} \qquad \text{if } s = 1,$$

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)} \wedge Y_{1,j}^{(\alpha)} = Y_{1,j^\dagger}^{(\beta_2)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^{2n}} \qquad \text{if } s = 2.$$

By using the above bounds, the probability that two collisions occur between the α-th and β_1-th queries and between the α-th and β_1-th queries is

$$\begin{aligned} &\leq \max\left\{4\ell_{\beta_2} \cdot \frac{2\omega \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}}, 4\ell_\alpha \cdot \frac{2\omega^2 \max\{\ell_\alpha, \ell_{\beta_1}\} \max\{\ell_\alpha, \ell_{\beta_2}\}}{2^{2n}}, \right. \\ &\qquad \left. 4\ell_\alpha\ell_{\beta_2} \cdot \frac{2\omega \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}}, 4\ell_\alpha\ell_{\beta_2} \cdot \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^{2n}}\right\} \\ &\leq \frac{8\omega^2 \ell_\alpha \max\{\ell_\alpha, \ell_{\beta_2}\} \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}}. \end{aligned}$$

Summing the bound for each tuple $(\nu, \alpha, \beta_1, \beta_2)$, we have the following bound:

$$\begin{aligned} \Pr[\mathsf{fail}_{\mathsf{sample}}^{*1}] &\leq \sum_{\nu \in [u]} \sum_{\alpha, \beta_1, \beta_2 \in \mathcal{Q}^{[\nu]}} \frac{8\omega^2 \ell_\alpha \max\{\ell_\alpha, \ell_{\beta_2}\} \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}} \\ &\leq \sum_{\nu \in [u]} \frac{8\omega^2 \sigma_\nu^3}{2^{2n}} \\ &\leq \frac{8\omega^2 \sigma_{\mathsf{u}}^2 \sigma}{2^{2n}}. \end{aligned}$$

Evaluation of $\Pr[\mathsf{fail}_{\mathsf{sample}}^{*2}]$. For each $\nu \in [u], \alpha \in \mathcal{Q}_{\mathsf{coll},1}^{[\nu]} \cup \mathcal{Q}_{\mathsf{non\text{-}coll}}^{[\nu]}$, initially, $\mathsf{sample}_\alpha = 0$. If two collisions occur with plaintext or ciphertext blocks of the α-th query, then $\mathsf{sample}_\alpha \geq 2$. As the evaluation of $\Pr[\mathsf{fail}_{\mathsf{sample}}^{2}]$, the probability of the two collisions can be bounded by

using by using the AXU/regular hash functions, n-bit random values, or M_0/C_0, and we have the following bound.

$$\Pr[\mathsf{fail}^{*2}_{\mathsf{sample}}] \leq \frac{3\omega^2\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{4\omega^2\ell\sigma_{\mathsf{u}}^2\sigma}{2^{2n}}.$$

6.7 Evaluation for Good Transcript: Lower-Bounding $\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]}$

As explained in Section 5.2, our resampling-based proof increases the number of chances of satisfying the target good transcript, i.e., $\Pr[\mathsf{T}_I = \tau]$ is increased, since our algorithm resamples the internal values $Y^{(\alpha)}_{1,0}, X^{(\alpha)}_{2,0}$, and $(Y^{(\alpha)}_{i,j})_{i\in[2],j\in[r_\alpha]}$ if a collision occurs. On the other hand, in the real world, the internal values are defined by RPs $(\dot{\pi}^{\pm}_\nu, \ddot{\pi}^{\pm}_\nu)_{\nu\in[u]}$, and in the ideal world, the internal values are chosen uniformly at random from $\{0,1\}^n$. Hence, the sampling space of the ideal world is larger than that of the real world. Similarly to the evaluation for good transcripts of the security proof of SoP, by using the "budget" of the ideal world, the impact of the increase can be canceled out, and we can prove that $\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]} \geq 1 - \frac{4q_{\mathsf{u}}\sigma_{\mathsf{u}}\sigma}{2^{2n}}$. The detailed evaluation is given in Section 10.

6.8 Conclusion

Summing the mu-SPRP-advantage, the bound in Eq. (1) for bad transcripts, and the bound $\frac{4q_{\mathsf{u}}\sigma_{\mathsf{u}}\sigma}{2^{2n}}$ for good transcripts, we obtain the bound in Theorem 1.

7 Security of HCTR2 and HCTR2-TwKD

In this proof, we assume that pad has the following structures that captures the feature of the hash function of HCTR2. Let $\mathsf{zp}_n : \{0,1\}^* \to \{0,1\}^{n*}$ be a padding function such that for an input $D \in \{0,1\}^*$, the output is defined as $\mathsf{zp}_n \coloneqq D\|0^{n|D|_n-|D|}$. Let $\mathsf{ozp}_n : \{0,1\}^* \to \{0,1\}^{n*}$ be a padding function such that for an input $D \in \{0,1\}^*$, the output is defined as $\mathsf{ozp}_n(D) \coloneqq D\|10^{n|D\|1|_n-|D|}$. Let $\mathsf{pad}_{\mathsf{pf}} : [0, 2^{n-3}-1] \times \{0,1\} \to \{0,1\}^n\backslash\{0^n\}$ be an injective function.

- pad is an injective padding function such that for an input pair $(T, D) \in \mathcal{T} \times \mathcal{M}$, the output is defined as
  - if n **divides** $|D|$ then $\mathsf{pad}(T, M) \coloneqq (\mathsf{pad}_{\mathsf{pf}}(|T|, 1)\|\mathsf{zp}_n(T)\|D)$, and
  - otherwise $\mathsf{pad}(T, D) \coloneqq \mathsf{poly}_H((\mathsf{pad}_{\mathsf{pf}}(|T|, 0)\|\mathsf{zp}_n(T)\|\mathsf{ozp}_n(D))$.

In this proof, we use the following property.

Definition 3 (Almost 2XOR Universal). *$\{\mathsf{H}_H\}_{H\in\mathcal{H}}$ is ω-A2XU if for any $Y \in \{0,1\}^n$, pairs (T, M), $(T', M') \in \mathcal{T} \times \mathcal{M}$, and $C \in \{0,1\}^{|M|}$ such that $\mathsf{lsb}_{|\mathsf{pad}(T',M')|}(\mathsf{pad}(T, M) \oplus \mathsf{pad}(T, C)) \neq \mathsf{pad}(T', M')$,*

$$\Pr[H \xleftarrow{\$} \mathcal{H}, \mathsf{H}_H(T, M) \oplus \mathsf{H}_H(T, C) \oplus \mathsf{H}_H(T', M') = Y] \leq \frac{\omega\left(\max\{|T|_n + |M|_n, |T'|_n + |M'|_n\} + 1\right)}{2^n}.$$

The HCTR2's hash with POLYVAL is 1-A2XU.

The following theorem shows the mu-STPRP-security bound of HCTR2.

Theorem 2 (Security of HCTR2). *Assume that* $\{\mathsf{H}_H\}_{H\in\mathcal{H}}$ *is* ω*-regular,* ω*-AXU, and* ω*-A2XU. For any adversary* $\mathbf{A}$ *running in time* t *and making* q *queries to* u *users such that the total number of* n*-bit data blocks in all queries is* σ *and the maximum number of data blocks per user is* σ_{u}*, we have*

$$\mathbf{Adv}^{\mathsf{mu\text{-}vil\text{-}stprp}}_{\mathsf{HCTR2}}(\mathbf{A}) \le \frac{3\omega\sigma_{\mathsf{u}}\sigma}{2^n} + \frac{4n\sigma}{2^n} + \mathbf{Adv}^{\mathsf{mu\text{-}sprp}}_{E}(u, \sigma, t + O(\sigma)).$$

The proof is given in Section 11. An overview of the proof is given after the following theorem.

Regarding HCTR2-TwKD, we use mu-PRF-secure functions for TwKD. Then, for each user $\nu \in [u]$, TwKD with a fresh tweak generates a fresh key of HCTR2. Hence, the maximum number of data blocks per tweak σ_{tw} becomes the maximum number of data blocks per key of HCTR2. Replacing σ_{u} in the above bound with σ_{tw}, we obtain the following bound of HCTR2-TwKD.

Theorem 3 (Security of HCTR2-TwKD). *Assume that* $\{\mathsf{H}_H\}_{H\in\mathcal{H}}$ *is* ω*-regular,* ω*-AXU, and* ω*-A2XU. For any adversary* $\mathbf{A}$ *running in time* t *and making* q *queries to* u *users such that the total number of* n*-bit data blocks in all queries is* σ*, the maximum number of* n*-bit data blocks per pair of user and tweak is* σ_{tw}*, and the total number of pairs of user and tweak is at most* N_{tw}*, we have*

$$\begin{aligned}\mathbf{Adv}^{\mathsf{mu\text{-}vil\text{-}stprp}}_{\mathsf{HCTR2\text{-}TwKD}}(\mathbf{A}) \le \frac{3\omega\sigma_{\mathsf{tw}}\sigma}{2^n} + \frac{4n\sigma}{2^n} + \mathbf{Adv}^{\mathsf{mu\text{-}sprp}}_{E}(N_{tw}, \sigma, t + O(\sigma)) \\ + \mathbf{Adv}^{\mathsf{mu\text{-}prf}}_{F}(u, q, t + O(\sigma)).\end{aligned}$$

Overview of the Proof of Theorem 2. The bound in Theorem 2 can be obtained by simplifying the proof of CHCTR2. The proof of CHCTR2 considers two collisions of internal plaintext blocks or ciphertext blocks for BBB security. On the other hand, for the proof of HCTR2, we can consider only a single collision within the same user for birthday-bound security. By the birthday analysis with ω-regular, ω-AXU, and ω-A2XU, for each user $\nu \in [u]$, the collision probability is at most $\frac{3\omega\sigma_\nu^2}{2^n} + \frac{4n\sigma}{2^n}$ where σ_ν is the number of n-bit data blocks in queries to the ν-th user. Summing the bound for each user, we obtain the bound in Theorem 2.

8 Software Performance Evaluation

This section benchmarks optimized software implementations of the proposed methods and the state-of-the-art modes. We instantiate the BBB-secure modes (CHCTR2, HCTR2-TwKD, and *bbb-ddd-AES*) using the 128-bit primitives, AES-256 and POLYVAL. Meanwhile, the birthday-bounded HCTR2 is instantiated with the 256-bit block cipher Rijndael-256-256, which benefits from AES instructions and is being considered for standardization by NIST [27, 36]. For instantiating HCTR2 with 256-bit primitives, we use a simple 256-bit universal hash function that concatenates two POLYVAL instances with different hash keys [25]. This comparison aims at comparing different approaches for achieving more than 64-bit security [6], and the security levels are not strictly aligned: HCTR2 with Rijndael-256-256 achieves 128-bit security while the others have $\frac{2}{3} \cdot 128$-bit security.

Our evaluation considers tweaks shorter than 128 bits to align with *bbb-ddd-AES*. In particular, we use CENC [23] for deriving 256-bit keys in HCTR2-TwKD, and the key derivation function F_L is given by

$$F_L : T \mapsto (E_L(00||T) \oplus E_L(01||T))||(E_L(00||T) \oplus E_L(10||T)),$$

wherein L is a 256-bit key, E_L is encryption with AES-256, and T is a 126-bit tweak.

8.1 Implementation

The software implementations of the target schemes are optimized using dedicated instructions for the x86 and ARM instruction sets. The descriptions below summarizes the techniques for implementing each primitive on the x86 platform. The implementations on the ARM platform are based on the same techniques using the corresponding dedicated instructions.

AES-256. AES-256 can be realized simply with AES-NI [14], which implements AES rounds, and our AES-256 implementation comprises of fifteen AVX instructions: (i) thirteen `vaesenc`, (ii) one `vaesenclast`, and (iii) one `vpxor` for the first key addition.

POLYVAL. The carry-less multiplication instruction [16], `vpclmulqdq`, is used for implementing FMUL in POLYVAL and the 256-bit hash function. FMUL involves textbook multiplication for obtaining a 256-bit polynomial and polynomial reduction to bring it back to a 128-bit polynomial. Our FMUL implementation uses aggregated polynomial reduction [24] to run several textbook multiplications in parallel and reduce the cost of the reduction step.

Rijndael-256-256. Rijndael-256-256 also benefits from AES-NI because it uses the same SubBytes and MixColumns as AES. By following the technique by Gueron [15], we use the shuffling (`vpshufb`) and blending (`vpblendvb`) instructions to fill the gap between AES and Rijndael-256-256, i.e., ShiftRows over a 256-bit state.

Our software implementation unrolls the loops to improve performance by saturating pipelined execution units in the target processors. The implementations process four and eight consecutive message blocks at once in the x86 and ARM processors, respectively, which are the minimum unrolling factors with which the cpb performance reaches plateaus.

We implement two versions of *bbb-ddd-AES* with and without precomputation. Encryption and decryption of *bbb-ddd-AES* require a series of tweak-dependent masks given by $E_{K_2}(2||T||i)$, where K_2 is a key, T is a tweak, and i is a block index. When using the same tweak T repeatedly, we can improve performance by precomputing a table containing $E_{K_2}(2||T||i)$ for possible i [11]. This technique significantly impacts the performance by reducing the number of BC calls for each message block, as summarized in Table 1. Meanwhile, such an optimization is not always available; in the case of disk encryption, for example, the tweak is uncontrollable because that depends on a sector number.

8.2 Evaluation

We determine cpb performance by recording the values in the hardware performance monitors while the target implementation processes a 64 kB message. This procedure is performed 200 times, and the average is taken as the representative value. Table 2 shows the benchmark results for eight x86 processors from Intel and AMD, as well as four ARM processors from Apple and AWS. The evaluation environment includes both local machines and the Amazon AWS EC2 instances in the cloud [1].

By comparing CHCTR2 with AES-256 and HCTR2 with Rijndael-256-256, which achieve more than 64-bit security without any restrictions on the tweak, CHCTR2 outperforms the other in all the processor models. This result can be explained by the number of instructions as follows. For processing a 256-bit message chunk, CHCTR2 requires four AES-256 calls, involving 56 ($= 14 \times 4$) AES rounds, i.e., instructions. HCTR2, on the other hand, uses more instructions because a single Rijndael-256-256 needs 84 instructions for 28 ($= 14 \times 2$) AES rounds, 28 `vpblendvb` instructions, and 28 `vpshufb` instructions. Moreover, CHCTR2

Table 2. Cycles-per-byte (cpb) performances of the VIL-STPRP schemes instantiated for more than 64-bit security in twelve processor models. The entries with † are measured on Amazon AWS instances on the cloud. *bbb-ddd-AES* has two implementations with and without precomputation (PC).

Processor Model	Architecture	CHCTR2	HCTR2	HCTR2-TwKD	*bbb-ddd-AES*	
					w/o PC	w/ PC
Intel Core Ultra 7 265K	Arrow Lake-S	1.840	2.562	1.077	1.541	1.177
Intel Xeon w5-3423	Sapphire Rapids	1.894	2.536	1.102	1.513	1.089
Intel Xeon 8488C†	Sapphire Rapids	1.944	2.604	1.128	1.516	1.089
Intel Xeon 8375C†	Ice Lake	2.604	3.149	1.499	1.889	1.446
AMD Ryzen 5 9600X	Zen5	2.731	3.417	1.639	2.128	1.640
AMD Ryzen 5 7600X	Zen4	2.771	3.404	1.637	2.123	1.661
AMD EPYC 9R14†	Zen4	2.784	3.441	1.670	2.139	1.683
AMD EPYC 7R13†	Zen3	2.435	3.635	1.507	2.259	1.708
Apple M2	Firestorm-M2	1.480	1.811	0.792	1.095	0.861
Apple M1	Firestorm-M1	1.411	1.762	0.840	1.172	0.866
AWS Graviton4†	Neoverse-V2	1.489	1.986	0.873	1.184	0.953
AWS Graviton3†	Neoverse-V1	1.409	1.932	0.826	1.157	0.867

invokes a smaller number of FMUL calls: CHCTR2 and HCTR2 use 6 and 8 FMUL calls for processing a 256-bit chunk, respectively.

Next, we compare HCTR2-TwKD and *bbb-ddd-AES*, which achieve BBB security when the same tweak is not repeated too often. First, Table 2 shows that HCTR2-TwKD outperforms *bbb-ddd-AES* without precomputation in all the processor models. Second, HCTR2-TwKD still achieves a similar performance compared with *bbb-ddd-AES* using precomputation. These results are aligned with the number of primitive calls shown in Table 1. In summary, HCTR2-TwKD is better than *bbb-ddd-AES* when the tweak cannot be repeated. Even when the tweak is repeated, HCTR2-TwKD achieves a similar performance without paying the memory cost for storing the precomputed values while maintaining backward compatibility with HCTR2.

9 Proof of Theorem 1: Upper-Bounding $\mathbf{Pr}[\mathsf{fail}_{\mathsf{perm}} \wedge \neg \mathsf{bad}]$

We recall the structures of plaintext and ciphertext blocks of CHCTR2, where query indexes are omitted.

$$
\begin{aligned}
X_{1,0} &= M_0 \oplus \mathsf{H}_1^{[\nu]}(T, M_*),\\
X_{2,0} &= Y_{1,0} \oplus \mathsf{H}_1^{[\nu]}(T, R) \oplus \mathsf{H}_2^{[\nu]}(T, R),\\
X_{1,j} &= X_{1,0} \oplus Y_{1,0} \oplus A_j \text{ where } Y_{1,0} = X_{2,0} \oplus \mathsf{H}_1^{[\nu]}(T, R) \oplus \mathsf{H}_2^{[\nu]}(T, R)),\\
X_{2,j} &= X_{2,0} \oplus Y_{2,0} \oplus A_j \text{ where } Y_{2,0} = C_0 \oplus \mathsf{H}_2^{[\nu]}(T, C_*).
\end{aligned}
$$

9.1 Evaluation of $\mathbf{Pr}[\mathsf{fail}^{*1}_{\mathsf{sample}}]$

Fix $\nu \in [u]$ and $\alpha \in \mathcal{Q}^{[\nu]}$. We evaluate the probability that $\mathsf{fail}^{*1}_{\mathsf{sample}}$ occurs in the sampling procedure of the α-th query.

Fist Condition of Initializing $\mathsf{sample}_\alpha = 1$. First, sample_α is initialized to 1 by the following condition.

– $\mathsf{C}_{1,0} \Leftrightarrow \exists \beta_1 \in \mathcal{Q}^{[\nu]}$ s.t. $\alpha \neq \beta_1 \wedge r_{\beta_1} \geq r_\alpha \wedge \left(X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)} \vee Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}\right)$.

Let $\mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$ be a collision set with the query index α. Let $s \in [2]$ be the round index for the collision in $\mathsf{C}_{1,0}$: $s = 1$ if $X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}$; $s = 2$ if $Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}$.

Second Conditions of Incrementing sample_α. We show the second conditions for each query that increment sample_α. For the non-longest query index α, the following conditions increment sample_α by 1 in Algorithm 4.

– $\mathsf{C}_{1,1} \Leftrightarrow X_{3-s,0}^{(\alpha)} \in \mathcal{X}_{3-s}^{[\nu]}$.
– $\mathsf{C}_{1,2} \Leftrightarrow Y_{3-s,0}^{(\alpha)} \in \mathcal{Y}_{3-s}^{[\nu]}$.
– $\mathsf{C}_{1,3} \Leftrightarrow \exists j \in [r_\alpha]$ s.t. $X_{3-s,j}^{(\alpha)} \in \mathcal{X}_{3-s}^{[\nu]} \cup \{X_{3-s,0}^{(\alpha)}\}$.
– $\mathsf{C}_{1,4} \Leftrightarrow \exists j \in [r_\alpha]$ s.t. $Y_{3-s,j}^{(\alpha)} \in \mathcal{Y}_{3-s}^{[\nu]} \cup \{Y_{3-s,j'}^{(\alpha)} \mid j' \in [0, j-1]\}$.

Note that for $\mathsf{C}_{1,1}$, each element in $\mathcal{X}_{3-s}^{[\nu]}$ is defined before $X_{3-s,0}^{(\alpha)}$ is defined, and the same condition holds for the sets in the other conditions, i.e., for $\mathsf{C}_{1,2}$, each element in $\mathcal{Y}_{3-s}^{[\nu]}$ is defined before $Y_{3-s,0}^{(\alpha)}$ is defined; for $\mathsf{C}_{1,3}$ each element in $\mathcal{X}_{3-s}^{[\nu]}$ is defined before $X_{3-s,j}^{(\alpha)}$ is defined; for $\mathsf{C}_{1,4}$, each element in $\mathcal{Y}_{3-s}^{[\nu]}$ is defined before $Y_{3-s,j}^{(\alpha)}$ is defined.

Evaluations. We evaluate $\Pr[\mathsf{fail}_{\mathsf{sample}}^{*1}]$ by using the first and second conditions.

$\underline{\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,1}}$. Fix $s \in [2]$, $\beta_1 \in \mathcal{Q}^{[\nu]}$, and $X_{3-s,j}^{(\beta_2)} \in \mathcal{X}_{3-s}^{[\nu]}$. We evaluate the probabilities that the following collisions occur under the assumption that the other conditions $\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,l}$ for $l \neq 1$ do not occur.

$$\begin{cases} X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)} & \text{if } s = 1 \\ Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)} & \text{if } s = 2 \end{cases} \text{(from } \mathsf{C}_{1,0}\text{)}, \qquad X_{3-s,0}^{(\alpha)} = X_{3-s,j}^{(\beta_2)} \text{ (from } \mathsf{C}_{1,1}\text{)}.$$

We consider the following cases.

– $s = 1 \wedge (\beta_2, *) \notin \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$.
  • $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

  • $\mathsf{C}_{1,1}$. Since $X_{2,0}^{(\alpha)} = Y_{1,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$ and $Y_{1,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

– $s = 1 \wedge (\beta_2, *) \in \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$.
  In this case, $\beta_1 = \beta_2$. Let $v \coloneqq \max\{\alpha, \beta_2\}$. We consider the following cases.
  • The v-th query is an encryption one.
    ∗ $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

* $\mathsf{C}_{1,1}$. Since $X_{2,0}^{(\alpha)}$ is defined by using $C_0^{(v)}$ that is chosen uniformly at random from at least $2^n - q$ elements in $\{0,1\}^n$, we have

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j}^{(\beta_2)}] \leq \frac{2}{2^n},$$

assuming that $q \leq 2^{n-1}$.

- The v-th query is a decryption one.
  * $\mathsf{C}_{1,0}$. Since $X_{1,0}^{(v)}$ is defined by using $M_0^{(v)}$ that is chosen uniformly at random from at least $2^n - q$ elements in $\{0,1\}^n$, we have

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}] \leq \frac{2}{2^n},$$

assuming that $q \leq 2^{n-1}$.
  * $\mathsf{C}_{1,1}$. Since $X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_2)} \Rightarrow Y_{1,0}^{(\alpha)} = Y_{1,0}^{(\beta_2)} \Rightarrow (T^{(\alpha)}, R^{(\alpha)}) \neq (T^{(\beta_2)}, R^{(\beta_2)})$, the AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j}^{(\beta_2)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_2}\}}{2^n}.$$

– $s = 2$.
- $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

- $\mathsf{C}_{1,1}$. By $\neg$2coll, $j \geq 1$. Since $X_{1,j}^{(\beta_2)}$ is defined by using $Y_{1,0}^{(\beta_2)}$ where $Y_{1,0}^{(\beta_2)} = X_{2,0}^{(\beta_2)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\beta_2)}, R^{(\beta_2)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\beta_2)}, R^{(\beta_2)})$ and $Y_{1,0}^{(\beta_2)}$ or $X_{2,0}^{(\beta_2)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following hound:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

$\underline{\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,2}}$ Fix $s \in [2]$, $\nu \in [u]$, $\beta_1 \in \mathcal{Q}^{[\nu]}$, and $Y_{3-s,j}^{(\beta_2)} \in \mathcal{Y}_{3-s}^{[\nu]}$. We evaluate the probabilities that the following collisions occur under the assumption that the other conditions $\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,l}$ for $l \neq 2$ do not occur.

$$\begin{cases} X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)} & \text{if } s = 1 \\ Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)} & \text{if } s = 2 \end{cases} \text{(from } \mathsf{C}_{1,0}\text{)}, \qquad Y_{3-s,0}^{(\alpha)} = Y_{3-s,j}^{(\beta_2)} \text{ (from } \mathsf{C}_{1,4}\text{)}.$$

We consider the following cases.

– $s = 1$.
- $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

- $\mathsf{C}_{1,2}$. In this case, $j \geq 1$. Since $Y_{2,j}^{(\beta_2)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

– $s = 2 \wedge (\beta_2, *) \notin \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$.
  - $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

  - $\mathsf{C}_{1,2}$. Since $Y_{1,0}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$ and $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

– $s = 2 \wedge (\beta_2, *) \in \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$.
  In this case, $\beta_1 = \beta_2$.
  - $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

  - $\mathsf{C}_{1,2}$. We consider the following cases.
    * If $j = 0$, then since $Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)} \wedge \beta_1 = \beta_2 \Rightarrow (T^{(\alpha)}, R^{(\alpha)}) \neq (T^{(\beta_2)}, R^{(\beta_2)})$, the AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,0}^{(\beta_2)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_2}\}}{2^n}.$$

    * If $j \geq 1$, then since $Y_{1,j}^{(\beta_2)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

$\underline{\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,3}}$ Fix $\beta_1 \in \mathcal{Q}^{[\nu]}$, $s \in [2]$, $j \in [r_\alpha]$, and $X_{3-s,j^\dagger}^{(\beta_2)} \in \mathcal{X}_{3-s}^{[\nu]} \cup \{X_{3-s,0}^{(\alpha)}\}$. We evaluate the probabilities that the following collisions occur under the assumption that the other conditions $\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,l}$ for $l \neq 3$ do not occur.

$$\begin{cases} X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)} & \text{if } s = 1 \\ Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)} & \text{if } s = 2 \end{cases} \text{(from } \mathsf{C}_{1,0}\text{)}, \qquad X_{3-s,j}^{(\alpha)} = X_{3-s,j^\dagger}^{(\beta_2)} \text{ (from } \mathsf{C}_{1,3}\text{)}.$$

We consider the following cases.

– $s = 1 \wedge (\beta_2, *) \notin \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$.

  - $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

  - $\mathsf{C}_{1,3}$. Since $X_{2,j}^{(\alpha)}$ is defined by using $Y_{1,0}^{(\alpha)}$ that is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j^\dagger}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

– $s = 1 \wedge \beta_2 = \alpha$.
In this case, $j^\dagger = 0$.
  • $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

  • $\mathsf{C}_{1,3}$. Since $X_{2,j}^{(\alpha)} = X_{2,j^\dagger}^{(\beta_2)} \Rightarrow Y_{2,0}^{(\alpha)} = A_j \Rightarrow \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, C_*^{(\alpha)}) = C_0^{(\alpha)} \oplus A_j$, the regular hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j^\dagger}^{(\beta_2)}] \leq \frac{\omega \ell_\alpha}{2^n}.$$

– $s = 1 \wedge \beta_2 \neq \alpha \wedge (\beta_2, *) \in \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$.
In this case, $\beta_2 = \beta_1$. Let $v \in \max\{\alpha, \beta_1\}$. We consider the following cases.
  • The v-th query is an encryption one.
    ∗ $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

    ∗ $\mathsf{C}_{1,1}$. Since $X_{2,0}^{(\alpha)}$ is defined by using $C_0^{(a)}$ that is chosen uniformly at random from at least $2^n - q$ elements in $\{0,1\}^n$, we have

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j}^{(\beta_2)}] \leq \frac{2}{2^n},$$

    assuming that $q \leq 2^{n-1}$.
  • The v-th query is a decryption one.
    ∗ $\mathsf{C}_{1,0}$. Since $X_{1,0}^{(a)}$ is defined by using $M_0^{(v)}$ that is chosen uniformly at random from at least $2^n - q$ elements in $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}] \leq \frac{2}{2^n},$$

    assuming that $q \leq 2^{n-1}$.
    ∗ $\mathsf{C}_{1,1}$. Since $X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_2)} \Rightarrow Y_{1,0}^{(\alpha)} = Y_{1,0}^{(\beta_2)} \Rightarrow (T^{(\alpha)}, R^{(\alpha)}) \neq (T^{(\beta_2)}, R^{(\beta_2)})$, the AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j^\dagger}^{(\beta_2)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_2}\}}{2^n}.$$

– $s = 2 \wedge (\beta_2, *) \notin \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma]$.

  • $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

  • $\mathsf{C}_{1,3}$. Since $X_{1,j}^{(\alpha)}$ is defined by using $X_{2,0}^{(\alpha)}$ that is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j^\dagger}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

– $s = 2 \wedge (\beta_2, *) \in \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma] \wedge \alpha = \beta_2$.
In this case, $j^\dagger = 0$.
  - $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

  - $\mathsf{C}_{1,3}$. Since $X_{1,j}^{(\alpha)} = X_{1,j^\dagger}^{(\beta_2)} \Rightarrow Y_{1,0}^{(\alpha)} = A_j, Y_{1,0}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$, and $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j^\dagger}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

– $s = 2 \wedge (\beta_2, *) \in \mathcal{G}_{\mathsf{coll}}^{[\nu]}[\gamma] \wedge \alpha \neq \beta_2$.
In this case, $\beta_2 = \beta_1$. Let $v \in \max\{\alpha, \beta_1\}$. We consider the following cases.
  - The v-th query is an encryption one.
    * $\mathsf{C}_{1,0}$. Since $Y_{2,0}^{(v)}$ is defined by using $C_0^{(v)}$ that is chosen uniformly at random from at least $2^n - q$ elements in $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}] \leq \frac{2}{2^n},$$

      assuming that $q \leq 2^{n-1}$.
    * $\mathsf{C}_{1,3}$. The AXU hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j^\dagger}^{(\beta_2)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_2}\}}{2^n}.$$

  - The v-th query is a decryption one.
    * $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

    * $\mathsf{C}_{1,3}$. Since $X_{1,0}^{(v)}$ is defined by using $M_0^{(v)}$ that is chosen uniformly at random from at least $2^n - q$ elements in $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j^\dagger}^{(\beta_2)}] \leq \frac{2}{2^n},$$

      assuming that $q \leq 2^{n-1}$.

$\underline{\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,4}}$. Fix $s \in [2]$, $\nu \in [u]$, $\alpha, \beta_1 \in \mathcal{Q}^{[\nu]}$, $j \in [r_\alpha]$, and $Y_{3-s,j^\dagger}^{(\beta_2)} \in \mathcal{Y}_{3-s}^{[\nu]} \cup \{Y_{3-s,j'}^{(\alpha)} \mid j' \in [0, j-1]\}$ such that $\alpha \neq \beta_1$. We evaluate the probabilities that the following collisions occur under the assumption that the other conditions $\mathsf{C}_{1,0} \wedge \mathsf{C}_{1,l}$ for $l \neq 4$ do not occur.

$$\begin{cases} X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)} & \text{if } s = 1 \\ Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)} & \text{if } s = 2 \end{cases} \text{(from } \mathsf{C}_{1,0}\text{)}, \qquad Y_{3-s,j}^{(\alpha)} = Y_{3-s,j^\dagger}^{(\beta_2)} \text{ (from } \mathsf{C}_{1,4}\text{)},$$

We consider the following cases.

– $s = 1$.

- $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_1^{[\nu]}$ ensures that

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

- $\mathsf{C}_{1,4}$. Since $Y_{2,j}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{2,j}^{(\alpha)} = Y_{2,j^\dagger}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

– $s = 2$.

- $\mathsf{C}_{1,0}$. The AXU hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,0}^{(\beta_1)}] \leq \frac{\omega \max\{\ell_\alpha, \ell_{\beta_1}\}}{2^n}.$$

- $\mathsf{C}_{1,4}$. Since $Y_{1,j}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j^\dagger}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Bound of $\Pr[\mathsf{fail}^{*1}_{\mathsf{sample}}]$. The above evaluations show that for each of α_1, β_1, and β_2, the probability that a collision occurs between the edge values of the α-th query and the β_1-th query and a collision occurs between the edge values of the α-th query and the β_2-th query is at most

$$4 \cdot \frac{2\omega^2 \ell_\alpha \max\{\ell_\alpha, \ell_{\beta_2}\} \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}}.$$

We thus have the following bound:

$$\begin{aligned}
\Pr[\mathsf{fail}^{*1}_{\mathsf{sample}}] &\leq \sum_{\nu \in [u]} \sum_{\alpha, \beta_1, \beta_2 \in \mathcal{Q}^{[\nu]}} \frac{8\omega^2 \ell_\alpha \max\{\ell_\alpha, \ell_{\beta_2}\} \max\{\ell_\alpha, \ell_{\beta_1}, \ell_{\beta_2}\}}{2^{2n}} \\
&\leq \sum_{\nu \in [u]} \frac{8\omega^2 \sigma_\nu^3}{2^{2n}} \\
&\leq \frac{8\omega^2 \sigma_{\mathsf{u}}^2 \sigma}{2^{2n}}.
\end{aligned}$$

9.2 Evaluation of $\Pr[\mathsf{fail}^{*2}_{\mathsf{sample}}]$

Fix $\nu \in [u]$ and $\alpha \in \mathcal{Q}^{[\nu]}$. We evaluate the probability that $\mathsf{fail}^{*2}_{\mathsf{sample}}$ occurs in the sampling procedure of the α-th query.

Events for $\mathsf{fail}^{*2}_{\mathsf{sample}}$. By Algorithm 3, if two two of the following events occur in the sampling procedure of the α-th query, then $\mathsf{fail}^{2}_{\mathsf{sample}}$ occurs.

– $\mathsf{C}_{2,1} \Leftrightarrow X_{1,0}^{(\alpha)} \in \mathcal{X}_1^{[\nu]}$
– $\mathsf{C}_{2,2} \Leftrightarrow Y_{2,0}^{(\alpha)} \in \mathcal{Y}_2^{[\nu]}$.

- $\mathsf{C}_{2,3} \Leftrightarrow X_{1,0}^{(\alpha)} \notin \mathcal{X}_1^{[\nu]} \wedge Y_{1,0}^{(\alpha)} \in \mathcal{Y}_1^{[\nu]}$
- $\mathsf{C}_{2,4} \Leftrightarrow Y_{2,0}^{(\alpha)} \notin \mathcal{Y}_2^{[\nu]} \wedge X_{2,0}^{(\alpha)} \in \mathcal{X}_2^{[\nu]}$.
- $\mathsf{C}_{2,5} \Leftrightarrow \exists j \in [r_\alpha]$ s.t. $X_{1,j}^{(\alpha)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$
- $\mathsf{C}_{2,6} \Leftrightarrow \exists j \in [r_\alpha]$ s.t. $X_{2,j}^{(\alpha)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$.
- $\mathsf{C}_{2,7} \Leftrightarrow \exists i \in [2], j \in [r_\alpha]$ s.t. $(X_{i,j}^{(\alpha)} \notin \mathcal{X}_i^{[\nu]} \cup \{X_{i,0}^{(\alpha)}\})$
  $\wedge (Y_{i,j}^{(\alpha)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\})$.

Note that for $\mathsf{C}_{2,1}$, each element in $\mathcal{X}_1^{[\nu]}$is defined before $X_{1,0}^{(\alpha)}$ is defined, and the same condition holds for the sets in the other conditions.

Evaluations. For $(l_1, l_2) \in [7] \times [7]$, let $(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2})$ be an event that C_{2,l_1} occurs and sample_α is incremented by 1 at the first sampling of the α-th query, and then C_{2,l_2} occurs and sample_α is incremented by 1 at the first or second sampling. We evaluate $\Pr[\mathsf{fail}^{*2}_{\mathsf{sample}}]$ by using the events $(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2})$.

$\underline{(\mathsf{C}_{2,1}, \mathsf{C}_{2,1})}$ Since $X_{1,0}^{(\alpha)}$ is not resampled, this case does not occur.

$\underline{(\mathsf{C}_{2,1}, \mathsf{C}_{2,2})}$ Fix $X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]}$ and $Y_{2,j_2}^{(\beta_2)} \in \mathcal{Y}_2^{[\nu]}$ such that $j_1, j_2 \geq 1$ (due to $\alpha \in \mathcal{Q}_{\mathsf{non\text{-}coll}}$). We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (1, 2)\}$ do not hold.

$$X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \text{ (from } \mathsf{C}_{2,1}) \qquad\qquad Y_{2,0}^{(\alpha)} = Y_{2,j_2}^{(\beta_2)} \text{ (from } \mathsf{C}_{2,2}),$$

- $\mathsf{C}_{2,1}$. Since $X_{1,j_1}^{(\beta_1)} = X_{1,0}^{(\beta_1)} \oplus Y_{1,0}^{(\beta_1)} \oplus A_{j_1}$, $Y_{1,0}^{(\beta_1)} = X_{2,0}^{(\beta_1)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\beta_1)}, R^{(\beta_1)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\beta_1)}, R^{(\beta_1)})$, and $Y_{1,0}^{(\beta_1)}$ or $X_{2,0}^{(\beta_1)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:
$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$
- $\mathsf{C}_{2,2}$. Since $Y_{2,j_2}^{(\beta_2)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:
$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

  Hence, for each pair $(X_{1,j_1}^{(\beta_1)}, Y_{2,j_2}^{(\beta_2)})$, we have the following bound:
$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge Y_{2,0}^{(\alpha)} = Y_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,1}, \mathsf{C}_{2,3})}$ Since $\mathsf{C}_{2,1}$ and $\mathsf{C}_{2,3}$ are mutually exclusive, this case does not occur.

$\underline{(\mathsf{C}_{2,1}, \mathsf{C}_{2,4})}$ Fix $X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]}$ such that $j_1 \geq 1$ (due to $\alpha \in \mathcal{Q}_{\mathsf{non\text{-}coll}}$). We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (1, 4)\}$ do not hold.

$$X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \text{ (from } \mathsf{C}_{2,1}) \qquad\qquad X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ (from } \mathsf{C}_{2,4}).$$

– $\mathsf{C}_{2,1}$. The evaluation is the same as that of $(\mathsf{C}_{2,1}, \mathsf{C}_{2,2})$, and we have the following bound:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,4}$. Since $X_{2,0}^{(\alpha)} = Y_{1,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$ and $Y_{1,0}^{(\alpha)}$ or $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each pair $(X_{1,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,1}, \mathsf{C}_{2,5})}$ Fix $j \in [r_\alpha], X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]}$, and $X_{1,j_2}^{(\beta_2)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$ such that $j_1 \geq 1$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (1,5)\}$ do not hold.

$$X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \text{ (from } \mathsf{C}_{2,1}), \qquad X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \text{ (from } \mathsf{C}_{2,5}).$$

– $\mathsf{C}_{2,1}$. The evaluation is the same as that of $(\mathsf{C}_{2,1}, \mathsf{C}_{2,2})$, and we have

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,5}$. We consider the following cases.

- If $\beta_2 = \alpha$, then $j_2 = 0$ and $X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \Leftrightarrow Y_{1,0}^{(\alpha)} = A_j$. Since $Y_{1,0}^{(\alpha)}$ $(= Y_{1,j_1}^{(\beta_1)})$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

- If $\beta_2 \neq \alpha$, then since $X_{1,j}^{(\alpha)} = X_{1,0}^{(\alpha)} \oplus Y_{1,0}^{(\alpha)} \oplus A_j$ and $Y_{1,0}^{(\alpha)}$ $(= Y_{1,j_1}^{(\beta_1)})$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(j, X_{1,j_1}^{(\beta_1)}, X_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,1}, \mathsf{C}_{2,6})}$ Fix $j \in [r_\alpha]$, $X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$ such that $j_1 \geq 1$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (1,6)\}$ do not hold.

$$X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \text{ (from } \mathsf{C}_{2,1}), \qquad X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ (from } \mathsf{C}_{2,6}).$$

– $\mathsf{C}_{2,1}$. The evaluation is the same as that of $(\mathsf{C}_{2,1}, \mathsf{C}_{2,2})$, and we have

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,6}$. We consider the following cases.
  • If $(\beta_2, j_2) = (\alpha, 0)$, then

$$X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \Rightarrow Y_{2,0}^{(\alpha)} = A_j \Rightarrow \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, C_*^{(\alpha)}) = C_0^{(\alpha)} \oplus A_j,$$

  and the regular hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \le \frac{\omega\ell_\alpha}{2^n}.$$

  • If $(\beta_2, j_2) \neq (\alpha, 0)$, then since

$$X_{2,j}^{(\alpha)} = \begin{cases} Y_{1,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) & \text{if } j = 0, \\ X_{2,0}^{(\alpha)} \oplus Y_{2,0}^{(\alpha)} \oplus A_j & \text{if } j \ge 1, \end{cases}$$

  and $Y_{1,0}^{(\alpha)}$ $(= Y_{1,j_1}^{(\beta_1)})$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,i_2}^{(\beta_2)}] \le \frac{1}{2^n}.$$

  Hence, for each tuple $(j, X_{1,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$,

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \le \begin{cases} \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

$\underline{(\mathsf{C}_{2,1}, \mathsf{C}_{2,7})}$ Fix $i \in [2]$, $j \in [r_\alpha]$, $X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]}$ and $Y_{i,j_2}^{(\beta_2)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\})$ such that $j_1 \ge 1$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (1, 7)\}$ do not hold.

$$X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \text{ (from } \mathsf{C}_{2,1}), \qquad Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)} \text{ (from } \mathsf{C}_{2,7}).$$

– $\mathsf{C}_{2,1}$. The evaluation is the same as that of $(\mathsf{C}_{2,1}, \mathsf{C}_{2,2})$, and we have the following bound:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \le \frac{1}{2^n}.$$

– $\mathsf{C}_{2,7}$. Since $Y_{i,j}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \le \frac{1}{2^n}.$$

  Hence, for each tuple $(i, j, X_{1,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{1,0}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \le \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,2}, \mathsf{C}_{2,1})}$ Since $\mathsf{C}_{2,2}$ does not occur before $\mathsf{C}_{2,1}$, this case does not occur.

$\underline{(\mathsf{C}_{2,2}, \mathsf{C}_{2,2})}$ Since $Y_{2,0}^{(\alpha)}$ is not resampled, this case does not occur.

$\underline{(\mathsf{C}_{2,2}, \mathsf{C}_{2,3})}$ Fix $Y_{2,j_1}^{(\beta_1)} \in \mathcal{Y}_2^{[\nu]}$ and $Y_{1,j_2}^{(\beta_2)} \in \mathcal{Y}_1^{[\nu]}$ such that $j_1 \geq 1$ (due to $\alpha \in \mathcal{Q}_{\mathsf{non\text{-}coll}}$). We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (2,3)\}$ do not hold.

$$Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)} \text{ (from } \mathsf{C}_{2,2}), \qquad Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)} \text{ (from } \mathsf{C}_{2,3}).$$

– $\mathsf{C}_{2,2}$. Since $Y_{2,j_1}^{(\beta_1)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,3}$. Since $Y_{1,0}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$, and $Y_{1,0}^{(\alpha)}$ or $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each pair $(X_{1,j_1}^{(\beta_1)}, Y_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)} \wedge Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,2}, \mathsf{C}_{2,4})}$ Since the conditions of $\mathsf{C}_{2,2}$ and $\mathsf{C}_{2,4}$ are mutually exclusive, this case does not occur.

$\underline{(\mathsf{C}_{2,2}, \mathsf{C}_{2,5})}$ Fix $j \in [r_\alpha]$, $Y_{2,j_1}^{(\beta_1)} \in \mathcal{Y}_2^{[\nu]}$, and $X_{1,j_2}^{(\beta_2)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$ such that $j_1 \geq 1$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (2,5)\}$ do not hold.

$$Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)} \text{ (from } \mathsf{C}_{2,2}), \qquad X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \text{ (from } \mathsf{C}_{2,5}).$$

– $\mathsf{C}_{2,2}$. Since $Y_{2,j_1}^{(\beta_1)}$ is chosen uniformly at random from $\{0,1\}^n$, we have

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,5}$. We consider the following cases.

- If $\beta_2 = \alpha$, then $j_2 = 0$, $X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \Rightarrow Y_{1,0}^{(\alpha)} = A_j$, and $Y_{1,0}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$. Since $Y_{1,0}^{(\alpha)}$ or $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

- If $\beta_2 \neq \alpha$, then since $X_{1,j}^{(\alpha)} = X_{1,0}^{(\alpha)} \oplus Y_{1,0}^{(\alpha)} \oplus A_j$, $Y_{1,0}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$, and $Y_{1,0}^{(\alpha)}$ or $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each j, $Y_{2,j_1}^{(\beta_1)}$, and $X_{1,j_2}^{(\beta_2)}$, we have the following bound:

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)} \wedge X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,2}, \mathsf{C}_{2,6})}$ Fix $j \in [r_\alpha]$, $Y_{2,j_1}^{(\beta_1)} \in \mathcal{Y}_2^{[\nu]}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$ such that $j_1 \geq 1$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (2, 6)\}$ do not hold.

$$Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)} \text{ (from } \mathsf{C}_{2,2}), \qquad X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ (from } \mathsf{C}_{2,6}).$$

– $\mathsf{C}_{2,2}$. Since $Y_{2,j_1}^{(\beta_1)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,6}$. We consider the following cases.

- If $\beta_2 = \alpha$, then $j_2 = 0$ and $X_{2,j}^{(\alpha)} = X_{2,0}^{(\beta_2)} \Rightarrow \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, C_*^{(\alpha)}) = C_0^{(\alpha)} \oplus A_j$. The regular hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{\omega\ell_\alpha}{2^n}.$$

- If $\beta_2 \neq \alpha$ and $\beta_1 = \beta_2$, then since $Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)} \Rightarrow X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}$, we have

$$\begin{aligned} X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} &\Leftrightarrow X_{2,0}^{(\alpha)} \oplus Y_{2,0}^{(\alpha)} \oplus A_j = X_{2,j_1}^{(\beta_1)} \oplus A_{j_1} \oplus A_{j_2} \\ &\Leftrightarrow Y_{2,0}^{(\alpha)} = A_j \oplus A_{j_1} \oplus A_{j_2} \\ &\Leftrightarrow \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, C_*^{(\alpha)}) = A_j \oplus A_{j_1} \oplus A_{j_2} \oplus C_0^{(\alpha)}. \end{aligned}$$

  The regular hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$

- If $\beta_2 \neq \alpha$ and $\beta_1 \neq \beta_2$, then since $X_{2,j_2}^{(\beta_2)} = X_{2,0}^{(\beta_2)} \oplus Y_{2,0}^{(\beta_2)} \oplus A_{j_2}$, $X_{2,0}^{(\beta_2)} = Y_{1,0}^{(\beta_2)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\beta_2)}, R^{(\beta_2)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\beta_2)}, R^{(\beta_2)})$, and $Y_{1,0}^{(\beta_2)}$ or $X_{2,0}^{(\beta_2)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(j, Y_{2,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)} \wedge X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_2, j_2) \neq (\alpha, 0) \text{ and } \beta_1 = \beta_2, \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_2, j_2) \neq (\alpha, 0) \text{ and } \beta_1 \neq \beta_2. \end{cases}$$

$\underline{(\mathsf{C}_{2,2}, \mathsf{C}_{2,7})}$ Fix $i \in [2]$, $j \in [r_\alpha]$, $Y_{2,j_1}^{(\beta_1)} \in \mathcal{Y}_2^{[\nu]}$ and $Y_{i,j_2}^{(\beta_2)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\})$ such that $j_1 \geq 1$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (2, 7)\}$ do not hold.

$$Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)} \text{ (from } \mathsf{C}_{2,2}), \qquad Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)} \text{ (from } \mathsf{C}_{2,7}).$$

– $\mathsf{C}_{2,2}$. Since $Y_{2,j_1}^{(\beta_1)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,7}$. Since $Y_{i,j}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each $X_{2,j_1}^{(\beta_1)}$ and $X_{2,j_2}^{(\beta_2)}$, we have the following bound:

$$\Pr[Y_{2,0}^{(\alpha)} = Y_{2,j_1}^{(\beta_1)} \wedge X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,3}, \mathsf{C}_{2,1})}$ The conditions of $\mathsf{C}_{2,3}$ and $\mathsf{C}_{2,1}$ are mutually exclusive, this case does not occur.

$\underline{(\mathsf{C}_{2,3}, \mathsf{C}_{2,2})}$ Since $\mathsf{C}_{2,3}$ does not occur before $\mathsf{C}_{2,2}$, this case does not occur.

$\underline{(\mathsf{C}_{2,3}, \mathsf{C}_{2,3})}$ Fix $Y_{1,j_1}^{(\beta_1)}, Y_{1,j_2}^{(\beta_2)} \in \mathcal{Y}_1^{[\nu]}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (3,3)\}$ do not hold.

$$Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,3}),$$
$$Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,3}).$$

Since $Y_{1,0}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$ and $Y_{1,0}^{(\alpha)}$ or $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$ in the first and second samplings, we have the following bounds:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}, \qquad \Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each pair $(Y_{1,j_1}^{(\beta_1)}, Y_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)} \wedge Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,3}, \mathsf{C}_{2,4})}$ Fix $Y_{1,j_1}^{(\beta_1)} \in \mathcal{Y}_1^{[\nu]}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (3,4)\}$ do not hold.

$$Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,3}),$$
$$X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,4}).$$

– $\mathsf{C}_{2,3}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,4}$. Since $X_{2,0}^{(\alpha)} = Y_{1,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$ and $Y_{1,0}^{(\alpha)}$ or $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$ in the second sampling of the α-th query, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each pair $(Y_{1,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)} \wedge X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,3}, \mathsf{C}_{2,5})}$ Fix $j \in [r_\alpha]$, $Y_{1,j_1}^{(\beta_1)} \in \mathcal{Y}_1^{[\nu]}$ and $X_{1,j_2}^{(\beta_2)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (3, 5)\}$ do not hold.

$$Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,3}),$$
$$X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,5}).$$

– The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,5}$. We consider the following cases.
  • If $\beta_2 = \alpha$, then $j_2 = 0$, $X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \Rightarrow Y_{1,0}^{(\alpha)} = A_j$, and $Y_{1,0}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$. Since $Y_{1,0}^{(\alpha)}$ or $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

  • If $\beta_2 \neq \alpha$, then since $X_{1,j}^{(\alpha)} = X_{1,0}^{(\alpha)} \oplus Y_{1,0}^{(\alpha)} \oplus A_j$, $Y_{1,0}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)})$, and $Y_{1,0}^{(\alpha)}$ or $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(j, Y_{1,j_1}^{(\beta_1)}, X_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)} \wedge X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})}$ Fix $j \in [r_\alpha]$, $Y_{1,j_1}^{(\beta_1)} \in \mathcal{Y}_1^{[\nu]}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (3, 6)\}$ do not hold.

$$Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,3}),$$
$$X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,6}).$$

– $\mathsf{C}_{2,3}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,6}$. We consider the following cases.
  • If $\beta_2 = \alpha$, then $j_2 = 0$ and $X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \Rightarrow \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, C_*^{(\alpha)}) = C_0^{(\alpha)} \oplus A_j$. The regular hash function $\mathsf{H}_2^{[\nu]}$ ensures that

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{\omega \ell_\alpha}{2^n}.$$

- If $\beta_2 \neq \alpha$, then since $X_{2,j}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus Y_{2,0}^{(\alpha)} \oplus A_j$, $X_{2,0}^{(\alpha)} = X_{2,0}^{(\alpha)} \oplus Y_{2,0}^{(\alpha)} \oplus A_j$, and $X_{2,0}^{(\alpha)}$ or $Y_{1,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$ in the second sampling, we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(j, Y_{1,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)} \wedge X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

$\underline{(\mathsf{C}_{2,3}, \mathsf{C}_{2,7})}$ Fix $i \in [2]$, $j \in [r_\alpha]$, $Y_{1,j_1}^{(\beta_1)} \in \mathcal{Y}_1^{[\nu]}$ and $Y_{i,j_2}^{(\beta_2)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\})$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (3,7)\}$ do not hold.

$$Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,3}),$$
$$Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,7}).$$

– $\mathsf{C}_{2,3}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,4})$, and we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,7}$. Since $Y_{i,j}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(i, j, Y_{1,j_1}^{(\beta_1)}, Y_{i,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_1}^{(\beta_2)} \wedge Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,4}, \mathsf{C}_{2,1})}$ Since $\mathsf{C}_{2,4}$ does not occur before $\mathsf{C}_{2,1}$, this case does not occur.

$\underline{(\mathsf{C}_{2,4}, \mathsf{C}_{2,2})}$ Since $\mathsf{C}_{2,4}$ does not occur before $\mathsf{C}_{2,2}$, this case does not occur.

$\underline{(\mathsf{C}_{2,4}, \mathsf{C}_{2,3})}$ Fix $X_{2,j_1}^{(\beta_1)} \in \mathcal{X}_2^{[\nu]}$ and $Y_{1,j_2}^{(\beta_2)} \in \mathcal{Y}_1^{[\nu]}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (4,3)\}$ do not hold.

$$X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,4}),$$
$$Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,3}).$$

– $\mathsf{C}_{2,4}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,4})$, and we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,3}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each pair $(X_{2,j_1}^{(\beta_1)}, Y_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \wedge Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,4}, \mathsf{C}_{2,4})}$ Fix $X_{2,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (4,4)\}$ do not hold.

$$X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,4}),$$
$$X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,4}).$$

The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,4})$, and we have the following bounds:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}, \qquad \Pr[X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each pair $(X_{2,j_1}^{(\beta_1)}, X_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \wedge X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,4}, \mathsf{C}_{2,5})}$ Fix $j \in [r_\alpha], X_{2,j_1}^{(\beta_1)} \in \mathcal{X}_2^{[\nu]}$, and $X_{1,j_2}^{(\beta_2)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (4,5)\}$ do not hold.

$$X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,4}),$$
$$X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,5}).$$

– $\mathsf{C}_{2,4}$. The evaluation is the same as that of $(\mathsf{C}_{2,4}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,5}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,5})$, and we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(j, X_{2,j_1}^{(\beta_2)}, X_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \wedge X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,4}, \mathsf{C}_{2,6})}$ Fix $j \in [r_\alpha], X_{2,j_1}^{(\beta_1)} \in \mathcal{X}_2^{[\nu]}$, and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (4,6)\}$ do not hold.

$$X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,4}),$$
$$X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,6}).$$

– $\mathsf{C}_{2,4}$. The evaluation is the same as that of $(\mathsf{C}_{2,4}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,6}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})$, and we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega \ell_\alpha}{2^n} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{1}{2^n} & \text{if } (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

Hence, for each tuple $(j, X_{2,j_1}^{(\beta_2)}, X_{1,j_2}^{(\beta)})$, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \wedge X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

$\underline{(\mathsf{C}_{2,4}, \mathsf{C}_{2,7})}$ Fix $i \in [2]$, $j \in [r_\alpha]$, $X_{2,j_1}^{(\beta_1)} \in \mathcal{X}_2^{[\nu]}$, and $Y_{i,j_2}^{(\beta_2)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\}$). We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (4, 7)\}$ do not hold.

$$X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,4}),$$
$$Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,7}).$$

– $\mathsf{C}_{2,4}$. The evaluation is the same as that of $(\mathsf{C}_{2,4}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,7}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,7})$, and we have the following bound:

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(i, j, X_{2,j_1}^{(\beta_1)}, X_{i,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j}^{(\beta_1)} \wedge Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,5}, \mathsf{C}_{2,1})}$ Since $\mathsf{C}_{2,5}$ does not occur before $\mathsf{C}_{2,1}$, this case does not occur.

$\underline{(\mathsf{C}_{2,5}, \mathsf{C}_{2,2})}$ Since $\mathsf{C}_{2,5}$ does not occur before $\mathsf{C}_{2,2}$, this case does not occur.

$\underline{(\mathsf{C}_{2,5}, \mathsf{C}_{2,3})}$ Fix $j \in [r_\alpha]$, $X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$ and $Y_{1,j_2}^{(\beta_2)} \in \mathcal{Y}_1^{[\nu]}$ such that $j_1 \geq 1$ (due to $\alpha \in \mathcal{Q}_{\mathsf{non\text{-}coll}}$). We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (5, 3)\}$ do not hold.

$$X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,5}),$$
$$Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,3}).$$

Note that these collisions occur in the first sampling.

– $\mathsf{C}_{2,5}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,5})$, and we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,3}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(j, X_{1,j_1}^{(\beta_1)}, Y_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,5}, \mathsf{C}_{2,4})}$ Fix $j \in [r_\alpha]$, $X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (5, 4)\}$ do not hold.

$$X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \text{ in the first sampling (from(from } \mathsf{C}_{2,5}),$$
$$X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ in the second sampling (from(from } \mathsf{C}_{2,4}).$$

– $\mathsf{C}_{2,5}$. The evaluation is the same as that of $(\mathsf{C}_{2,5}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,4}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,4})$, and we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each $(j, X_{1,j_1}^{(\beta_1)}, X_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,5}, \mathsf{C}_{2,5})}$ Fix $j, j' \in [r_\alpha]$, $X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (5, 5)\}$ do not hold.

$$X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,5}),$$
$$X_{1,j'}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,5}).$$

These evaluations are the same as that of $(\mathsf{C}_{2,5}, \mathsf{C}_{2,3})$, and we have the following bounds:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}, \qquad \Pr[X_{1,j'}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(j, j', X_{1,j_1}^{(\beta_1)}, X_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge X_{1,j'}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,5}, \mathsf{C}_{2,6})}$ Fix $j, j' \in [r_\alpha]$, $X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (5, 6)\}$ do not hold.

$$X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,5}),$$
$$X_{2,j'}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,6}).$$

– $\mathsf{C}_{2,5}$. The evaluation is the same as that of $(\mathsf{C}_{2,5}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,6}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})$, and we have the following bound:

$$\Pr[X_{2,j'}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^n} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{1}{2^n} & \text{if } (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

Hence, for each tuple $(j, j', X_{1,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge X_{2,j'}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

$\underline{(\mathsf{C}_{2,5}, \mathsf{C}_{2,7})}$ Fix $i \in [2]$, $j, j^\dagger \in [r_\alpha]$, $X_{1,j_1}^{(\beta_1)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$ and $Y_{i,j_2}^{(\beta_2)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\}$). We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (5, 7)\}$ do not hold.

$$X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}, \qquad Y_{i,j^\dagger}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}.$$

– $\mathsf{C}_{2,5}$. The evaluation is the same as that of $(\mathsf{C}_{2,5}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,7}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,7})$, and we have the following bound:

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(j, j^\dagger, X_{1,j_1}^{(\beta)}, Y_{i,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_1}^{(\beta_1)} \wedge Y_{i,j^\dagger}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,6}, \mathsf{C}_{2,1})}$ Since $\mathsf{C}_{2,6}$ does not occur before $\mathsf{C}_{2,1}$, this case does not occur.

$\underline{(\mathsf{C}_{2,6}, \mathsf{C}_{2,2})}$ Since $\mathsf{C}_{2,6}$ does not occur before $\mathsf{C}_{2,2}$, this case does not occur.

$\underline{(\mathsf{C}_{2,6}, \mathsf{C}_{2,3})}$ Fix $j \in [r_\alpha]$, $X_{2,j_1}^{(\beta_1)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$ and $Y_{1,j_2}^{(\beta_2)} \in \mathcal{Y}_1^{[\nu]}$. We evaluate the probabilities that the following collisions occur under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (6, 3)\}$ do not hold.

$$\begin{aligned} &X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,6}), \\ &Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,3}). \end{aligned}$$

– $\mathsf{C}_{2,6}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})$, and we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^n} & \text{if } (\beta_1, j_1) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_1, j_1) \neq (\alpha, 0). \end{cases}$$

– $\mathsf{C}_{2,3}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \le \frac{1}{2^n}.$$

Hence, for each tuple $(j, X_{2,j_1}^{(\beta)}, Y_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \wedge Y_{1,0}^{(\alpha)} = Y_{1,j_2}^{(\beta_2)}] \le \begin{cases} \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_1, j_1) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_1, j_1) \neq (\alpha, 0). \end{cases}$$

$\underline{(\mathsf{C}_{2,6}, \mathsf{C}_{2,4})}$ Fix $j \in [r_\alpha]$, $X_{2,j_1}^{(\beta_1)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]}$ such that $j_2 \ge 1$. We evaluate the probabilities that the following collisions occur under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (6, 4)\}$ do not hold.

$$X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,6})$$
$$X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,4}).$$

– $\mathsf{C}_{2,6}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})$, and we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \le \begin{cases} \dfrac{\omega\ell_\alpha}{2^n} & \text{if } (\beta_1, j_1) = (\alpha, 0), \\ \dfrac{1}{2^n} & \text{if } (\beta_1, j_1) \neq (\alpha, 0). \end{cases}$$

– $\mathsf{C}_{2,4}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,4})$, and we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \le \frac{1}{2^n}.$$

Hence, for each tuple $(j, X_{2,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \wedge X_{2,0}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \le \begin{cases} \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_1, j_1) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_1, j_1) \neq (\alpha, 0). \end{cases}$$

$\underline{(\mathsf{C}_{2,6}, \mathsf{C}_{2,5})}$ Fix $j \in [r_\alpha]$, $X_{2,j_1}^{(\beta_1)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$ and $X_{1,j_2}^{(\beta_2)} \in \mathcal{X}_1^{[\nu]}$ such that $j_2 \ge 1$. We evaluate the probabilities that the following collisions occur under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (6, 5)\}$ do not hold.

$$X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,6})$$
$$X_{1,0}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,5}).$$

– $\mathsf{C}_{2,6}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})$, and we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \le \begin{cases} \dfrac{\omega\ell_\alpha}{2^n} & \text{if } (\beta_1, j_1) = (\alpha, 0), \\ \dfrac{1}{2^n} & \text{if } (\beta_1, j_1) \neq (\alpha, 0). \end{cases}$$

– $\mathsf{C}_{2,5}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,5})$, and we have the following bound:

$$\Pr[X_{1,j}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \le \frac{1}{2^n}.$$

Hence, for each tuple $(j, X_{2,j_1}^{(\beta)}, X_{1,j_2}^{(\beta)})$, we have the following bound:

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j}^{(\beta_1)} \wedge X_{1,0}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_1, j_1) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_1, j_1) \neq (\alpha, 0). \end{cases}$$

$\underline{(\mathsf{C}_{2,6}, \mathsf{C}_{2,6})}$ Fix $j, j' \in [r_\alpha]$, $X_{2,j_1}^{(\beta_1)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$ and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$. We evaluate the probabilities that the following collisions occur under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (6, 6)\}$ do not hold.

$$\begin{aligned} X_{2,j}^{(\alpha)} &= X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,6}) \\ X_{2,j'}^{(\alpha)} &= X_{2,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,6}). \end{aligned}$$

The evaluations are the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})$, and we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^n} & \text{if } (\beta_1, j_1) = (\alpha, 0), \\ \dfrac{1}{2^n} & \text{if } (\beta_1, j_1) \neq (\alpha, 0), \end{cases}$$

$$\Pr[X_{2,j'}^{(\alpha)} = X_{2,j_2}^{(\beta_1)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^n} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{1}{2^n} & \text{if } (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

Hence, for each tuple $(j, j', X_{2,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \wedge X_{2,j'}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{(\omega\ell_\alpha)^2}{2^{2n}} & \text{if } (\beta_1, j_1) = (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_1, j_1) \neq (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\alpha, 0) = (\beta_1, j_1) \neq (\beta_2, j_2), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_1, j_1) \neq (\alpha, 0), (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

$\underline{(\mathsf{C}_{2,6}, \mathsf{C}_{2,7})}$ Fix $i \in [2]$, $j, j^\dagger \in [r_\alpha]$, $X_{2,j_1}^{(\beta_1)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$ and $Y_{i,j_2}^{(\beta_2)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\}$). We evaluate the probabilities that the following collisions occur under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (6, 7)\}$ do not hold.

$$\begin{aligned} X_{2,j}^{(\alpha)} &= X_{2,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,6}) \\ Y_{i,j^\dagger}^{(\alpha)} &= Y_{i,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,7}). \end{aligned}$$

– $\mathsf{C}_{2,6}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})$, and we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)}] \leq \begin{cases} \dfrac{\omega\ell_\alpha}{2^n} & \text{if } (\beta_1, j_1) = (\alpha, 0), \\ \dfrac{1}{2^n} & \text{if } (\beta_1, j_1) \neq (\alpha, 0). \end{cases}$$

– $\mathsf{C}_{2,7}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,7})$, and we have the following bound:

$$\Pr[Y_{i,j^\dagger}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(i, j, j^\dagger, X^{(\beta)}_{2,j_1}, X^{(\beta)}_{1,j_2})$, we have the following bound:

$$\Pr[X^{(\alpha)}_{2,j} = X^{(\beta_1)}_{2,j_1} \wedge Y^{(\alpha)}_{i,j^\dagger} = Y^{(\beta_2)}_{i,j_2}] \le \begin{cases} \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_1, j_1) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_1, j_1) \neq (\alpha, 0). \end{cases}$$

$(\mathsf{C}_{2,7}, \mathsf{C}_{2,1})$ Since $\mathsf{C}_{2,7}$ does not occur before $\mathsf{C}_{2,1}$ occurs, this case does not occur.

$(\mathsf{C}_{2,7}, \mathsf{C}_{2,2})$ Since $\mathsf{C}_{2,7}$ does not occur before $\mathsf{C}_{2,2}$ occurs, this case does not occur.

$(\mathsf{C}_{2,7}, \mathsf{C}_{2,3})$ Fix $i \in [2], j \in [r_\alpha]$, $Y^{(\beta_1)}_{i,j_1} \in \mathcal{Y}^{[\nu]}_i \cup \{Y^{(\alpha)}_{i,j'} \mid j' \in [0, j-1]\})$ and $Y^{(\beta_2)}_{1,j_2} \in \mathcal{Y}^{[\nu]}_1$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (7, 3)\}$ do not hold.

$$\begin{aligned} Y^{(\alpha)}_{i,j} &= Y^{(\beta_1)}_{i,j_1} \text{ in the first sampling (from } \mathsf{C}_{2,7}) \\ Y^{(\alpha)}_{1,0} &= Y^{(\beta_2)}_{1,j_2} \text{ in the second sampling (from } \mathsf{C}_{2,3}). \end{aligned}$$

– $\mathsf{C}_{2,7}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,7})$, and we have the following bound:

$$\Pr[Y^{(\alpha)}_{i,j} = Y^{(\beta_1)}_{i,j_1}] \le \frac{1}{2^n}.$$

– $\mathsf{C}_{2,3}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,3})$, and we have the following bound:

$$\Pr[Y^{(\alpha)}_{1,0} = Y^{(\beta_2)}_{1,j_2}] \le \frac{1}{2^n}.$$

Hence, for each tuple $(i, j, Y^{(\beta_1)}_{i,j_1}, Y^{(\beta_2)}_{1,j_2})$, we have the following bound:

$$\Pr[Y^{(\alpha)}_{i,j} = Y^{(\beta_1)}_{i,j_1} \wedge Y^{(\alpha)}_{1,0} = Y^{(\beta_2)}_{1,j_2}] \le \frac{1}{2^{2n}}.$$

$(\mathsf{C}_{2,7}, \mathsf{C}_{2,4})$ Fix $i \in [2], j \in [r_\alpha]$, $Y^{(\beta_1)}_{i,j_1} \in \mathcal{Y}^{[\nu]}_i \cup \{Y^{(\alpha)}_{i,j'} \mid j' \in [0, j-1]\})$, and $X^{(\beta_2)}_{2,j_2} \in \mathcal{X}^{[\nu]}_2$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (7, 4)\}$ do not hold.

$$\begin{aligned} Y^{(\alpha)}_{i,j} &= Y^{(\beta_1)}_{i,j_1} \text{ in the first sampling (from } \mathsf{C}_{2,7}) \\ X^{(\alpha)}_{2,0} &= X^{(\beta_2)}_{2,j_2} \text{ in the second sampling (from } \mathsf{C}_{2,4}). \end{aligned}$$

– $\mathsf{C}_{2,7}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,7})$, and we have the following bound:

$$\Pr[Y^{(\alpha)}_{i,j} = Y^{(\beta_1)}_{i,j_1}] \le \frac{1}{2^n}.$$

– $\mathsf{C}_{2,4}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,4})$, and we have the following bound:

$$\Pr[X^{(\alpha)}_{2,0} = X^{(\beta_2)}_{2,j_2}] \le \frac{1}{2^n}.$$

Hence, for each tuple $(i, j, Y^{(\beta_1)}_{i,j_1}, X^{(\beta_2)}_{2,j_2})$, we have the following bound:

$$\Pr[Y^{(\alpha)}_{i,j} = Y^{(\beta_1)}_{i,j_1} \wedge X^{(\alpha)}_{2,0} = X^{(\beta_2)}_{2,j_2}] \le \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,7}, \mathsf{C}_{2,5})}$ Fix $i \in [2], j, j^\dagger \in [r_\alpha]$, $Y_{i,j_1}^{(\beta_1)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\})$, and $X_{1,j_2}^{(\beta_2)} \in \mathcal{X}_1^{[\nu]} \cup \{X_{1,0}^{(\alpha)}\}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (7, 5)\}$ do not hold.

$$Y_{i,j}^{(\alpha)} = Y_{i,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,7})$$
$$X_{1,j'}^{(\alpha)} = X_{1,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,5}).$$

– $\mathsf{C}_{2,7}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,7})$, and we have the following bound:

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,5}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,5})$, and we have the following bound:

$$\Pr[X_{1,j^\dagger}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(i, j, j^\dagger, Y_{i,j_1}^{(\beta_1)}, X_{1,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_1}^{(\beta_1)} \wedge X_{1,j^\dagger}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{1}{2^{2n}}.$$

$\underline{(\mathsf{C}_{2,7}, \mathsf{C}_{2,6})}$ Fix $i \in [2], j, j^\dagger \in [r_\alpha]$, $Y_{i,j_1}^{(\beta_1)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\})$, and $X_{2,j_2}^{(\beta_2)} \in \mathcal{X}_2^{[\nu]} \cup \{X_{2,0}^{(\alpha)}\}$. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (7, 6)\}$ do not hold.

$$Y_{i,j}^{(\alpha)} = Y_{i,j_1}^{(\beta_1)} \text{ in the first sampling (from } \mathsf{C}_{2,7})$$
$$X_{2,j'}^{(\alpha)} = X_{2,j_2}^{(\beta_2)} \text{ in the second sampling (from } \mathsf{C}_{2,6}).$$

– $\mathsf{C}_{2,7}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,7})$, and we have the following bound:

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_1}^{(\beta_1)}] \leq \frac{1}{2^n}.$$

– $\mathsf{C}_{2,6}$. The evaluation is the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})$, and we have the following bound:

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega \ell_\alpha}{2^n} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{1}{2^n} & \text{if } (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

Hence, for each tuple $(i, j, j^\dagger, Y_{i,j_1}^{(\beta_1)}, X_{2,j_2}^{(\beta_2)})$, we have the following bound:

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_1}^{(\beta_1)} \wedge X_{2,j^\dagger}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{\omega \ell_\alpha}{2^{2n}} & \text{if } (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{1}{2^{2n}} & \text{if } (\beta_2, j_2) \neq (\alpha, 0). \end{cases}$$

$\underline{(\mathsf{C}_{2,7}, \mathsf{C}_{2,7})}$ Fix $i_1, i_2 \in [2], j, j' \in [r_\alpha]$, $Y^{(\beta_1)}_{i_1,j_1} \in \mathcal{Y}_i^{[\nu]} \cup \{Y^{(\alpha)}_{i_1,j^\dagger} \mid j^\dagger \in [0, j-1]\})$, and $Y^{(\beta_2)}_{i_2,j_2} \in \mathcal{Y}_i^{[\nu]} \cup \{Y^{(\alpha)}_{i_2,j^\ddagger} \mid j^\ddagger \in [0, j-1]\})$, where $Y^{(\alpha)}_{i_1,j}$ is defined in the first sampling and $Y^{(\alpha)}_{i_2,j'}$ is defined in the second sampling. We evaluate the probabilities of the following collisions under the assumption that the other conditions $\{(\mathsf{C}_{2,l_1}, \mathsf{C}_{2,l_2}) \mid (l_1, l_2) \neq (7,7)\}$ do not hold.

$$Y^{(\alpha)}_{i,j} = Y^{(\beta_1)}_{i,j_1} \text{ in the first sampling (from } \mathsf{C}_{2,7})$$
$$Y^{(\alpha)}_{i_2,j'} = Y^{(\beta_2)}_{i_2,j_2} \text{ in the second sampling (from } \mathsf{C}_{2,7}).$$

The evaluations are the same as that of $(\mathsf{C}_{2,3}, \mathsf{C}_{2,7})$, and we have the following bound:

$$\Pr[Y^{(\alpha)}_{i_1,j} = Y^{(\beta_1)}_{i_1,j_1}] \leq \frac{1}{2^n}, \qquad \Pr[Y^{(\alpha)}_{i_2,j'} = Y^{(\beta_2)}_{i_2,j_2}] \leq \frac{1}{2^n}.$$

Hence, for each tuple $(i_1, i_2, j, j', Y^{(\beta_1)}_{i_1,j_1}, Y^{(\beta_2)}_{i_2,j_2})$, we have the following bound:

$$\Pr[Y^{(\alpha)}_{i,j} = Y^{(\beta_1)}_{i,j_1} \wedge Y^{(\alpha)}_{i_2,j'} = Y^{(\beta_2)}_{i_2,j_2}] \leq \frac{1}{2^{2n}}.$$

Bound of $\mathbf{Pr}[\mathsf{fail}^{*2}_{\mathsf{sample}}]$. In summary, the above evaluations offer the following bounds.

- For $(\mathsf{C}_{2,1}, \mathsf{C}_{2,6})$ with $(\beta_2, j_2) = (\alpha, 0)$,
$$\Pr[X^{(\alpha)}_{1,0} = X^{(\beta_1)}_{1,j_1} \wedge X^{(\alpha)}_{2,j} = X^{(\beta_2)}_{2,j_2}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$
- For $(\mathsf{C}_{2,2}, \mathsf{C}_{2,6})$ with $(\beta_2, j_2) = (\alpha, 0)$ or $(\beta_2, j_2) \neq (\alpha, 0) \wedge \beta_1 = \beta_2$,
$$\Pr[Y^{(\alpha)}_{2,0} = Y^{(\beta_1)}_{2,j_1} \wedge X^{(\alpha)}_{2,j} = X^{(\beta_2)}_{2,j_2}] \leq \frac{\omega\ell_\alpha}{2^{2n}}$$
- For $(\mathsf{C}_{2,3}, \mathsf{C}_{2,6})$ with $(\beta_2, j_2) = (\alpha, 0)$,
$$\Pr[Y^{(\alpha)}_{1,0} = Y^{(\beta_1)}_{1,j_1} \wedge X^{(\alpha)}_{2,j} = X^{(\beta_2)}_{2,j_2}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$
- For $(\mathsf{C}_{2,4}, \mathsf{C}_{2,6})$ with $(\beta_2, j_2) = (\alpha, 0)$,
$$\Pr[X^{(\alpha)}_{2,0} = X^{(\beta_1)}_{2,j_1} \wedge X^{(\alpha)}_{2,j} = X^{(\beta_2)}_{2,j_2}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$
- For $(\mathsf{C}_{2,5}, \mathsf{C}_{2,6})$ with $(\beta_2, j_2) = (\alpha, 0)$,
$$\Pr[X^{(\alpha)}_{1,j} = X^{(\beta_1)}_{1,j_1} \wedge X^{(\alpha)}_{2,j'} = X^{(\beta_2)}_{2,j_2}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$
- For $(\mathsf{C}_{2,6}, \mathsf{C}_{2,3})$ with $(\beta_1, j_1) = (\alpha, 0)$,
$$\Pr[X^{(\alpha)}_{2,j} = X^{(\beta_1)}_{2,j_1} \wedge Y^{(\alpha)}_{1,0} = Y^{(\beta_2)}_{1,j_2}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$
- For $(\mathsf{C}_{2,6}, \mathsf{C}_{2,4})$ with $(\beta_1, j_1) = (\alpha, 0)$,
$$\Pr[X^{(\alpha)}_{2,0} = X^{(\beta_1)}_{2,j_1} \wedge X^{(\alpha)}_{2,0} = X^{(\beta_2)}_{2,j_2}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$

– For $(\mathsf{C}_{2,6}, \mathsf{C}_{2,5})$ with $(\beta_1, j_1) = (\alpha, 0)$,

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,j}^{(\beta_1)} \wedge X_{1,0}^{(\alpha)} = X_{1,j_2}^{(\beta_2)}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$

– For $(\mathsf{C}_{2,6}, \mathsf{C}_{2,6})$ with $(\beta_1, j_1) = (\beta_2, j_2) = (\alpha, 0)$, $(\beta_1, j_1) \neq (\beta_2, j_2) = (\alpha, 0)$, or $(\alpha, 0) = (\beta_1, j_1) \neq (\beta_2, j_2)$,

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \wedge X_{2,j'}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \begin{cases} \dfrac{(\omega\ell_\alpha)^2}{2^{2n}} & \text{if } (\beta_1, j_1) = (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\beta_1, j_1) \neq (\beta_2, j_2) = (\alpha, 0), \\ \dfrac{\omega\ell_\alpha}{2^{2n}} & \text{if } (\alpha, 0) = (\beta_1, j_1) \neq (\beta_2, j_2). \end{cases}$$

– For $(\mathsf{C}_{2,6}, \mathsf{C}_{2,7})$ with $(\beta_1, j_1) = (\alpha, 0)$,

$$\Pr[X_{2,j}^{(\alpha)} = X_{2,j_1}^{(\beta_1)} \wedge Y_{i,j^\dagger}^{(\alpha)} = Y_{i,j_2}^{(\beta_2)}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$

– For $(\mathsf{C}_{2,7}, \mathsf{C}_{2,6})$ with $(\beta_2, j_2) = (\alpha, 0)$,

$$\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j_1}^{(\beta_1)} \wedge X_{2,j^\dagger}^{(\alpha)} = X_{2,j_2}^{(\beta_2)}] \leq \frac{\omega\ell_\alpha}{2^{2n}}.$$

– For the other cases, the collision probabilities are respectively at most

$$\frac{1}{2^{2n}}.$$

Summing the above bonds for each possible case, we have the following bound:

$$\begin{aligned} \Pr[\mathsf{fail}^{*2}_{\mathsf{sample}}] &\leq \sum_{\nu\in[u]} \left(\sum_{\alpha\in\mathcal{Q}^{[\nu]}} \ell_\alpha^2 \cdot \left(\frac{\omega\ell_\alpha}{2^n}\right)^2 + 7 \cdot \sum_{\substack{\alpha,\beta_1\in\mathcal{Q}^{[\nu]} \\ \text{s.t. } \alpha\neq\beta_1}} \ell_\alpha^2\ell_{\beta_1} \cdot \frac{\omega\ell_\alpha}{2^{2n}} \right. \\ &\qquad \left. + 6 \cdot \sum_{\alpha,\beta_2\in\mathcal{Q}^{[\nu]}\text{s.t. } \alpha\neq\beta_2} \ell_\alpha\ell_{\beta_2} \cdot \frac{\omega\ell_\alpha}{2^{2n}} + \sum_{\alpha,\beta_1,\beta_2\in\mathcal{Q}^{[\nu]}} \ell_\alpha^2\ell_{\beta_1}\ell_{\beta_2} \cdot \frac{1}{2^{2n}} \right) \\ &\leq \sum_{\nu\in[u]} \frac{3\omega^2\sigma_\nu^3 + 4\omega^2\ell\sigma_\nu^3}{2^{2n}} \\ &\leq \frac{3\omega^2\sigma_{\mathsf{u}}^2\sigma}{2^{2n}} + \frac{4\omega^2\ell\sigma_{\mathsf{u}}^2\sigma}{2^{2n}}. \end{aligned}$$

10 Proof of Theorem 1: Lower-Bounding $\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]}$

In this section, we show that

$$\frac{\Pr[\mathsf{T}_R = \tau]}{\Pr[\mathsf{T}_I = \tau]} \geq 1 - \frac{4q_{\mathsf{u}}\sigma_{\mathsf{u}}\sigma}{2^{2n}}.$$

Fix a good transcript τ that consists of the hash keys $\mathcal{H}$ and internal pairs $(\mathcal{P}_1^{[\nu]\star})_{\nu\in[q]}$ and $(\mathcal{P}_2^{[\nu]\star})_{\nu\in[q]}$. We use the symbol "$\star$" for values and sets corresponding to the good transcript τ. For a set $\mathcal{L}$, let $\mathsf{T}_R \vdash \mathcal{L}$ (resp. $\mathsf{T}_I \vdash \mathcal{L}$) be an event that T_R (resp. T_I) satisfies all elements in $\mathcal{L}$.

10.1 Evaluation of $\mathbf{Pr}[\mathsf{T}_{\boldsymbol{W}} \vdash \mathcal{H}^\star]$ for $\boldsymbol{W} \in \{\boldsymbol{R}, \boldsymbol{I}\}$

In the real world, for each $i \in [2]$, the hash key $H_i^{[\nu]}$ is defined as $H_i^{[\nu]} = \pi_i(\mathtt{bin}_n(0))$ and chosen uniformly at random from $\{0,1\}^n$ by the RP π_i. In the ideal world, each hash key $H_i^{[\nu]}$ is chosen uniformly at random from $\{0,1\}^n$. We thus have

$$\Pr[\mathsf{T}_R \vdash \mathcal{H}^\star] = \Pr[\mathsf{T}_I \vdash \mathcal{H}^\star].$$

10.2 Evaluation of $\mathbf{Pr}[\mathsf{T}_{\boldsymbol{W}} \vdash (\boldsymbol{\mathcal{P}}_{\boldsymbol{1}}^{[\boldsymbol{\nu}]\star}, \boldsymbol{\mathcal{P}}_{\boldsymbol{2}}^{[\boldsymbol{\nu}]\star})_{\boldsymbol{\nu}\in[\boldsymbol{u}]} \mid \mathsf{T}_{\boldsymbol{W}} \vdash \boldsymbol{\mathcal{H}}^\star]$ for $\boldsymbol{W} \in \{\boldsymbol{R}, \boldsymbol{I}\}$

Fix $\nu \in [u]$ and $\alpha \in \mathcal{Q}^{[\nu]}$. We evaluate the real-world and ideal-world probabilities for the α-th query. Note that by $\neg\mathsf{hit}_0$, for each $\nu \in [u]$, the pairs $(\mathtt{bin}_n(0), H_1^{[\nu]})$ and $(\mathtt{bin}_n(0), H_2^{[\nu]})$ are not in $\mathcal{P}_1^{[\nu]\star}$ and $\mathcal{P}_2^{[\nu]\star}$ respectively.

In the ideal world, if $\alpha \in \mathcal{Q}_{\mathsf{coll}}^{[\nu]}$, then each value for the α-th query is sampled once. On the other hand, if $\alpha \in \mathcal{Q}_{\mathsf{non\text{-}coll}}^{[\nu]}$, then each value of the α-th query is sampled at most twice. Hence, the probability that sampled values satisfy $\mathcal{P}_1^{(\alpha)\star}$ and $\mathcal{P}_2^{(\alpha)\star}$ for $\alpha \in \mathcal{Q}_{\mathsf{non\text{-}coll}}^{[\nu]}$ is greater than the probability for $\alpha \in \mathcal{Q}_{\mathsf{coll}}^{[\nu]}$. We thus assume that for each $\nu \in [u]$, $\mathcal{Q}_{\mathsf{coll}}^{[\nu]} = \emptyset$ and $\mathcal{Q}_{\mathsf{non\text{-}coll}}^{[\nu]} = \mathcal{Q}^{[\nu]}$.

For $i \in [2]$, let

$$z_i^{(\alpha)} \coloneqq \Big|\{(X_{i,j}^{(\alpha)\star}, Y_{i,j}^{(\alpha)\star}) \mid (X_{i,j}^{(\alpha)\star}, Y_{i,j}^{(\alpha)\star}) \in \mathcal{P}_i^{(\alpha)\star} \wedge \forall \beta \in \mathcal{Q}_{<\alpha}^{[\nu]} : (X_{i,j}^{(\alpha)\star}, Y_{i,j}^{(\alpha)\star}) \notin \mathcal{P}_i^{(\beta)\star}]\}\Big|$$

be the number of pairs of $\dot{\pi}_\nu$ (if $i = 1$); $\ddot{\pi}_\nu$ (if $i = 2$) defined by the α-th query, and

$$z_i^{(<\alpha)} \coloneqq \sum_{\beta \in \mathcal{Q}_{<\alpha}^{[\nu]}} z_i^{(\alpha)}$$

be the number of internal pairs of $\dot{\pi}_\nu$ (if $i = 1$); $\ddot{\pi}_\nu$ (if $i = 2$) defined by queries to the ν-th user before the α-th query. Let

$$\mathcal{T}^{(<\alpha)} \coloneqq \left\{\beta \in \mathcal{Q}_{<\alpha}^{[\nu]} \mid T^{(\beta)} = T^{(\alpha)}\right\} \text{ and } t^{(<\alpha)} \coloneqq \left|\mathcal{T}^{(<\alpha)}\right|.$$

Real-World Probability. In the real world, by $\neg\mathsf{fail}_{\mathsf{sample}}$, for each $\alpha \in \mathcal{Q}^{[\nu]}$, we have

$$\Pr[\mathsf{T}_R \vdash \mathcal{P}^{(\alpha)\star}] = \left(\prod_{i \in [z_1^{(\alpha)}]} \frac{1}{2^n - z_1^{(<\alpha)} - i}\right)\left(\prod_{i \in [z_2^{(\alpha)}]} \frac{1}{2^n - z_2^{(<\alpha)} - i}\right).$$

Ideal-World Probability. In this evaluation, we use the hat symbol to rephrase a query index $\alpha \in \mathcal{Q}^{[\nu]}$ to $\hat{\alpha} \in [q_\nu]$ which is the query index to the ν-th user, i.e., the $\hat{\alpha}$-th query to the ν-th user is equal to the α-th query made by an adversary. For $\alpha \in \mathcal{Q}^{[\nu]}$, if the α-th query is an encryption (resp. a decryption) one, then $C^{(\alpha)}$ (resp. $M^{(\alpha)}$) is chosen uniformly at random from at least $2^{m_\alpha} - t^{(<\alpha)}$ elements of m_α bits. The remaining m_α^+-bit value $C_+^{(\alpha)}$ or $M_+^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^{m_\alpha^+}$. For each $\alpha \in \mathcal{Q}^{[\nu]}$, we have

$$\Pr[\mathsf{T}_I \vdash \{(T^{(\alpha)\star}, \hat{M}^{(\alpha)\star}, \hat{C}^{(\alpha)\star})\}] \le \frac{1}{2^{m_\alpha^+}(2^{m_\alpha} - t^{(<\alpha)})} = \frac{1}{2^{(r_\alpha+1)n} - t^{(<\alpha)}2^{m_\alpha^+}}.$$

We assume that $\mathsf{T}_I \vdash \{(T^{(\alpha)\star}, \hat{M}^{(\alpha)\star}, \hat{C}^{(\alpha)\star}) \mid \alpha \in \mathcal{Q}^{[\nu]}\}$.

We next evaluate $\Pr[\forall i \in [0, r_\alpha] : Y_{1,i}^{(\alpha)} = Y_{1,i}^{(\alpha)\star}]$, the probability that the remaining values satisfy the target values in τ. Note that the assumption $\mathsf{T}_I \vdash \{(T^{(\alpha)\star}, \hat{M}^{(\alpha)\star}, \hat{C}^{(\alpha)\star})\}$ ensures that $\forall i \in [0, r_\alpha] : Y_{1,i}^{(\alpha)} = Y_{1,i}^{(\alpha)\star} \Rightarrow \forall i \in [0, r_\alpha] : Y_{2,i}^{(\alpha)} = Y_{2,i}^{(\alpha)\star}$, and the number of the remaining new ciphertext blocks $Y_{1,i}^{(\alpha)}$ is $z_1^{(\alpha)} + z_2^{(\alpha)} - (r_\alpha + 1)$. For $Y_{1,j}^{(\alpha)}$, if the input is new, then $Y_{1,j}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$ and we have $\Pr[Y_{1,j}^{(\alpha)} = Y_{1,j}^{(\alpha)\star}] \leq \frac{1}{2^n}$. Since each fresh value is sampled at most twice, we consider the following two cases.

- **Case 1**: The first sampling of the α-th query succeeds.
  The probability that $\forall i \in [0, r_\alpha] : Y_{1,i}^{(\alpha)} = Y_{1,i}^{(\alpha)\star}$ is at most
  $$\left(\frac{1}{2^n}\right)^{z_1^{(\alpha)} + z_2^{(\alpha)} - (r_\alpha + 1)}.$$
- **Case 2**: The first sampling of the α-th query fails and the second sampling succeeds. The second sampling is performed if a collision with $Y_{1,0}^{(\alpha)}, X_{2,0}^{(\alpha)}$ or $Y_{i,j}^{(\alpha)}$ $((i,j) \in [2] \times [r_\alpha])$ occurs. We evaluate the collision probabilities below.
  - We evaluate the probability that a collision with $Y_{i,j}^{(\alpha)}$ occurs for some $(i,j) \in [2] \times [r_\alpha]$, i.e.,
    $$p_1 \coloneqq \Pr\left[\exists (i,j) \in [2] \times [r_\alpha] \text{ s.t. } Y_{i,j}^{(\alpha)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\})\right].$$
    For $Y_{i,j'}^{(\beta)} \in \mathcal{Y}_i^{[\nu]} \cup \{Y_{i,j'}^{(\alpha)} \mid j' \in [0, j-1]\})$ such that $Y_{i,j'}^{(\beta)}$ is defined before $Y_{i,j}^{(\alpha)}$ is defined, since $Y_{i,j}^{(\alpha)}$ is an n-bit random value or defined by using an n-bit random value $Y_{3-i,j}^{(\alpha)}$, we have $\Pr[Y_{i,j}^{(\alpha)} = Y_{i,j'}^{(\beta)}] \leq \frac{1}{2^n}$ and
    $$p_1 \leq \sum_{i \in [2]} \frac{z_i^{(<\alpha)}(z_i^{(\alpha)} - 1) + \binom{z_i^{(\alpha)}}{2}}{2^n}.$$
  - We evaluate the probability that a collision with $Y_{1,0}^{(\alpha)}$ occurs, i.e.,
    $$p_2 \coloneqq \Pr\left[Y_{1,0}^{(\alpha)} \in \mathcal{Y}_1^{[\nu]}\right].$$
    Since $Y_{1,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have
    $$p_2 \leq \frac{z_1^{(<\alpha)}}{2^n}.$$
  - We evaluate the probability that a collision with $X_{2,0}^{(\alpha)}$ occurs and no collision with $Y_{1,0}^{(\alpha)}$ occurs, i.e.,
    $$p_3 \coloneqq \Pr\left[X_{2,0}^{(\alpha)} \in \mathcal{X}_2^{[\nu]} \wedge Y_{1,0}^{(\alpha)} \notin \mathcal{Y}_1^{[\nu]}\right].$$
    We consider the following cases.
    * For $X_{2,j}^{(\beta)} \in \mathcal{X}_2^{[\nu]}$ such that $j \geq 1$ or $T^{(\alpha)} \neq T^{(\beta)}$, since $X_{2,0}^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$ or defined by using an n-bit random value $Y_{1,0}^{(\alpha)}$, we have
      $$\Pr\left[X_{2,0}^{(\alpha)} = X_{2,j}^{(\beta)} \mid j \geq 1 \vee T^{(\alpha)} \neq T^{(\beta)}\right] \leq \frac{1}{2^n}.$$

∗ For $X_{2,j}^{(\beta)} \in \mathcal{X}_2^{[\nu]}$ such that $j = 0$ and $T^{(\alpha)} = T^{(\beta)}$, the collision has the following structure:

$$\begin{aligned} X_{2,0}^{(\alpha)} = X_{2,0}^{(\beta)} \Leftrightarrow Y_{1,0}^{(\alpha)} &\oplus \mathsf{H}_1^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\alpha)}, R^{(\alpha)}) \\ &= Y_{1,0}^{(\beta)} \oplus \mathsf{H}_1^{[\nu]}(T^{(\beta)}, R^{(\beta)}) \oplus \mathsf{H}_2^{[\nu]}(T^{(\beta)}, R^{(\beta)}). \end{aligned}$$

If no collision with $Y_{1,0}^{(\alpha)}$ occurs, then $Y_{1,0}^{(\alpha)} \neq Y_{1,0}^{(\beta)}$ must hold. On the other hand, $X_{2,0}^{(\alpha)} = X_{2,0}^{(\beta)} \wedge R^{(\alpha)} = R^{(\beta)} \wedge T^{(\alpha)} = T^{(\beta)} \Rightarrow Y_{1,0}^{(\alpha)} = Y_{1,0}^{(\beta)}$. Hence, $R^{(\alpha)} = R^{(\beta)}$ must hold. Since $|Y_{1,0}^{(\alpha)}| + |R^{(\alpha)}| = (r_\alpha + 1)n - m_\alpha^+$ and there is at least one choice for $Y_{1,0}^{(\alpha)} = Y_{1,0}^{(\beta)}$ in the $2^{(r_\alpha+1)n-m_\alpha^+}$ pairs of $(Y_{1,0}^{(\alpha)}, R^{(\alpha)})$, we have

$$\Pr[X_{2,0}^{(\alpha)} = X_{2,0}^{(\beta)} \mid T^{(\alpha)} = T^{(\beta)}] \leq \frac{2^{r_\alpha n - m_\alpha^+} - 1}{2^{(r_\alpha+1)n-m_\alpha^+}}.$$

By using the above bounds, we have the following bound:

$$\begin{aligned} p_3 &\leq \frac{z_2^{(<\alpha)} - t^{(<\alpha)}}{2^n} + \frac{t^{(<\alpha)}(2^{r_\alpha n - m_\alpha^+} - 1)}{2^{(r_\alpha+1)n-m_\alpha^+}} \\ &= \frac{z_2^{(<\alpha)} - \frac{t^{(<\alpha)}}{2^{r_\alpha n - m_\alpha^+}}}{2^n}. \end{aligned}$$

Hence, the probability that The first sampling of the α-th query fails and the second sampling succeeds: $\forall i \in [0, r_\alpha] : Y_{1,i}^{(\alpha)} = Y_{1,i}^{(\alpha)\star}$ is at most

$$\begin{aligned} &\left(\frac{1}{2^n}\right)^{z_1^{(\alpha)}+z_2^{(\alpha)}-(r_\alpha+1)} \cdot (p_1 + p_2 + p_3) \\ &\leq \left(\frac{1}{2^n}\right)^{z_1^{(\alpha)}+z_2^{(\alpha)}-(r_\alpha+1)} \cdot \left(\left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right) - \frac{\frac{t^{(<\alpha)}}{2^{r_\alpha n - m_\alpha^+}}}{2^n}\right) \end{aligned}$$

By the above bounds, we have the following bound:

$$\begin{aligned} \Pr[\mathsf{T}_I \vdash \mathcal{P}^{(\alpha)\star}] \leq \frac{1}{2^{(r_\alpha+1)n} - t^{(<\alpha)} 2^{m_\alpha^+}} &\cdot \left(\frac{1}{2^n}\right)^{z_1^{(\alpha)}+z_2^{(\alpha)}-(r_\alpha+1)} \\ &\times \left(1 + \left(\left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right) - \frac{\frac{t^{(<\alpha)}}{2^{r_\alpha n - m_\alpha^+}}}{2^n}\right)\right). \end{aligned}$$

Deriving the Lower Bound. By using the above bounds, we have the following bound:

$$
\begin{aligned}
&\frac{\Pr[\mathsf{T}_R \vdash \mathcal{P}^{(\alpha)\star}]}{\Pr[\mathsf{T}_I \vdash \mathcal{P}^{(\alpha)\star}]} \\
&\geq \left(\prod_{i\in[z_1^{(\alpha)}]} \frac{1}{2^n - z_1^{(<\alpha)} - (i-1)}\right)\left(\prod_{i\in[z_2^{(\alpha)}]} \frac{1}{2^n - z_2^{(<\alpha)} - (i-1)}\right) \\
&\quad \times \left(\frac{1}{2^{(r_\alpha+1)n} - t^{(<\alpha)}2^{m_\alpha^+}} \cdot \left(\frac{1}{2^n}\right)^{z_1^{(\alpha)}+z_2^{(\alpha)}-(r_\alpha+1)}\right)^{-1} \\
&\qquad \times \left(1 + \left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right) - \frac{\frac{t^{(<\alpha)}}{2^{r_\alpha n - m_\alpha^+}}}{2^n}\right)^{-1} \\
&= \left(\left(\prod_{i\in[z_1^{(\alpha)}]} \left(1 - \frac{z_1^{(<\alpha)} + (i-1)}{2^n}\right)\right)\left(\prod_{i\in[z_2^{(\alpha)}]} \left(1 - \frac{z_2^{(<\alpha)} + (i-1)}{2^n}\right)\right)\right) \\
&\quad \times \left(1 + \frac{t^{(<\alpha)}2^{m_\alpha^+}}{2^{n(r_\alpha+1)} - t^{(<\alpha)}2^{m_\alpha^+}}\right)\left(1 + \left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right) - \frac{\frac{t^{(<\alpha)}}{2^{r_\alpha n - m_\alpha^+}}}{2^n}\right)^{-1}.
\end{aligned}
$$

By using the relation: $(1-a)(1+b) \leq 1-a+b$ for any $a, b \geq 0$ and the following inequations

$$
\begin{aligned}
&\star \sum_{i\in[z_1^{(\alpha)}]} \frac{z_1^{(<\alpha)} + (i-1)}{2^n} + \sum_{i\in[z_2^{(\alpha)}]} \frac{z_2^{(<\alpha)} + (i-1)}{2^n} = \sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}, \\
&\star \frac{t^{(<\alpha)}2^{m_\alpha^+}}{2^{n(r_\alpha+1)} - t^{(<\alpha)}2^{m_\alpha^+}} \leq \frac{\frac{t^{(<\alpha)}}{2^{r_\alpha n - m_\alpha^+}}}{2^n}, \\
&\star \left(1 + \frac{t^{(<\alpha)}2^{m_\alpha^+}}{2^{n(r_\alpha+1)} - t^{(<\alpha)}2^{m_\alpha^+}}\right)\left(1 + \left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right) - \frac{\frac{t^{(<\alpha)}}{2^{r_\alpha n - m_\alpha^+}}}{2^n}\right) \\
&\quad \leq 1 + \left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right) + \frac{t^{(<\alpha)}2^{m_\alpha^+}}{2^{n(r_\alpha+1)} - t^{(<\alpha)}2^{m_\alpha^+}} \cdot \left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right)
\end{aligned}
$$

we have the following bound:

$$
\begin{aligned}
&\frac{\Pr[\mathsf{T}_R \vdash \mathcal{P}^{(\alpha)\star}]}{\Pr[\mathsf{T}_I \vdash \mathcal{P}^{(\alpha)\star}]} \\
&\geq \left(1 - \left(\sum_{i\in[z_1^{(\alpha)}]} \frac{z_1^{(<\alpha)} - (i-1)}{2^n}\right) - \left(\sum_{i\in[z_2^{(\alpha)}]} \frac{z_2^{(<\alpha)} - (i-1)}{2^n}\right)\right. \\
&\qquad \left. + \left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right) + \frac{t^{(<\alpha)}2^{m_\alpha^+}}{2^{n(r_\alpha+1)} - t^{(<\alpha)}2^{m_\alpha^+}} \cdot \left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right)\right)^{-1} \\
&\geq 1 - \frac{t^{(<\alpha)}2^{m_\alpha^+}}{2^{n(r_\alpha+1)} - t^{(<\alpha)}2^{m_\alpha^+}} \cdot \left(\sum_{i\in[2]} \frac{z_i^{(<\alpha)} z_i^{(\alpha)} + \binom{z_i^{(\alpha)}}{2}}{2^n}\right),
\end{aligned}
$$

and

$$\begin{aligned}\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]} &\geq \prod_{\nu\in[u]}\prod_{\alpha\in\mathcal{Q}^{[\nu]}}\frac{\Pr[\mathsf{T}_R\vdash\mathcal{P}^{(\alpha)\star}]}{\Pr[\mathsf{T}_I\vdash\mathcal{P}^{(\alpha)\star}]}\\ &\geq \prod_{\nu\in[u]}\prod_{\alpha\in\mathcal{Q}^{[\nu]}}\left(1-\frac{t^{(<\alpha)}2^{m_\alpha^+}}{2^{n(r_\alpha+1)}-t^{(<\alpha)}2^{m_\alpha^+}}\cdot\left(\sum_{i\in[2]}\frac{z_i^{(<\alpha)}z_i^{(\alpha)}+\binom{z_i^{(\alpha)}}{2}}{2^n}\right)\right)\\ &\geq 1-\sum_{\nu\in[u]}\sum_{\alpha\in\mathcal{Q}^{[\nu]}}\frac{t^{(<\alpha)}2^{m_\alpha^+}}{2^{n(r_\alpha+1)}-t^{(<\alpha)}2^{m_\alpha^+}}\cdot\left(\sum_{i\in[2]}\frac{z_i^{(<\alpha)}z_i^{(\alpha)}+\binom{z_i^{(\alpha)}}{2}}{2^n}\right).\end{aligned}$$

By using the following inequation with the assumption $t^{(<\alpha)}\leq 2^{n-1}$:

$$\frac{t^{(<\alpha)}2^{m_\alpha^+}}{2^{n(r_\alpha+1)}-t^{(<\alpha)}2^{m_\alpha^+}}\leq\frac{t^{(<\alpha)}}{2^n-t^{(<\alpha)}}\leq\frac{2q_{\mathsf{u}}}{2^n},$$

we have

$$\begin{aligned}\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]} &\geq 1-\sum_{\nu\in[u]}\sum_{\alpha\in\mathcal{Q}^{[\nu]}}\frac{2q_{\mathsf{u}}}{2^n}\cdot\left(\sum_{i\in[2]}\frac{z_i^{(<\alpha)}z_i^{(\alpha)}+0.5(z_i^{(\alpha)})^2}{2^n}\right)\\ &\geq 1-\sum_{\nu\in[u]}\sum_{\alpha\in\mathcal{Q}^{[\nu]}}\frac{2q_{\mathsf{u}}}{2^n}\cdot\left(\sum_{i\in[2]}\frac{(z_i^{(<\alpha)}+0.5z_i^{(\alpha)})z_i^{(\alpha)}}{2^n}\right)\\ &\geq 1-\sum_{\nu\in[u]}\sum_{\alpha\in\mathcal{Q}^{[\nu]}}\frac{2q_{\mathsf{u}}}{2^n}\cdot\frac{\sigma_{\mathsf{u}}}{2^n}\cdot\sum_{i\in[2]}z_i^{(\alpha)}\\ &\geq 1-\frac{4q_{\mathsf{u}}\sigma_{\mathsf{u}}\sigma}{2^{2n}}.\end{aligned}$$

11 Proof of Theorem 2

Without loss of generality, we assume that the adversary is deterministic and makes no repeated or trivial query.

We first replace the BCs $(E^{\pm}_{K^{[\nu]}})_{\nu\in[u]}$ with RPs $(\pi^{\pm}_\nu)_{\nu\in[u]}$ where $\forall\nu\in[u]:\pi_\nu\xleftarrow{\$}\mathsf{Perm}(n)$ and $\pi^{\pm}\coloneqq(\pi_\nu,\pi_\nu^{-1})$, thereby introducing the advantage function $\mathbf{Adv}^{\mathsf{mu\text{-}sprp}}_E(u,\sigma,t+O(\sigma))$ in the bound. Hence, the following evaluation considers the oracles $\mathcal{O}_{\mathsf{real}}=(\mathsf{HCTR2}[\pi_\nu,\mathsf{H}],\ \mathsf{HCTR2}^{-1}[\pi^{\pm}_\nu,\mathsf{H}])_{\nu\in[u]}$ as the real-world ones.

11.1 Definitions

We use the same definitions and notations given in the proof of CHCTR2 in Section 6.1. Let q_ν be the number of queries to the ν-th user. Without loss of generality, we assume that $\forall\nu\in[u]:q_\nu\geq 1$. Let σ_ν be the total number of BC calls by queries to the ν-th user. Note that $\sigma=\sum_{\nu\in[u]}\sigma_\nu$. We call the stage that an adversary makes queries the "query stage" and the stage after the query stage the "decision stage".

For $\alpha\in[q]$, values corresponding to the α-th query are denoted by using the superscript symbol (α), e.g., $M^{(\alpha)}$, $T^{(\alpha)}$, $X_i^{(\alpha)}$, $Y_i^{(\alpha)}$, etc. Let $t_\alpha\coloneqq|T^{(\alpha)}|_n$. Let $m_\alpha\coloneqq|M^{(\alpha)}|$ be the length of the plaintext $M^{(\alpha)}$. Let $r_\alpha=\lceil|M_*^{(\alpha)}|/n\rceil$ be the block length of the right part of the plaintext $M^{(\alpha)}$. Let $m_\alpha^+\coloneqq n(r_\alpha+1)-m_\alpha$ be the number of truncated bits in CTR for

the α-th query and $\ell_\alpha = 2(r_\alpha + 1)$ the total number of RP calls in $\mathsf{CHCTR2}$ in the α-th query. Let $r_{\mathsf{max}} := \max\{r_\alpha \mid \alpha \in [q]\}$. For the plaintext $M_*^{(\alpha)}$ (resp. ciphertext $C_*^{(\alpha)}$) of the right part, let $M_i^{(\alpha)}$ (resp. $C_i^{(\alpha)}$) be the i-th n-bit plaintext (resp. ciphertext) block, i.e., $M_*^{(\alpha)} = M_1^{(\alpha)} \| \cdots \| M_{r_\alpha}^{(\alpha)}$ and $C_*^{(\alpha)} = C_1^{(\alpha)} \| \cdots \| C_{r_\alpha}^{(\alpha)}$. Note that the length of the last block is $\leq n$. Let $\mathsf{u}_\alpha \in [u]$ be the user index corresponding to the α-th query, i.e., if the query is to the ν-th user, then $\mathsf{u}_\alpha = \nu$. For $\nu \in [u]$, let $\mathcal{Q}^{[\nu]} := \{\alpha \in [q] \mid \mathsf{u}_\alpha = \nu\}$ be query indexes for the ν-th user. For $\alpha \in [q]$, let $\mathsf{query}^{(\alpha)}$ be a predicate for the α-th query: if the α-th query is an encryption one, then $\mathsf{query}^{(\alpha)} := \mathsf{enc}$, and else $\mathsf{query}^{(\alpha)} := \mathsf{dec}$. For $\nu \in [u]$, let $\mathcal{P}^{[\nu]} := \{(X_i^{(\alpha)}, Y_i^{(\alpha)}) \mid \alpha \in \mathcal{Q}^{[\nu]}, i \in [0, r_\alpha]\}$ be the set of all input-output tuples of $\pi_\nu^{\pm}$ for the ν-th user.

11.2 Sampling Algorithm for HCTR2

In the decision stage of the ideal world, we define dummy hash keys $(H^{[\nu]})_{\nu \in [u]}$ and dummy internal pairs of plaintext and ciphertext blocks $\{(X_i^{(\alpha)}, Y_i^{(\alpha)}) \mid \alpha \in [q], i \in [0, r_\alpha]\}$ according the structure of $\mathsf{HCTR2}$.

For each $\nu \in [u]$, the hash key is defined as $H^{[\nu]} \xleftarrow{\$} \{0,1\}^n$. For each $\alpha \in [q]$, the dummy internal pairs are defined as follows.

1. $Z_1^{(\alpha)} \leftarrow \mathsf{H}_{H^{[\nu]}}(T^{(\alpha)}, M_*^{(\alpha)})$; $Z_2^{(\alpha)} \leftarrow \mathsf{H}_{H^{[\nu]}}(T^{(\alpha)}, C_*^{(\alpha)})$
2. $X_0^{(\alpha)} \leftarrow Z_1 \oplus M_0$; $Y_0^{(\alpha)} \leftarrow Z_2 \oplus C_0$; $IV^{(\alpha)} \leftarrow X_0^{(\alpha)} \oplus Y_0^{(\alpha)}$
3. **for** $j \in [r_\alpha - 1]$ **do** $X_j^{(\alpha)} \leftarrow \mathsf{add}(IV^{(\alpha)}, j)$; $X_j^{(\alpha)} \leftarrow M_j^{(\alpha)} \oplus C_j^{(\alpha)}$ **end for**
4. $X_{r_\alpha}^{(\alpha)} \leftarrow \mathsf{add}(IV^{(\alpha)}, r_\alpha)$; $Y^* \xleftarrow{\$} \{0,1\}^{n - |M_{r_\alpha}^{(\alpha)}|}$; $Y_{r_\alpha}^{(\alpha)} \leftarrow (M_{r_\alpha}^{(\alpha)} \oplus C_{r_\alpha}^{(\alpha)}) \| Y^*$

11.3 Adversary's View

In the decision stage, all the user's keys and all internal pairs defined for each query are revealed to the adversary $\mathbf{A}$. Then, $\mathbf{A}$'s transcript τ consists of

- (dummy) user's keys $\mathcal{H} = \left\{H^{[\nu]} \mid \nu \in [u]\right\}$,
- query-response tuples $\left\{(T^{(\alpha)}, M^{(\alpha)}, C^{(\alpha)} \mid \alpha \in [q]\right\}$, and
- (dummy) pairs $\left\{(X_i^{(\alpha)}, Y_i^{(\alpha)} \mid \alpha \in [q], i \in [0, r_\alpha]\right\}$.

11.4 Good and Bad Transcripts

We define three conditions on bad transcripts in the following. The set of bad transcripts $\mathcal{T}_{\mathsf{bad}}$ is a subset of $\mathcal{T}$ such that one of the conditions holds. The set of good transcripts is defined as $\mathcal{T}_{\mathsf{good}} := \mathcal{T} \backslash \mathcal{T}_{\mathsf{bad}}$, which is the subset of $\mathcal{T}$ such that no condition holds.

hit_0. The condition is defined as

$$\mathsf{hit}_0 \Leftrightarrow \exists \alpha \in [q], i \in [0, r_\alpha] \text{ s.t. } X_i^{(\alpha)} = \mathtt{bin}_n(0) \vee Y_i^{(\alpha)} = H^{[\mathsf{u}_\alpha]}.$$

In the condition, for some α, the hash key $H^{[\mathsf{u}_\alpha]}$ is used in the process of $\mathsf{HCTR2}$ except for to the hash key generation. In other worlds, assuming that hit_0 does not occur, all hash keys are independent of the process of $\mathsf{HCTR2}$.

coll_X. The condition is defined as

$$\mathsf{coll}_X \Leftrightarrow \exists \nu \in [u], \alpha, \beta \in \mathcal{Q}^{[\nu]}, i \in [0, r_\alpha], j \in [0, r_\beta] \text{ s.t. } (\alpha, i) \neq (\beta, j) \wedge X_i^{(\alpha)} = X_j^{(\beta)}.$$

This condition considers a collision of plaintext blocks in the same user. Assuming that coll_X does not occur, for each user, the plaintext blocks are all distinct.

coll_Y. The condition is defined as

$$\mathsf{coll}_Y \Leftrightarrow \exists \nu \in [u], \alpha, \beta \in \mathcal{Q}^{[\nu]}, i \in [0, r_\alpha], j \in [0, r_\beta] \text{ s.t. } (\alpha, i) \neq (\beta, j) \wedge Y_i^{(\alpha)} = Y_j^{(\beta)}.$$

This condition considers a collision of ciphertext blocks in the same user. Assuming that coll_Y does not occur, for each user, the ciphertext blocks are all distinct.

11.5 Upper-Bounding $\Pr[\mathsf{T}_I \in \mathcal{T}_{\mathsf{bad}}]$

With the bad events hit_0, coll_X and coll_Y, we have

$$\begin{aligned}\Pr[\mathsf{T}_I \in \mathcal{T}_{\mathsf{bad}}] &\leq \Pr[\mathsf{hit}_0] + \Pr[\mathsf{coll}_X] + \Pr[\mathsf{coll}_Y] \\ &\leq \frac{\omega\sigma_{\mathsf{u}}\sigma}{2^n} + \frac{4n\sigma}{2^n} + \frac{\omega\sigma_{\mathsf{u}}\sigma}{2^n} + \frac{\omega\sigma_{\mathsf{u}}\sigma}{2^n} \\ &= \frac{3\omega\sigma_{\mathsf{u}}\sigma}{2^n} + \frac{4n\sigma}{2^n}.\end{aligned}$$

The bounds of $\Pr[\mathsf{hit}_0]$, $\Pr[\mathsf{coll}_X]$, and $\Pr[\mathsf{coll}_Y]$ are given below.

Evaluation of $\Pr[\mathsf{hit}_0]$. For each $\alpha \in [q]$ and $i \in [0, r_\alpha]$, $X_i^{(\alpha)}$ and $Y_0^{(\alpha)}$ are defined by using the regular hash $\mathsf{H}_{H^{[\mathsf{u}_\alpha]}}$, and we have the following bounds:

$$\Pr[X_i^{(\alpha)} = \mathtt{bin}_n(0)] \leq \frac{\omega(t_\alpha + r_\alpha + 1)}{2^n} \text{ and } \Pr[Y_0^{(\alpha)} = H^{[\mathsf{u}_\alpha]}] \leq \frac{\omega(t_\alpha + r_\alpha + 1)}{2^n}.$$

For each $\alpha \in [q]$ and $i \in [r_\alpha]$, since $Y_i^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_i^{(\alpha)} = H^{[\mathsf{u}_\alpha]}] \leq \frac{1}{2^n}.$$

Summing these bounds for each α, i, we have

$$\begin{aligned}\Pr[\mathsf{hit}_0] &\leq \sum_{\nu \in [u]} \sum_{\alpha \in \mathcal{Q}^{[\nu]}} \left(\frac{(r_\alpha + 1) \cdot \omega(t_\alpha + r_\alpha + 1)}{2^n} + \frac{\omega(t_\alpha + r_\alpha + 1)}{2^n} + \frac{r_\alpha + 1}{2^n} \right) \\ &\leq \sum_{\nu \in [u]} \frac{\omega\sigma_\nu^2}{2^n} \leq \frac{\omega\sigma_{\mathsf{u}}\sigma}{2^n}.\end{aligned}$$

Evaluation of $\Pr[\mathsf{coll}_X]$. We first define the following event for evaluating the collision $X_0^{(\alpha)} = X_j^{(\beta)}$. The event considers the case that for some query, the sum of the plaintext and the ciphertext of the right part has the structure of pad.

- $\mathsf{hit}_{\mathsf{pf}} : \exists \alpha \in [q], T' \in \mathcal{T}, D' \in \mathcal{M}$ s.t.
  - $M_*^{(\alpha)} \oplus C_*^{(\alpha)} = \mathsf{pad}_{\mathsf{pf}}(|T'|, 1) \| \mathsf{zp}_n(T') \| D'$ if $|D'| \bmod n = 0$, and

- $M_*^{(\alpha)} \oplus C_*^{(\alpha)} = \mathsf{pad}_{\mathsf{pf}}(|T'|, 0) \| \mathsf{zp}_n(T') \| \mathsf{ozp}_n(D')$ if $|D'| \bmod n \neq 0$.

For each $\alpha \in [q]$, there are at most $2n\ell_\alpha$ candidates for the prefix value $\mathsf{pad}_{\mathsf{pf}}(|T'|, b)$ where $b \in [2]$ and either $\mathsf{msb}_n(M_*^{(\alpha)})$ or $\mathsf{msb}_n(C_*^{(\alpha)})$ is chosen uniformly at random from at least $2^n - q$ elements, we have

$$\Pr[\mathsf{hit}_{\mathsf{pf}}] \leq \sum_{\alpha \in [q]} \frac{2n\ell_\alpha}{2^n - q} \leq \frac{4n\sigma}{2^n},$$

assuming that $q \leq 2^{n-1}$. Hereafter, we assume that $\mathbf{A}$ aborts just after $\mathsf{hit}_{\mathsf{pf}}$ occurs.

For each $\nu \in [u], \alpha, \beta \in \mathcal{Q}^{[\nu]}, j \in [0, r_\beta]$, we consider the following cases regarding the collision $X_0^{(\alpha)} = X_j^{(\beta)}$, where X_0 and X_i are respectively defined as

$$X_0 = M_0 \oplus \mathsf{H}_{H^{[\nu]}}(T, M_*) \text{ and } X_j = M_0 \oplus \mathsf{H}_{H^{[\nu]}}(T, M_*) \oplus C_0 \oplus \mathsf{H}_{H^{[\nu]}}(T, C_*) \oplus A_j.$$

– If $j = 0$, then $\alpha \neq \beta$ must holds, and by the AXU hash $\mathsf{H}_{H^{[\nu]}}$ we have

$$\Pr[X_0^{(\alpha)} = X_0^{(\beta)}] \leq \frac{\omega(\max\{t_\alpha + r_\alpha, t_\beta + r_\beta)\} + 1)}{2^n}.$$

– If $j \geq 1$, then we consider the following cases.
  - If $|\mathsf{pad}(T^{(\alpha)}, M_*^{(\alpha)})|_n \geq |\mathsf{pad}(T^{(\beta)}, M_*^{(\beta)})|_n$, then the prefix value $\mathsf{pad}_{\mathsf{pf}}(|T^{(\beta)}|, j)$ $(\neq 0^n)$ ensures that $\mathsf{lsb}_{|\mathsf{pad}(T^{(\beta)}, M^{(\beta)})|}(\mathsf{pad}(T^{(\alpha)}, M^{(\alpha)}) \oplus \mathsf{pad}(T^{(\alpha)}, C^{(\alpha)})) \neq \mathsf{pad}(T^{(\beta)}, M^{(\beta)})$ holds, and hence the A2XU hash ensures that

$$\Pr[X_0^{(\alpha)} = X_j^{(\beta)}] \leq \frac{\omega(t_\alpha + r_\alpha + 1)}{2^n}.$$

  - If $|\mathsf{pad}(T^{(\alpha)}, M_*^{(\alpha)})|_n < |\mathsf{pad}(T^{(\beta)}, M_*^{(\beta)})|_n$, then since $\mathsf{hit}_{\mathsf{pf}}$ does not occur, $\mathsf{lsb}_{|\mathsf{pad}(T^{(\beta)}, M^{(\beta)})|}(\mathsf{pad}(T^{(\alpha)}, M^{(\alpha)}) \oplus \mathsf{pad}(T^{(\alpha)}, C^{(\alpha)})) \neq \mathsf{pad}(T^{(\beta)}, M^{(\beta)})$ holds. Hence, the A2XU hash ensures that

$$\Pr[X_0^{(\alpha)} = X_i^{(\beta)}] \leq \frac{\omega(\max\{t_\alpha + r_\alpha, t_\beta + r_\beta\} + 1)}{2^n}.$$

By the above bounds, we have

$$\Pr[X_0^{(\alpha)} = X_i^{(\beta)}] \leq \frac{\omega(\max\{t_\alpha + r_\alpha, t_\beta + r_\beta)\} + 1)}{2^n}.$$

We next consider a collision $X_i^{(\alpha)} = X_j^{(\beta)}$, where $\nu \in [u], \alpha, \beta \in \mathcal{Q}^{[\nu]}$, $i \in [r_\alpha]$, and $j \in [r_\beta]$. Without loss of generality, we assume that $\alpha \leq \beta$. Since $M^{(\beta)}$ (resp. $C^{(\beta)}$) is chosen uniformly at random from at least $2^n - q$ elements in $\{0,1\}^n$, we have

$$\Pr\left[\begin{array}{l}\exists (i,j) \in [r_\alpha] \times [r_\beta] \text{ s.t.} \\ \quad X_i^{(\alpha)} = X_j^{(\beta)} \wedge (\alpha, i) \neq (\beta, j)\end{array}\right] \leq \begin{cases} 0 & \text{if } \alpha = \beta, \\ \dfrac{r_\alpha r_\beta}{2^n - q} \leq \dfrac{2 r_\alpha r_\beta}{2^n} & \text{if } \alpha \neq \beta, \end{cases}$$

assuming $q \leq 2^{n-1}$.

By the above bounds, we have

$$\begin{aligned}\Pr[\mathsf{coll}_X] &\le \frac{4n\sigma}{2^n} + \sum_{\nu\in[u]}\left(\sum_{\alpha\in\mathcal{Q}^{[\nu]}}\left(\frac{r_\alpha^2}{2^n} + \frac{\omega(t_\alpha + r_\alpha + 1)r_\alpha}{2^n}\right)\right. \\ &\qquad \left. + \sum_{\substack{\alpha,\beta\in\mathcal{Q}^{[\nu]}\\ \text{s.t. } \alpha<\beta}} \frac{\omega(\max\{t_\alpha + r_\alpha, t_\beta + r_\beta\} + 1)^2}{2^n}\right) \\ &\le \frac{4n\sigma}{2^n} + \sum_{\nu\in[u]}\left(\frac{\omega\sigma_\nu^2}{2^n}\right) \\ &\le \frac{4n\sigma}{2^n} + \frac{\omega\sigma_{\mathsf{u}}\sigma}{2^n}.\end{aligned}$$

Evaluation of $\Pr[\mathsf{coll}_Y]$. For each $\nu \in [u], \alpha, \beta \in \mathcal{Q}^{[\nu]}$, the AXU hash function $\mathsf{H}_{H^{[\nu]}}$ ensures that

$$\Pr[Y_0^{(\alpha)} = Y_0^{(\beta)}] \le \frac{\omega(\max\{t_\alpha + r_\alpha, t_\alpha + r_\alpha)\} + 1)}{2^n}.$$

For each $i \in [r_\alpha], j \in [0, r_\beta]$, since $Y_i^{(\alpha)}$ is chosen uniformly at random from $\{0,1\}^n$, we have the following bound:

$$\Pr[Y_i^{(\alpha)} = Y_j^{(\beta)}] \le \frac{1}{2^n}.$$

Summing the bounds for each each ν, α, β, we have the following bound:

$$\begin{aligned}\Pr[\mathsf{coll}_Y] &\le \sum_{\nu\in[u]}\sum_{\substack{\alpha,\beta\in\mathcal{Q}^{[\nu]}\\ \text{s.t. } \alpha<\beta}} \frac{\omega(\max\{t_\alpha + r_\alpha, t_\alpha + r_\alpha)\} + 1) + r_\alpha(r_\beta + 1)}{2^n} \\ &\le \sum_{\nu\in[u]} \frac{\omega\sigma_\nu^2}{2^n} \\ &\le \frac{\omega\sigma_{\mathsf{u}}\sigma}{2^n}.\end{aligned}$$

11.6 Lower-Bounding $\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]}$

For each good transcript τ, we show that $\frac{\Pr[\mathsf{T}_R=\tau]}{\Pr[\mathsf{T}_I=\tau]} \ge 1$. In this evaluation, we use the symbol "$\star$" for values and sets corresponding to a transcript τ. For a set $\mathcal{L}$, let $\mathsf{T}_R \vdash \mathcal{L}$ (resp. $\mathsf{T}_I \vdash \mathcal{L}$) be an event that T_R (resp. T_I) satisfies all elements in $\mathcal{L}$.

Fix a good transcript τ that consists of the hash keys $\mathcal{H}$ and pairs of plaintext and ciphertext blocks.

Evaluation of $\Pr[\mathsf{T}_W \vdash \mathcal{H}^\star]$ for $W \in \{R, I\}$. In the real world, each hash key $H^{[\nu]}$ is chosen uniformly at random from $\{0,1\}^n$ by $H^{[\nu]} = \pi(\mathtt{bin}_n(0))$. In the ideal world, each hash key $H_i^{[\nu]}$ is chosen uniformly at random from $\{0,1\}^n$. We thus have $\Pr[\mathsf{T}_R \vdash \mathcal{H}^\star] = \Pr[\mathsf{T}_I \vdash \mathcal{H}^\star]$.

Evaluation of $\Pr[\mathsf{T}_W \vdash (\mathcal{P}_1^{[\nu]\star}, \mathcal{P}_2^{[\nu]\star})_{\nu\in[u]} \mid \mathcal{H}^\star]$ for $W \in \{R, I\}$. For $\nu \in [u]$, let $\tau_\nu \coloneqq \{(X_i^{(\alpha)\star}, Y_i^{(\alpha)\star}) \mid \alpha \in \mathcal{Q}^{[\nu]}, i \in [0, r_\alpha]\}$ be the set of internal pairs of the ν-th user in τ, and $N_\nu \coloneqq |\tau_\nu|$.

Real-World Probability. By $\neg\mathsf{coll}_X$ and $\neg\mathsf{coll}_Y$, the real-world probability is not 0. We then have

$$\Pr[\mathsf{T}_R \vdash \tau_\nu] = \prod_{i\in[N_\nu]} \frac{1}{2^n - i}.$$

Ideal-World Probability. In the ideal world, for each $\alpha \in \mathcal{Q}^{[\nu]}$, if $\mathsf{query}_\alpha = \mathsf{enc}$ (resp. $\mathsf{query}_\alpha = \mathsf{dec}$), then $C^{(\alpha)}$ (resp. $M^{(\alpha)}$) is chosen uniformly at random from at least $2^{|M^{(\alpha)}|} - (\alpha - 1)$ elements such that $|M^{(\alpha)}| \geq n$ and the truncated value Y^* is chosen uniformly at random from $\{0,1\}^{n-|M^{(\alpha)}_{r_\alpha}|}$, thus each internal pair in the ideal world is defined from the space larger than the real world. We thus have

$$\Pr[\mathsf{T}_I \vdash \tau_\nu] \leq \Pr[\mathsf{T}_R \vdash \tau_\nu].$$

Deriving the Lower Bound. Using the above bounds, we have

$$\frac{\Pr[\mathsf{T}_R = \tau]}{\Pr[\mathsf{T}_I = \tau]} \geq \prod_{\nu\in[u]} \frac{\Pr[\mathsf{T}_R \vdash \tau_\nu]}{\Pr[\mathsf{T}_I \vdash \tau_\nu]} \geq 1.$$

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